

Agriculture

Currently, in the United States, approximately one quarter of farmland is planted with crops that are genetically modified. Most are resistant to certain insect pests because they contain genes from *Bacillus thuringiensis* (often referred to as Bt). These genes are for insecticidal proteins called δ endotoxins. For example, the proteins Cry1A and Cry1C from *Bacillus thuringiensis* protect the plants against larval forms of lepidopterans such as the European corn borer. Cry3A protects against coleopterans such as the Colorado potato beetle. In excess of fifty genetically altered crop plants have been approved for planting, including those protected against insect pests, frost, and premature ripening. Rice is being modified so that its vitamin A potential is maintained even after the husks are removed, a procedure done to allow for storage since the husks become rancid. That change alone will improve the health of millions of people throughout the world. Box 13.1 discussed some of the ethical concerns surrounding genetically modified crop plants.

Industry

Industrial applications of biotechnology include engineering bacteria to break down toxic wastes, modifying yeast to use cellulose to produce glucose and alcohol for fuel, using algae in mariculture (the cultivation of marine organisms in their natural environments) to produce both food and other useful substances, and developing better food processing methods and waste conversion. As an example, baker's yeast (*Saccharomyces cerevisiae*) has been modified with a plasmid that contains two cellulase genes, an endoglucanase and an exoglucanase, that convert cellulose to glucose. The yeast can then convert glucose to ethyl alcohol. These yeasts are now capable of digesting wood (cellulose) and converting it directly to alcohol. The potential exists to harvest the alcohol the yeast produces as a fuel to replace fossil fuels that are in dwindling supply and are polluting the planet.

As you can see, there is no one direction that biotechnology is taking. Many advances are being made that will probably affect every person's life in a beneficial way. Cautious optimism is certainly in order.

S U M M A R Y

STUDY OBJECTIVE 1: To look at the techniques of gene cloning 359–377

Recombinant DNA techniques revolve around the cloning of foreign DNA in a plasmid or phage. Cloned DNA can be amplified, expressed, and sequenced. Gene cloning techniques came about with the discovery of restriction endonucleases. Type II restriction endonucleases cleave DNA at palindromic regions, which have twofold symmetry.

Recombinant vectors can be constructed several different ways. Foreign and vector DNA can be made compatible by treating each with the same restriction endonuclease—each will then have the same sticky ends. If that does not work, T4 DNA ligase can join blunt ends. In a variation of this method, linkers containing restriction sites are added to vector and foreign DNA. These linkers are then treated with a restriction endonuclease that gives the DNA sticky ends.

DNA to be cloned can be synthesized from an RNA template (cDNA) or isolated by various techniques. If messenger RNA is available, it can be converted into a clonable complementary DNA with the enzyme reverse transcriptase. If DNA is to be isolated directly, it must be identified among all the other DNA fragments created. Locating a desirable piece of DNA is done with probes, complementary nucleic acids labeled with radioactivity or chemiluminescence. Southern blotting, a transfer technique, is used first,

followed by DNA-DNA or DNA-RNA hybridization and autoradiography. If the DNA is cloned first, as in the creation of a genomic library, probes can be created or expression of the cloned gene can be determined.

Eukaryotic vectors have been developed, including yeast plasmids, tumor virus vehicles in animals, and crown gall tumor plasmids in plants. Eukaryotes can be transfected by foreign DNA and express it in transgenic organisms. DNA can be injected, shot in on projectiles, electroporated, or introduced by viruses, plasmids, or liposomes. Knockout mice, lacking a specific gene, can be created.

STUDY OBJECTIVE 2: To examine the techniques of creating restriction maps 377–383

Restriction digests can be separated by electrophoresis, then used to construct a restriction map. This is a map of the DNA showing the location of restriction enzyme recognition sites. The genetic maps, generated by mating analysis, can then be superimposed on the restriction maps, locating regions of interest on the physical map. Restriction fragment length polymorphisms (RFLPs) provide a tool for locating genes through linkage analysis and are also valuable in forensic science. The polymerase chain reaction (PCR) is a technique used to rapidly amplify particular segments of DNA.

STUDY OBJECTIVE 3: To study the methods of DNA sequencing 383–390

DNA is usually sequenced by one of two methods. The dideoxy method developed by Sanger and his colleagues requires the synthesis of DNA in the presence of chain-terminating (dideoxy) nucleotides. Electrophoresis followed by autoradiography allows the sequence of nucleotides synthesized to be determined directly. Fluorescent labeling allows computerized sequence determinations. The phage ϕ X174 was sequenced in its entirety through the forerunner of this technique, the plus-and-minus method. Gilbert and Maxam's chemical method also is used widely.

STUDY OBJECTIVE 4: To look at the goals and methods of the Human Genome Project 390–397

The Human Genome Project is a massive, international effort to map and sequence all 3.3 billion bases of the human

genome. Initial success was announced in the spring of 2000. Modern linkage maps are being created of restriction sites, microsatellite markers, sequence-tagged sites, and single-nucleotide polymorphisms. They are being coordinated with physical maps created with overlapping contiguous clones of chromosomes. These techniques currently allow us to find genes of interest. The project also includes the sequencing of the genomes of other relevant organisms.

STUDY OBJECTIVE 5: To look at the practical benefits and human issues of genetic engineering 397–398

Genetic engineering is moving forward on a number of fronts. Medical, agricultural, and industrial applications are becoming widespread.

S O L V E D P R O B L E M S

PROBLEM 1: A piece of eukaryotic DNA is obtained by using a restriction endonuclease that leaves blunt ends (*Hae*III). How could we get this piece of DNA into a *Bam*HI site in plasmid pBR322, and how would we know when the foreign DNA has been cloned?

Answer: Since the two pieces of DNA (the eukaryotic piece and the plasmid) have different ends, they must be made compatible before cloning. The simplest way would be to attach blunt-ended linkers to the foreign DNA with phage T4 DNA ligase (see fig. 13.9). The linkers, of course, would have a *Bam*HI site within. After the linkers are attached to the foreign DNA, it would be treated with the *Bam*HI restriction enzyme, giving the foreign DNA *Bam*HI ends. The plasmid is then also treated with the restriction enzyme and the two (the foreign DNA and the cut plasmid) are now mixed together in the presence of *E. coli* DNA ligase, which seals up the plasmids, with or without cloned inserts (see fig. 13.6). Since they have compatible ends, some of the time, a piece of foreign DNA is inserted into a plasmid. The plasmids are then taken up by *E. coli* cells that are grown overnight in an incubator. The bacterial colonies are then replica-plated on media with the antibiotics ampicillin or tetracycline. Colonies that are resistant to ampicillin but sensitive to tetracycline are assumed to be bacteria containing plasmids with cloned inserts (see fig. 13.8).

PROBLEM 2: How does a reporter system work?

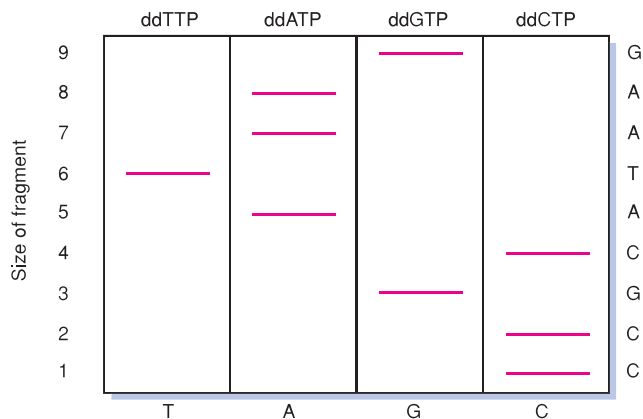
Answer: A reporter system is a genetically manipulated system that displays a particular phenotype or reaction

when a desired event has taken place. In this chapter, we discussed the firefly luciferase reporter system in which the desired result (transcription of a particular promoter) causes a transgenic tobacco plant to glow. Let us say that we are studying the control of transcription of a particular eukaryotic gene. We could attach the promoter of that gene to the firefly luciferase gene in a Ti plasmid by cloning techniques. The plasmid could then transfect tobacco plants, and we could continue our experiment to determine whether the promoter under study is active under various conditions. We would know whether it was active by watering the plants with luciferin. If the plant glows, then the luciferase gene product is present, which means that the promoter under question is active. In other words, the glowing of the plant “reports” the action of the promoter under question; the promoter is active because it allowed the transcription of the luciferase gene. We also discussed the green fluorescent protein reporter system.

PROBLEM 3: A piece of DNA has the sequence 3'-GGCG-TATTC-5'. It is sequenced using the dideoxy method. How many bands are found on the ladder gel? How many bands and of what size are found for each reaction mixture?

Answer: Since the piece of DNA is nine bases long, the total number of bands in all four lanes of a sequencing gel add up to nine (see fig. 13.33). By each reaction mixture, we mean the four reaction mixtures each with one of the dideoxynucleotides. In the reaction mixture with ddTTP,

chain termination occurs at the adenine in the piece of DNA; that is, a DNA segment was synthesized that is six bases long (see the following figure). In the reaction mixture with ddATP, chain termination occurs opposite each of the thymines, producing DNA segments of five, seven, and eight nucleotides. In the reaction mixture with ddGTP, chain termination occurs opposite the cytosines in positions three and nine. And, in the reaction mixture with ddCTP, chain termination occurs after synthesis of segments one, two, and four bases long. Note that the gel gives us the sequence of the complement strand of the original piece of single-stranded DNA.



PROBLEM 4: A linear DNA molecule 1,000 base pairs long is digested with the following restriction enzymes, producing the following results:

<i>Eco</i> RI	400 bp, 600 bp
<i>Bgl</i> II	250 bp, 750 bp
<i>Eco</i> RI + <i>Bgl</i> II	250, 350, 400 bp

Determine the restriction map.

Answer: Each enzyme alone produces two fragments, so the molecule has one site for each enzyme. Since we get different-sized fragments with each enzyme, the sites must be located asymmetrically along the DNA. Draw these sites:

*Eco*RI *Bgl*II
400 ↓ 600 and 250 ↓ 750

The *EcoRI* fragment that lacks a *BglIII* site should appear in the double digest. If *BglIII* cuts within the 400 base-pair fragment, we would expect to see 150, 250, and 600 base-pair fragments. We don't see this, so the *BglIII* site is not within the 400 base-pair *EcoRI* fragment. Thus, the map looks like this:

*Eco*RI *Bgl*II

400 ↓ 350 ↓ 250

EXERCISES AND PROBLEMS *

GENOMIC TOOLS

1. What specific properties of type II endonucleases make them useful in gene cloning?
2. The following is a double helix of DNA. What, if any, are potential restriction enzyme recognition sequences?

5'-TAGAATTCGACGGATCCGGGGCATGCAGATCA-3'
3'-ATCTTAAGCTGCCTAGGCCCGTACGTCTAGT-5'
3. Assuming a random arrangement of nucleotides on a piece of DNA, what is the probability that a restriction endonuclease whose recognition site consists of four bases (a four-cutter) will cut the DNA? What is the probability for a six-cutter? an eight-cutter?
4. Under what circumstances is a restriction endonuclease unsuitable for cloning a piece of foreign DNA?

5. What methods exist to create sticky ends or create ends for joining two incompatible pieces of DNA? When is each method favored?
6. Diagram a possible heteroduplex between two phage λ vectors, one with and one without a cloned insert, created by DNA-DNA hybridization.
7. What are the differences among plasmid, cosmid, expression vector, and YAC? Under what circumstances is each useful?
8. What are the steps by which messenger RNA can be converted into cDNA? How would we obtain radioactive cDNA? radioactive messenger RNA?
9. What is chromosome walking? When is it used?
10. How would we isolate a human alanine transfer RNA gene for cloning? How would we locate a clone with a human alanine transfer RNA gene in a genomic library?
11. What are the differences among Southern, western, northern, and dot blotting?

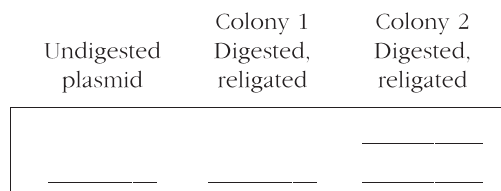
*Answers selected exercises and problems are on page A-14.

12. How would you develop a probe for a gene whose messenger RNA could not be isolated? How could an expression vector be used to isolate a cloned gene?
13. How are *E. coli* plasmids manipulated to survive in yeast? How can virus genomes, such as SV40 and phage λ , survive as functioning vectors when parts of their genomes are replaced by cloned DNA?
14. What methods are used to get foreign DNA into eukaryotic cells? What is transfection? What is a transgenic mouse?
15. Exonuclease III is an enzyme that sequentially removes bases from the 3' end of double-stranded DNA. The following two molecules, each 100 bp long, are digested with exonuclease III. Molecule 1 is completely digested; molecule 2 is only partially digested. Explain these results.

Molecule 1: CGTTCAG...
GCAAGTC...

Molecule 2: AAAAAAAAAA...
TTTTTTTTTT...

16. A plasmid that contains an *EcoRI* site within a gene for ampicillin resistance is cut with *EcoRI*, and then religated. This plasmid is used to transform *E. coli* cells, and the plasmid is reisolated from the ampicillin-resistant colonies. The reisolated plasmids from two different colonies are electrophoresed, and the results appear in the following figure.



How do you account for the two bands in colony 2?

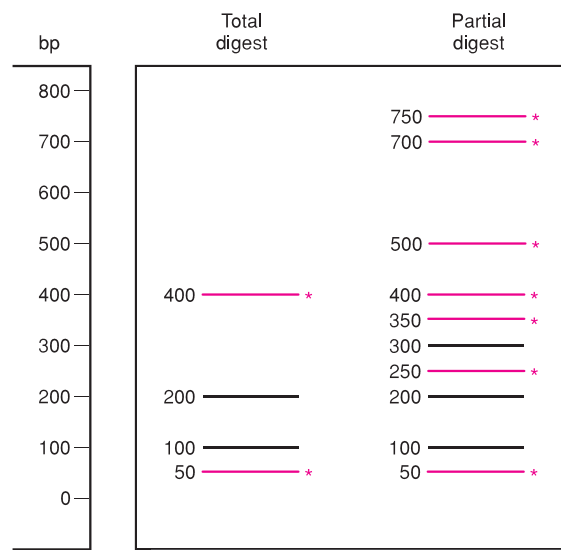
17. Most human genes contain one or more introns. Since bacteria cannot excise introns from nuclear messenger RNA (snRNPs are needed), how can bacteria be used to make large quantities of a human protein?

RESTRICTION MAPPING

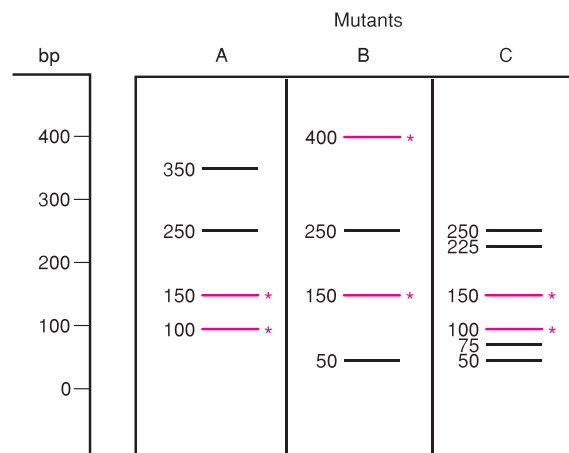
18. How are DNA fingerprints useful in forensic cases? Could they be used in paternity exclusion?
19. The following segment of DNA is cut four times by the restriction endonuclease *EcoRI* at the places shown. Diagram the gel banding that would result from electrophoresis of the total and partial digests. Note the end-labeled segments and regions where several segments form bands at the same place on the gel.



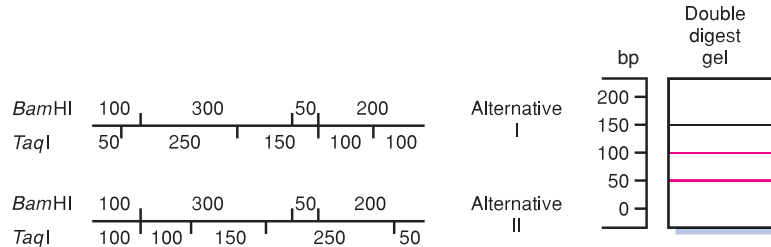
20. The following figure shows a gel of a total and partial digest of a DNA segment treated with *HindIII*. End-labeled segments are noted by asterisks. Draw the restriction map of the original segment.



21. Several mutants of the DNA segment shown in problem 19 were isolated. They gave the following gel patterns when the total digests were electrophoresed. Asterisks denote the end-labeled segments. Can you determine the nature of the mutations?



22. Restriction maps of a segment of DNA were worked out separately for *Bam*HI and *Taq*I. Two overlays of the maps are possible. The double-digest gel is shown in the following figure (asterisks denote end labels). Which overlay is correct?



23. A linear DNA molecule 1,000 bp long gives the following size fragments when treated with these restriction enzymes. Derive a restriction map.

EcoRI: 300 bp, 700 bp
BamHI: 150 bp, 200 bp, 250 bp, 400 bp
EcoRI + BamHI: 50 bp, 100 bp, 200 bp, 250 bp, 400 bp

24. A linear DNA molecule cut with *EcoRI* yields fragments of 3 kb, 4.2 kb, and 5 kb. What are the possible restriction maps?
25. You have double-stranded DNA that you radioactively label at the 5' ends. Digestion of this molecule with either *EcoRI* or *BamHI* yields the following fragments. The numbers are in kilobases (kb), and an asterisk indicates the fragments that are labeled.

EcoRI: 2.8, 4.6, 6.2*, 7.4, 8.0*
BamHI: 6.0*, 10.0*, 13.0

If unlabeled DNA is digested with both enzymes simultaneously, the following fragments appear: 1.0, 2.0, 2.8, 3.6, 6.0, 6.2, 7.4. What is the restriction map for the two enzymes?

26. A 12 kb DNA molecule cut with *EcoRI* yields one 12 kb fragment. When the original molecule is cut with *BamHI*, three fragments of 2 kb, 4.5 kb, and 5.5 kb are produced. When the fragment from *EcoRI* is treated with *BamHI*, four fragments of 2 kb, 2.5 kb, 3.0 kb, and 4.5 kb are produced. Draw a restriction map.
27. A plasmid 3 kb in length contains a gene for ampicillin resistance and a gene for tetracycline resistance. The plasmid has a single site for each of the following enzymes: *EcoRI*, *BglII*, *HindIII*, *PstI*, and *SalI*. If DNA is cloned into the *EcoRI* site, resistance to either antibiotic is not affected. DNA cloned into the *BglII*, *HindIII*, or *SalI* sites abolishes tetracycline resistance, and DNA inserted into the *PstI* site elimi-

nates ampicillin resistance. If the plasmid is digested completely with enzyme mixes, the following fragments result:

Mixture	Fragment Size (kb)
<i>EcoRI + PstI</i>	0.7, 2.3
<i>EcoRI + BglII</i>	0.3, 2.7
<i>EcoRI + HindIII</i>	0.08, 2.92
<i>EcoRI + SalI</i>	0.85, 2.15
<i>EcoRI + BglII + PstI</i>	0.3, 0.7, 2.00

Draw a restriction map of the plasmid, and indicate the locations of the resistance genes and the sites of enzymatic cleavage.

28. A gene has the following *EcoRI* restriction map (in kilobases):

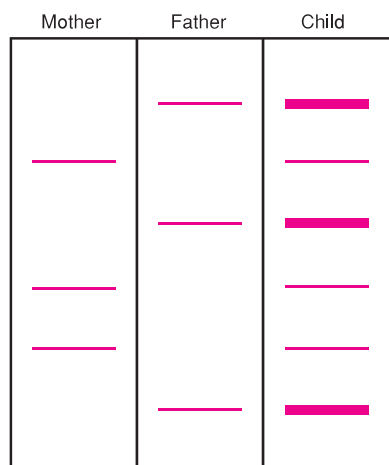
1.0 ↓ 0.7 ↓ 2.0

Draw the gel pattern expected from

- a. a mutant that has lost the site between the 1.0 and 0.7 kb fragments.
 b. a mutant that has a new site within the 2.0 kb fragment.
29. A DNA fragment 8 kb in size is labeled with ^{32}P at the 5' ends. It is then digested with *EcoRI*, *BglII*, or a mixture of both enzymes. The size of the fragments and the labeled fragments (*) appear as follows. Sizes are in kilobases.

	<i>EcoRI</i>	<i>BglII</i>	Mix
3.5	_____*		
3.0	_____	_____*	_____*
2.0		_____	_____
1.5		_____	
1.0	_____	_____*	_____
0.5	_____*	_____	_____*

Which of the following two maps is consistent with the results?



36. Provide, if possible, DNA sequences that can mark the termination of one gene and the initiation of another, given that the genes overlap in one, two, three, four, five, six, or seven bases.

37. Draw the expected gel pattern derived from the dideoxy sequencing method for a template strand with the following sequence:

5'-CAGCGAATGCGGAA-3'

38. A DNA strand with the sequence 3'-GACTATTCCGAAAC-5' is sequenced by the dideoxy method. If the reaction mixture contains all four radioactive deoxynucleotide triphosphates plus dideoxythymidine, what size labeled bands do you expect to see on the gel?

MAPPING AND SEQUENCING THE HUMAN GENOME

39. What is hypervariable DNA? a RFLP? a VNTR locus? microsatellite DNA? a sequence-tagged site? (See also RESTRICTION MAPPING)

PRACTICAL BENEFITS FROM GENE CLONING

40. Describe some areas of practical benefit from genetic engineering. Why might some people be concerned about its widespread use?

CRITICAL THINKING QUESTIONS

1. In the past, geneticists have used several different methods to splice pieces of DNA that do not have compatible "sticky ends." We mentioned blunt-end ligation and the addition of linkers containing specific restriction sites. Given that nucleotides can be added to the 3' ends of double-stranded DNA with the enzyme deoxynucleoside terminal transferase, can you see another way to create compatible ends on foreign and vehicle DNA?
2. The motion picture *Jurassic Park* was based on the premise that DNA of dinosaurs could be extracted from the blood-meals of mosquitoes preserved in amber and inserted into the genome of a frog, which would then produce living dinosaurs. Is this premise reasonable?

Suggested Readings for chapter 13 are on page B-11.

14

GENE EXPRESSION

Control in Prokaryotes and Phages

STUDY OBJECTIVES

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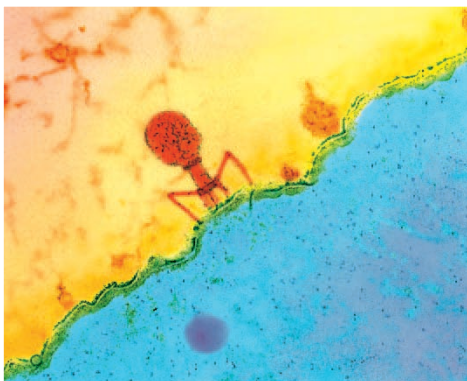
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Artificially colored transmission electron micrograph of a T4 bacteriophage attached to an *Escherichia coli* bacterium. (© Biozentrum, University of Basel/SPL/Photo

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Genes are transcribed into RNA, which, for the most part, is then translated into protein. Control mechanisms are exercised along the way. Without some control of gene expression, an *Escherichia coli* cell, for example, would produce all its proteins in large quantities all the time, and all the cells in a eukaryotic organism would be identical. Although most control mechanisms are negative (preventing something from happening), controls can also be positive (causing some action to occur or enhancing some action). This chapter is devoted to analyzing control processes in prokaryotes and phages; in chapter 16, we examine control processes in eukaryotes.

In the process leading from a sequence of nucleotides in DNA to a protein, control is exerted in many places. In general, control of gene expression can take place at the levels of transcription, translation, or protein functioning. The most efficient place to control gene expression is at the level of transcription.

One of the best-understood mechanisms exerts control of transcription, regulating the production of messenger RNA according to need. *E. coli* messenger RNAs are short-lived in vivo: They degrade enzymatically within about two minutes. A complete turnover (degradation and resynthesis) in the cell's messenger RNA occurs rapidly and continually, and this rapid turnover is a prerequisite for transcriptional control, a central feature of the regulation of prokaryotic gene expression.

THE OPERON MODEL

Not all of the proteins prokaryotes can produce are needed in all circumstances in the same quantities. For example, some metabolites, such as sugars, which the cell breaks down for energy and as a carbon source, may not always be present in the cell's environment. If a given metabolite is not present, enzymes for its breakdown are not useful, and synthesizing these enzymes is wasteful. If the cell produces enzymes for the degradation of a particular carbon source only when this carbon source is present in the environment, the enzyme system is known as an **inducible system**. Inducible enzymes are synthesized when the environment includes a substrate for those enzymes. The enzymes will then catabolize (break down) the substrate.

On the other hand, the enzymes in many synthetic pathways are in low concentration or absent when an adequate quantity of the end product of the pathway is already available to the cell. That is, if the cell encounters an abundance of the amino acid tryptophan in the envi-

ronment or if it is overproducing tryptophan, the cell stops the manufacture of tryptophan until a need arises again. A **repressible system** is a system of enzymes whose presence is repressed, stopping the production of the end product when it is no longer needed. Repressible systems are repressed by an excess of the end product of their synthetic (anabolic) pathway.

The best-studied inducible system is the ***lac* operon** in *E. coli*. Since the term *operon* refers to the control mechanism, we will defer a definition until we describe the mechanism.

LAC OPERON (INDUCIBLE SYSTEM)



Lactose Metabolism

Lactose (milk sugar—a disaccharide) is a β -galactoside that *E. coli* can use for energy and as a carbon source after it is broken down into glucose and galactose. The enzyme that performs the breakdown is **β -galactosidase** (fig. 14.1). (The enzyme can additionally convert lactose to allolactose, which, as we will see, is also important.) There are very few molecules of β -galactosidase in a wild-type *E. coli* cell grown in the absence of lactose. Within minutes after adding lactose to the medium, however, this enzyme appears in quantity within the bacterial cell. When the synthesis of β -galactosidase (encoded by the *lacZ*, or *z* gene) is induced, the production of two additional enzymes is also induced: **β -galactoside permease** (encoded by the *lacY*, or *y* gene) and **β -galactoside acetyltransferase** (encoded by the *lacA*, or *a* gene). The permease is involved in transporting lactose into the cell. The transferase is believed to protect the cell from the buildup of toxic products created by β -galactosidase acting on other galactosides. By acetylating galactosides other than lactose, the transferase prevents β -galactosidase from cleaving them.

The Regulator Gene



Not only are the three *lac* genes (*z*, *y*, *a*) induced together, but they are adjacent to one another in the *E. coli* chromosome; they are, in fact, transcribed on a single, polycistronic messenger RNA (fig. 14.2). Induction involves the protein product of another gene, called the **regulator gene**, or *i* gene (*lacI*). Although the regulator gene is located adjacent to the three other *lac* genes, it is a totally independent transcriptional entity. The regulator specifies a protein, called a **repressor**, that interferes with the transcription of the genes involved in lactose metabolism.

Lac Operon (Inducible System)

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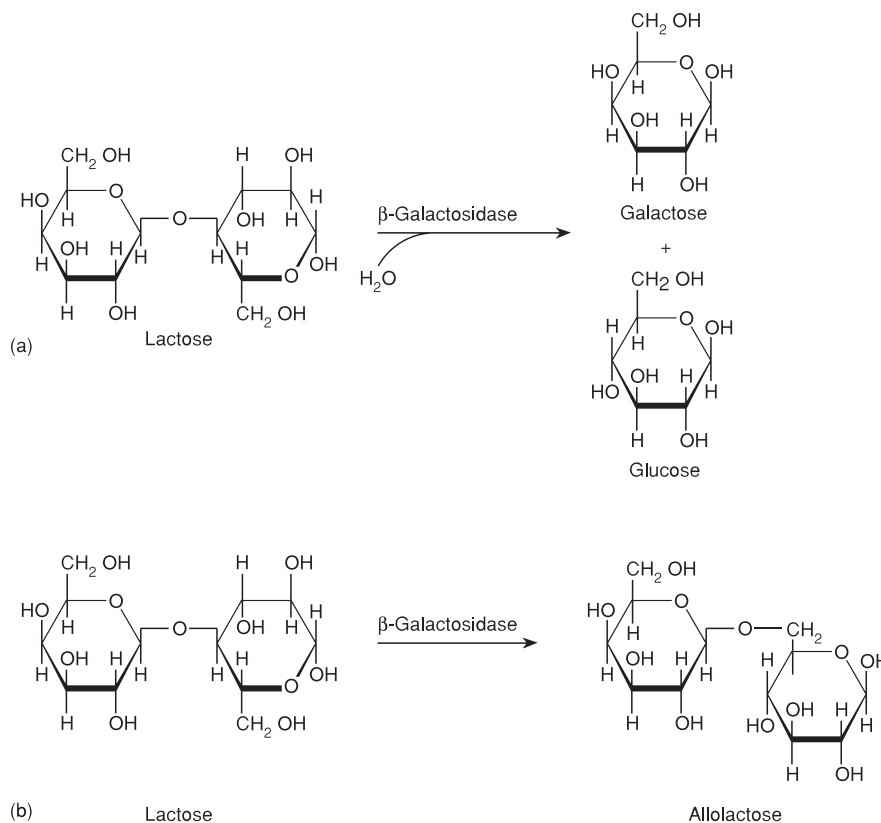


Figure 14.1 The enzyme β -galactosidase hydrolytically cleaves lactose into glucose and galactose (a). The enzyme can also convert lactose to allolactose (b).

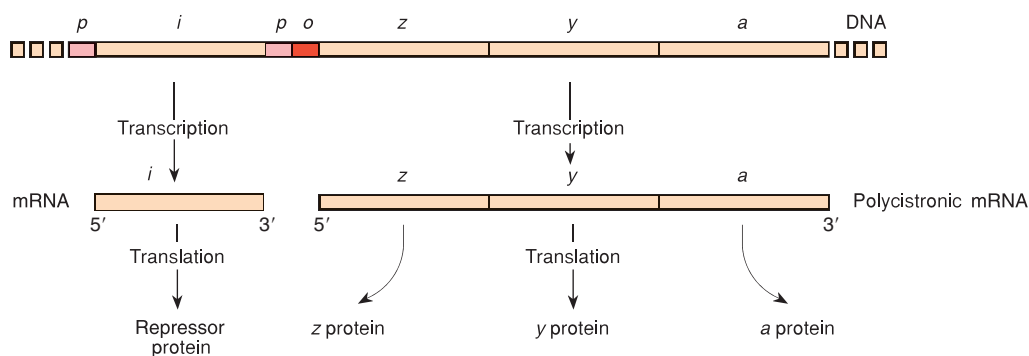


Figure 14.2 The *lac* operon is transcribed as a multigenic (polycistronic) mRNA. The *z*, *y*, and *a* indicate the *lacZ*, *lacY*, and *lacA* loci. The mRNA transcript is then translated as individual proteins. The *lac* operon regulator gene is denoted as *i*; the *o* stands for operator and the *p* for promoter. Both the operon and the regulator gene have their own promoters. (Source: Data from R. C. Dickson, et al., "Genetic regulation: The *lac* control region," *Science*, 187:27–35, January 10, 1975.)

The Operator

For the repressor protein to exert its influence over transcription, there must be a control element (receptor site) located near the beginning of the β -galactosidase (*lacZ*) gene. This control element is a region referred to as the **operator**, or operator site (fig. 14.2). The operator site is a sequence of DNA that the product of the regulator gene, the repressor, recognizes. When the repressor is bound to the operator, it either interferes with RNA polymerase binding or prevents the RNA polymerase from achieving the open complex (see chapter 10). In either case, transcription of the operon is prevented (fig. 14.3). The repressor is released when it combines with an *inducer*, a derivative of lactose called allolactose (see fig. 14.1).

Note that the promoter not only is recognized by RNA polymerase but also has other controlling elements in the immediate vicinity of the initiation site of transcription. We can now define an **operon** as a sequence of adjacent genes all under the transcriptional control of the same promoter and operator.

The nucleotide sequence of the *lac* operator region is shown in figure 14.4. The operator in figure 14.3 is referred to as the primary operator, o_1 , centered at +11. Two other operator sequences have been found. One, o_2 , is centered at +412. The third overlaps the C-terminal end of the *i* gene, is centered at -82, and is referred to as o_3 . The structure of the repressor and its interaction with the operator sites was worked out recently with X-ray crystallography. The functional repressor is a homotetramer of the protein product of

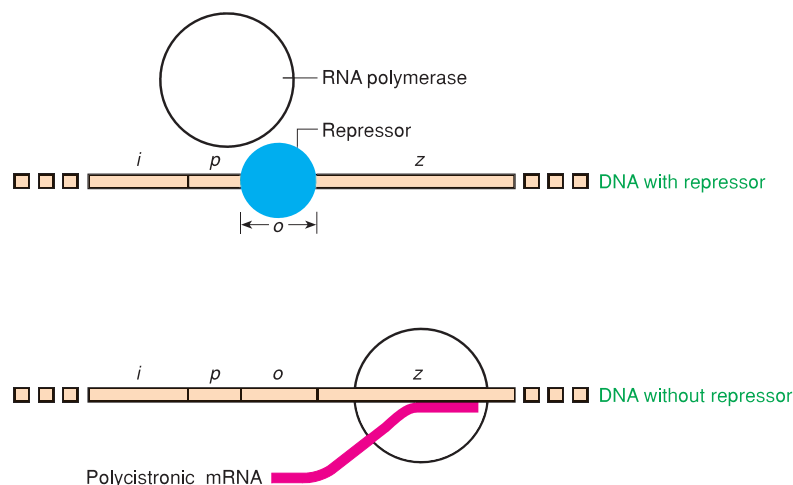


Figure 14.3 The repressor. By binding to the operator, the repressor either prevents RNA polymerase from binding to the promoter and transcribing the *lac* operon as shown, or prevents the polymerase from achieving the open configuration. In either case, transcription of the *lac* operon is prevented. When the repressor is not present, transcription takes place. The functional repressor is a tetramer.

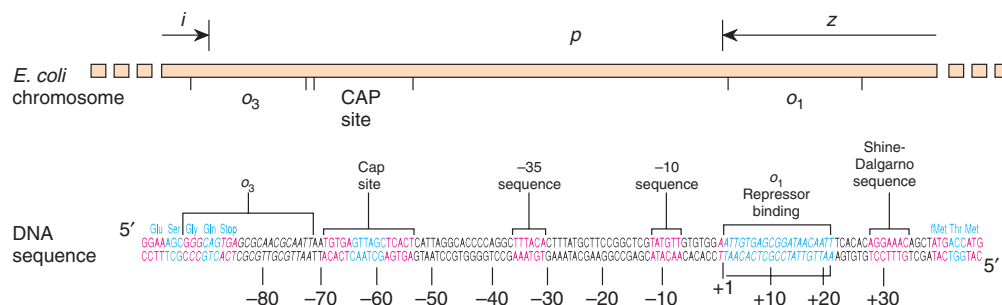


Figure 14.4 The *lac* operon promoter and operator regions. The CAP site is described later. The base sequence corresponds to the diagram above it. The terminal amino acids of the *i* gene are shown, as well as the initial amino acids of the *lacZ* gene. In addition, we picture the Shine-Dalgarno sequence of the DNA, the repressor-binding region (centered at around +10 of the gene), the -10 and -35 sequences of the promoter, and primary (o_1) and secondary (o_3) operator sites (see text). (Data from R. C. Dickson, et al., "Genetic regulation: The *lac* control region," *Science*, 187:27-35, January 10, 1975.)

the *i* gene; that is, it is formed from four identical copies of the repressor protein. Since each operator site has twofold symmetry, two repressor monomer proteins bind to each operator site. The monomer is shaped so that it fits into the major groove of the DNA to locate the exact base sequence of the operator; it then binds at that point through electrostatic forces. A tetramer can bind to two of the operator sites at the same time, presumably o_1 and o_3 or o_1 and o_2 . In the process, the DNA is formed into a loop (fig. 14.5).

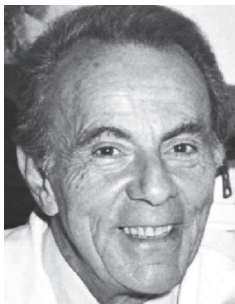
Induction of the Lac Operon



Under conditions of repression, before the operon can be “turned on” to produce lactose-utilizing enzymes, the repressor will have to be removed from the operator. The repressor is an **allosteric protein**; when it binds with one particular molecule, it changes the shape of the protein, which changes its ability to react with a second particular molecule. Here the first molecule is the inducer allolactose and the second molecule is the operator DNA. When allolactose is bound to the repressor, it causes the repressor to change shape and lose its affinity for operator sequences (fig. 14.5).

With allolactose bound to the repressor, the ability of the repressor to bind to the operator is greatly reduced, by a factor of 10^3 . Since no covalent bonds are involved, the repressor simply dissociates from the operator. After the repressor releases from the operator, RNA polymerase can now begin transcription. The three *lac* operon genes are then transcribed and subsequently translated into their respective proteins.

This system of control is very efficient. The presence of the lactose molecule permits transcription of the genes of the *lac* operon, which act to break down the lactose. After all the lactose is metabolized, the repressor returns to its original shape and can again bind to the operator. The system is “turned off.” Using very elegant genetic analysis, details of this system were worked out by François Jacob and Jacques Monod, who subsequently won 1965 Nobel prizes for their efforts.



François Jacob (1920–).
(Courtesy of Dr. François Jacob.)



Jacques Monod (1910–1976).
(Archives Photographiques, Musée
Pasteur.)

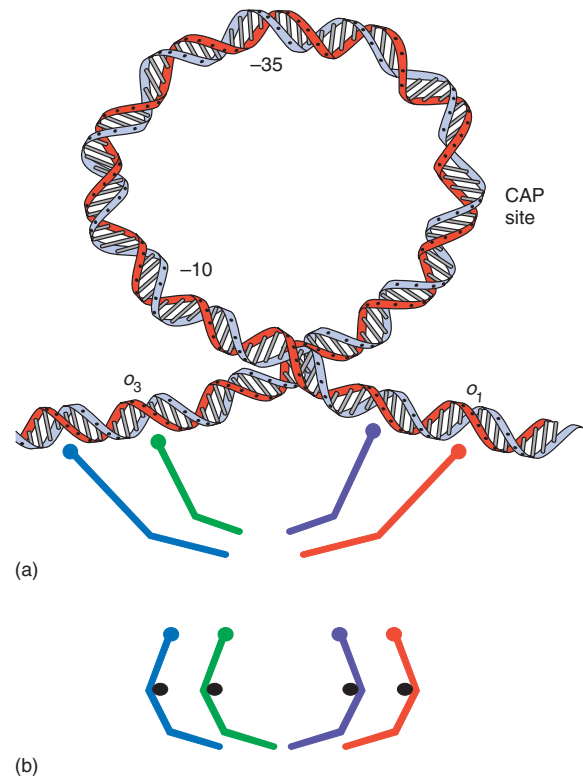


Figure 14.5 Because the *lac* operator DNA sequences are palindromes, each half can bind one repressor subunit. (a) The tetrameric repressor binds to o_1 and o_3 , causing the DNA in between to form a loop. Each of the subunits is shown in a different color. The round portion of the subunit in touch with the DNA is the N-terminal end of the repressor subunit; the C-terminal ends form tails that bind the subunits together. Also indicated are the CAP site and the -10 and -35 sequences. (b) When each of the subunits binds an allolactose molecule (black circles), the shape of the middle portion of the subunit changes, causing the subunit to fall free of the operators.

Lac Operon Mutants

Merozygote Formation

Discovery and verification of the *lac* operon system came about through the use of mutants and partial diploids of the *lac* operon well before DNA sequencing techniques had been developed. The structural (enzyme-specifying) genes of the *lac* operon, *z*, *y*, and *a*, all have known mutant forms in which the particular enzyme does not perform its function. These mutant forms are designated z^- , y^- , and a^- . The alleles for normal forms of the enzymes are z^+ , y^+ , and a^+ .

Partial diploids in *E. coli* can be created through sexual reproduction (chapter 7) because some strains of *E. coli* have

the *lac* operon incorporated into an F' factor. Since F^+ strains can pass the F' particle into F^- strains, *lac* operon diploids (also called merozygotes, or partial diploids) can be formed. By careful manipulation, various combinations of mutations can be looked at in the diploid state.

Constitutive Mutants

Constitutive mutants are mutants in which the three *lac* operon genes are transcribed at all times—that is, they are not turned off even in the absence of lactose. Inspection of figure 14.3 shows that constitutive production of the enzymes can come about in several ways. A defective repressor, produced by a mutant regulator gene, will not turn the system off, nor will a mutant op-

erator that will no longer bind the normal repressor. The regulator constitutive mutants are designated i^- ; the operator constitutive mutants are designated o^c . Both types of mutants produce the same phenotype: constitutive expression of the three *lac* operon genes.

When a new mutant is isolated, it is possible to determine whether it is caused by a regulator or operator mutation. For example, we can determine the exact location of a mutation on the bacterial chromosome by standard mapping techniques (see chapter 7) or, more recently, by DNA sequencing (see chapter 13). Alternatively, the Jacob and Monod model predicts different modes of action for the two types of mutations. In merozygotes, a constitutive operator mutation affects only the operon it is physically a part of. Operator muta-

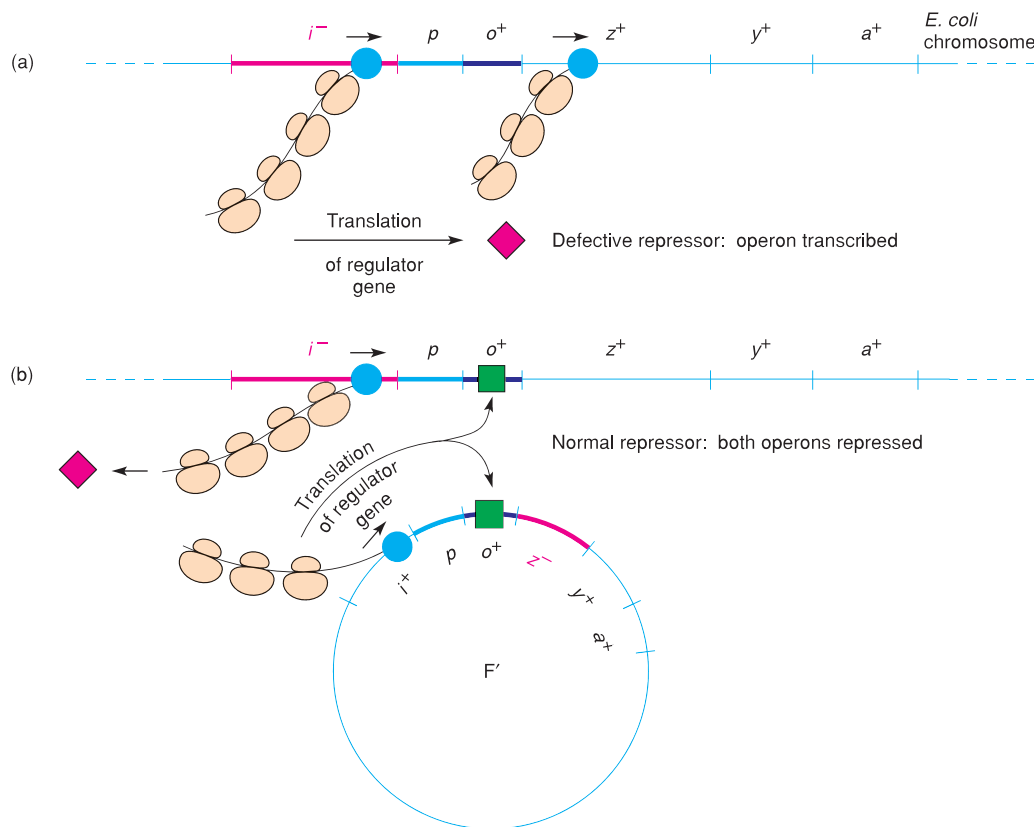


Figure 14.6 (a) A *lac* operon in *E. coli* with a mutation of the regulator gene (i^-). Transcription and translation of this gene yield a defective repressor; the cell thus has constitutive production of the *lac* operon. In (b), the wild-type regulator gene is introduced in an F' factor; there is both a bacterial chromosome and an F' factor, each containing a regulator gene. (The F' operon carries a mutant z allele, allowing us to keep track of the transcriptional control of the chromosomal operon only.) In this case, the phenotype is now normal (inducible) because enough repressor is produced by the F' allele (i^+), by transcription and translation, to bind to both operators. RNA polymerase is shown as solid spheres on the DNA; the wild-type repressor is shown as a green square; the mutant repressor, which cannot bind to the operator, is shown as a red diamond.

tions are therefore called **cis-dominant**. However, a constitutive *i*-gene mutation, since it works through an altered protein, is recessive to a wild-type regulator gene in the same cell, regardless of which operon (chromosomal or F' factor) the mutation is on. Constitutive regulator mutations are, therefore, **trans-acting**. (If two mutations are on the same piece of DNA, they are in the *cis* configuration. If they are on different pieces of DNA, they are in the *trans* configuration.) *Trans-acting* mutations usually work through a protein product that diffuses through the cytoplasm. *Cis-acting* mutants are changes in recognition sequences on the DNA.

In figure 14.6a, the bacterium has a regulator constitutive mutation (i^-); the cell has constitutive production

of the operon. If the wild-type regulator is introduced in an F' plasmid (fig. 14.6b), the normal (inducible) phenotype is restored because the F' i^+ allele is dominant to the chromosomal mutation—the i^+ regulates both the chromosomal and F' operons. Hence, both operons are inducible. We don't need to be concerned about the other components of the F' plasmid because it carries a z^- allele; only the activity of the chromosomal operon will be observed. In figure 14.7a, however, the chromosomal operon carries an operator constitutive mutation; the cell also has constitutive production of the operon. When a wild-type operator is introduced into the cell in an F' plasmid (fig. 14.7b), the cell still has the constitutive phenotype because the operator allele on the F'

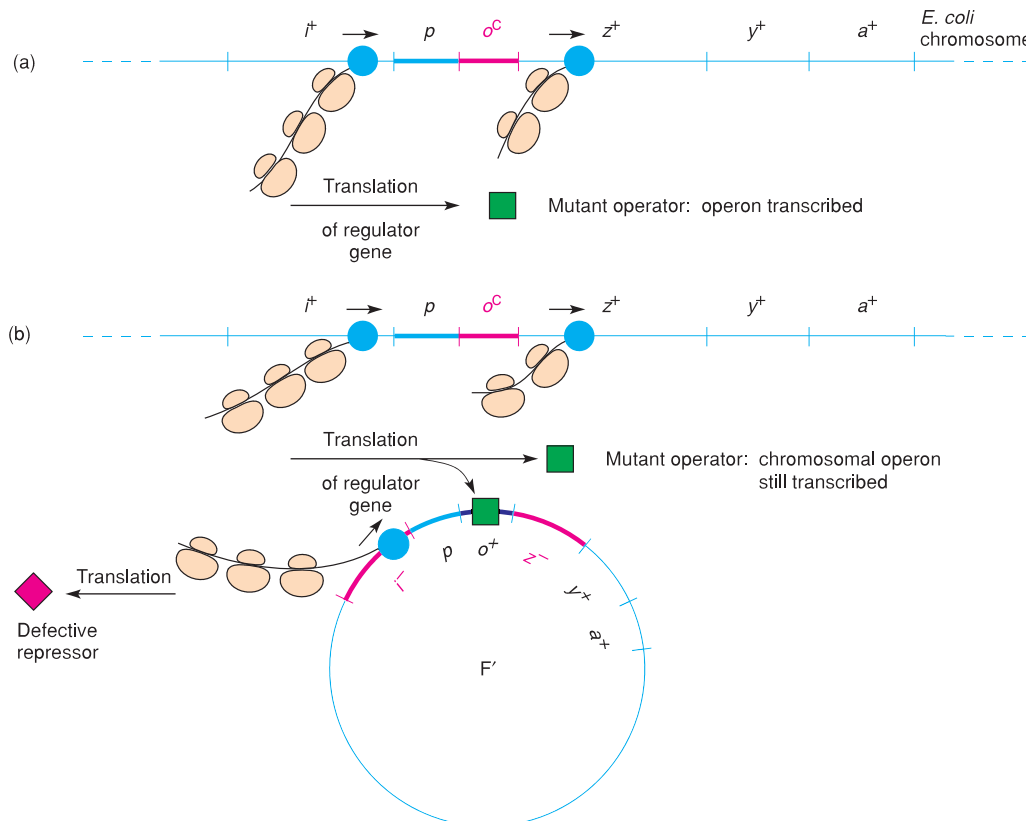


Figure 14.7 (a) A *lac* operon in *E. coli* with a mutation of the operator (o^c). The cell has a constitutive phenotype; the operator cannot bind the wild-type repressor protein, and thus transcription is continuous, even in the absence of lactose. The phenotype is unchanged even when a wild-type operator is introduced into the cell in an F' factor (b); there is both a bacterial chromosome and an F' factor, each containing an operator. (The F' operon carries mutant regulator and *z* alleles, allowing us to keep track of the transcriptional control of the chromosomal operon only.) The F' operator does not change the phenotype of the cell because the wild-type operator exerts no control over the chromosomal operator, which exerts a *cis-dominant* effect; another operator on another operon has no effect. RNA polymerase is shown as solid spheres on the DNA; the wild-type repressor is shown as a green square; the mutant repressor, which cannot bind to the operator, is shown as a red diamond.

plasmid does not control the bacterial operon; the *lac* operon on the bacterial chromosome will be continually transcribed. The chromosomal operon has a *cis-dominant* operator mutation that has a constitutive phenotype. Note, too, that only the bacterial chromosome determines the phenotype because the introduced F' plasmid has a α^- allele.

Other Lac Operon Control Mutations

Other mutations have also been discovered that support the Jacob and Monod operon model. A superrepressed mutation, i^s , was located. This mutation represses the operon even in the presence of large quantities of the inducer. Thus, the repressor seems to have lost the ability to recognize the inducer. Basically, the i -gene product is acting as a constant repressor rather than as an allosteric protein. In an i^s/i^+ merozygote, both operons are repressed because the i^s repressor binds to both operators. Another mutation, i^Q , produces much more of the repressor than normal and presumably represents a mutation of the promoter region of the i gene.

In 1966, W. Gilbert and B. Müller-Hill isolated the *lac* repressor and thereby provided the final proof of the validity of the model. At about the same time, M. Ptashne and his colleagues isolated the repressor for phage λ operons. Control of gene expression in phage λ is discussed later in this chapter.



Mark Ptashne (1940–).
(Courtesy of Dr. Mark
Ptashne.)

CATABOLITE REPRESSION

An interesting property of the *lac* operon and other operons that code for enzymes that catabolize certain sugars (e.g., arabinose, galactose) is that they are all repressed in the presence of glucose. That is, glucose is catabolized in preference to other sugars; the mechanism (**catabolite repression**) involves **cyclic AMP** (cAMP; fig. 14.8). In

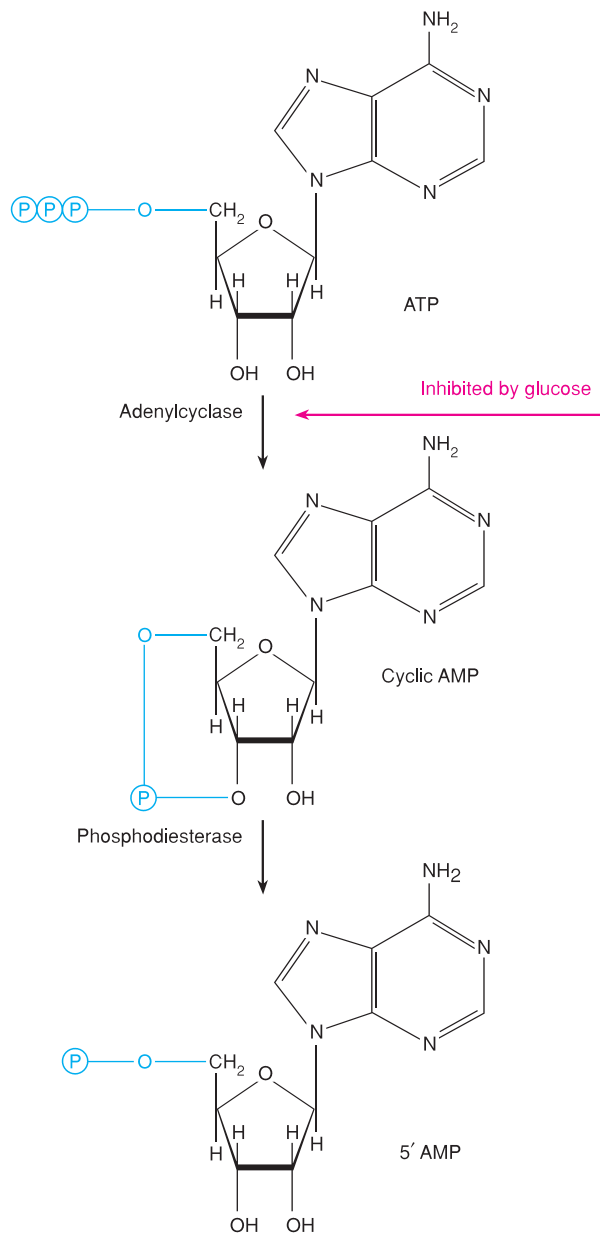


Figure 14.8 Structure of cyclic AMP (cAMP). Glucose uptake lowers the quantity of cyclic AMP in the cell by inhibiting the enzyme adenylcyclase, which converts ATP to cAMP.

eukaryotes, cAMP acts as a *second messenger*; an intracellular messenger regulated by certain extracellular hormones. Geneticists were surprised to discover cAMP in *E. coli*, where it works in conjunction with another regulatory protein, the **catabolite activator protein (CAP)**, to control the transcription of certain operons.

In the absence of glucose, cAMP combines with CAP, and the CAP-cAMP complex binds to a distal part of the promoter of operons with CAP sites (e.g., the *lac* operon; see fig. 14.4). This binding apparently enhances the affinity of RNA polymerase for the promoter, because without the binding of the CAP-cAMP complex to the promoter, the transcription rate is very low. The uptake of glucose by *E. coli* cells causes the loss of cAMP from the cell, probably by inhibiting adenylcyclase (fig. 14.8), and thus lowers the CAP-cAMP level. The transcription rate of operons with CAP sites will, therefore, be reduced (fig. 14.9). The same reduction of transcription rates is noticed in mutant strains of *E. coli* when this part of the distal end of the promoter is deleted. The binding of CAP-cAMP to the CAP site causes the DNA to bend more than 90 degrees (fig. 14.10). This bending, by itself, may enhance transcription, making the DNA more available to RNA polymerase.

In addition, at some point in the process of initiation of transcription, the CAP is in direct contact with RNA polymerase. This was shown by photo cross-linking studies in which the CAP was treated with a cross-linking agent that bound the α subunit of RNA polymerase when irradiated with UV light. For the two proteins to cross-link, they must be in direct contact during the initiation of transcription.

Catabolite repression is an example of positive regulation: Binding of the CAP-cAMP complex at the CAP site enhances the transcription rate of that transcriptional

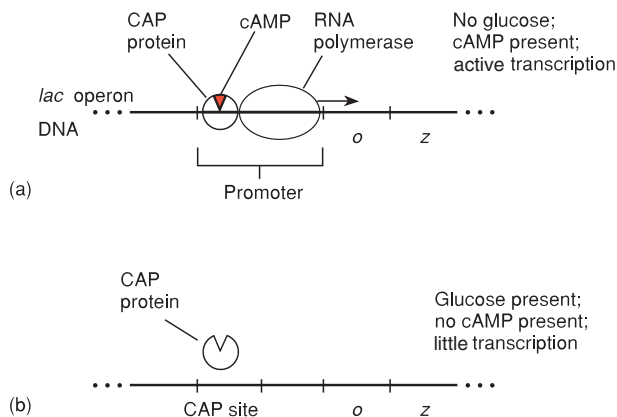


Figure 14.9 Catabolite repression. When cAMP is present in the cell (no glucose is present), it binds with CAP protein, and together they bind to the CAP site in various sugar-metabolizing operons, such as the *lac* operon shown here. The CAP-cAMP complex enhances the transcription of the operon. When glucose is present, it inhibits the formation of cAMP. Thus no CAP-cAMP complex forms, and transcription of the same operons is reduced.

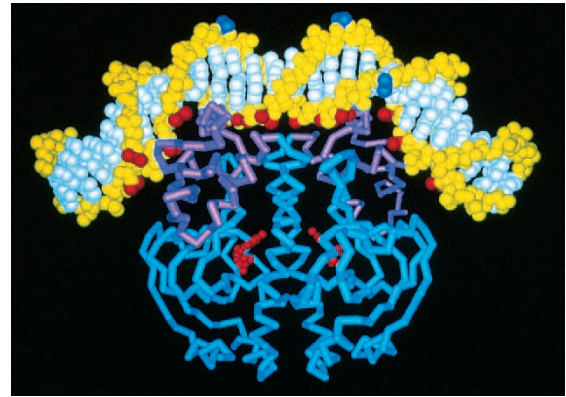


Figure 14.10 CAP-DNA interaction: model of cap protein and DNA. The cap site has twofold symmetry, like the operator. The cAMP-binding domain is dark blue, the DNA-binding domain is purple, and the cyclic AMP molecules within the protein are red. The DNA sugar-phosphate backbones are shown in yellow, the bases in light blue. DNA phosphates in red (on the double helix) are those whose modification interfere with CAP binding. DNA phosphates in dark blue (also on the double helix) are those especially prone to nuclease attack because of the bending of the DNA. (Courtesy of Thomas A. Steitz.)

unit. Thus, the *lac* operon is both positively and negatively regulated; the repressor exerts negative control, and the CAP-cAMP complex exerts positive control of transcription.

TRP OPERON (REPRESSIBLE SYSTEM)

The inducible operons are activated when the substrate that is to be catabolized enters the cell. Anabolic operons function in the reverse manner: They are turned off (repressed) when their end product accumulates beyond the needs of the cell. Two entirely different, although not mutually exclusive, mechanisms seem to control the transcription of repressible operons. The first mechanism follows the basic scheme of inducible operons and involves the end product of the pathway. The second mechanism involves secondary structure in messenger RNA transcribed from an attenuator region of the operon.

Tryptophan Synthesis

One of the best-studied repressible systems is the tryptophan, or *trp*, operon in *E. coli*. The *trp* operon contains

the five genes that code for the synthesis of the enzymes that build tryptophan, starting with chorismic acid (fig. 14.11). It has a promoter-operator sequence (*p*, *o*) as well as its own regulator gene (*trpR*).

Operator Control

In this repressible system, the product of the *trpR* gene, the repressor, is inactive by itself; it does not recognize the operator sequence of the *trp* operon. The repressor only becomes active when it combines with tryptophan. Thus, when tryptophan builds up, enough is available to bind with and activate the repressor. Tryptophan is thus referred to as the **corepressor**. The corepressor-repressor complex then recognizes the operator, binds to it, and prevents transcription by RNA polymerase.

After the available tryptophan in the cell is used up, the diffusion process causes tryptophan to leave the repressor, which then detaches from the *trp* operator. The transcription process no longer is blocked and can proceed normally (the operon is now **derepressed**). Transcription continues until enough of the various enzymes have been synthesized to again produce an excess of tryptophan. Some becomes available to bind to the repressor and make a functional complex, and the operon is again shut off and the process repeated, ensuring that tryptophan is being synthesized as needed (fig. 14.12). This regulation is modified, however, by the existence of the second mechanism for regulating repressible operons—attenuation.

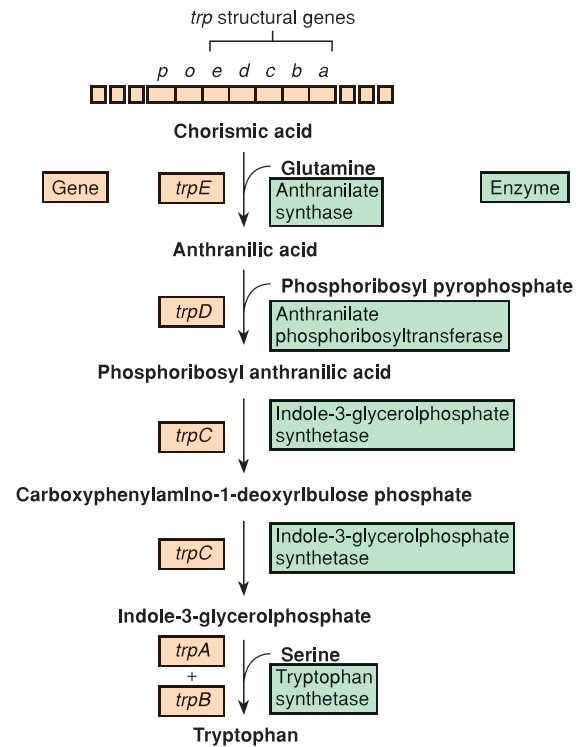


Figure 14.11 Genes of the tryptophan operon in *E. coli*. The enzymes they produce control the conversion of chorismic acid to tryptophan. The symbol *o* on the chromosome refers to the *trp* operator, which has its own repressor, the product of the *trpR* gene.

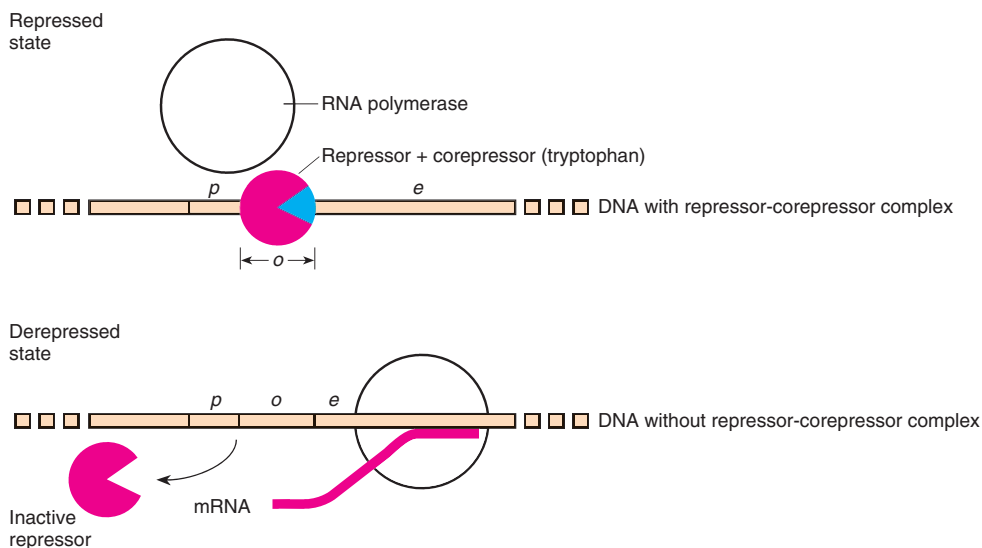


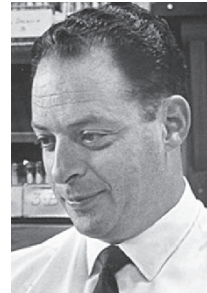
Figure 14.12 The repressor-corepressor complex binds at the operator and prevents the transcription of the *trp* operon in *E. coli*. Without the corepressor, the repressor cannot bind, and therefore transcription is not prevented. The blue wedge is the corepressor (two tryptophan molecules), and the partial red circle is the repressor.

TRP OPERON (ATTENUATOR-CONTROLLED SYSTEM)

Details of the second control mechanism of repressible operons have been elucidated primarily by C. Yanofsky and his colleagues, who worked with the tryptophan operon in *E. coli*. This type of operon control, control by an **attenuator region**, has been demonstrated for at least five other amino acid-synthesizing operons, including the leucine and histidine operons. This regulatory mechanism may be the same for most operons involved in the synthesis of an amino acid.

Leader Transcript

In the *trp* operon, an attenuator region lies between the operator and the first structural gene (fig. 14.13). The messenger RNA transcribed from the attenuator region, termed the **leader transcript**, has been sequenced, re-



Charles Yanofsky (1925–).
(Courtesy of Dr. Charles Yanofsky.)

vealing two surprising and interesting facts. First, four subregions of the messenger RNA have base sequences that are complementary to each other so that three different stem-loop structures can form in the messenger RNA (fig. 14.14). Depending on circumstances, regions 1–2 and 3–4 can form two stem-loop structures, or region 2–3 can form a single stem-loop. When one stem-loop structure is formed, the others are preempted.

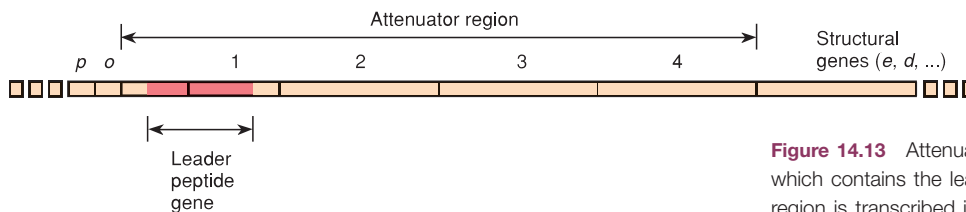


Figure 14.13 Attenuator region of the *trp* operon, which contains the leader peptide gene (red). This region is transcribed into the leader transcript.

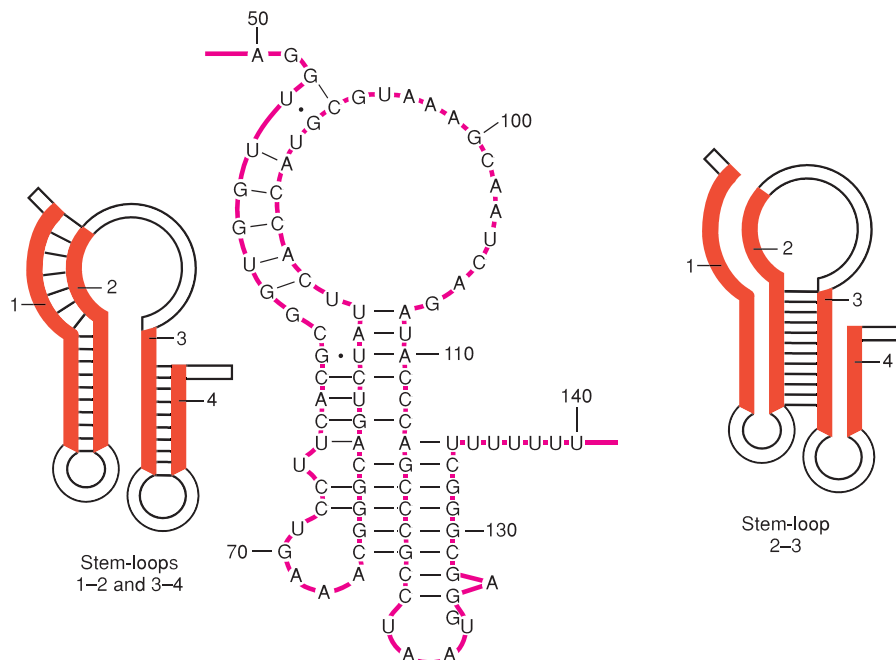


Figure 14.14 Nucleotide sequence of part of the leader transcript of the *trp* attenuator region (bases 50 to 140). Stem-loops 1–2 and 3–4, or stem-loop 2–3, can form because of complementarity of the nucleotides. All possible base pairings are shown in the middle of the figure. (From D. L. Oxender, et al., "Attenuation in the *Escherichia coli* tryptophan operon: Role of RNA secondary structure involving the tryptophan codon region," *Proceedings of the National Academy of Sciences*, 76:5524–28, 1979. Reprinted by permission.)

we will see, the particular combination of stem-loop structures determines whether transcription continues.

Leader Peptide Gene

The second fact obtained by sequencing the leader transcript is that there is a small gene coding information for a peptide from bases 27 to 68 (fig. 14.15). The gene for this peptide is referred to as the **leader peptide gene**. It codes for fourteen amino acids, including two adjacent tryptophans. These adjacent tryptophan codons are critically important in attenuator regulation. The proposed mechanism for this regulation follows.

Excess Tryptophan

Assuming that the operator site is available to RNA polymerase, transcription of the attenuator region will begin. As soon as the 5' end of the messenger RNA for the leader peptide gene has been transcribed, a ribosome attaches and begins translating this messenger RNA. Depending on the levels of amino acids in the cell,

three different outcomes can take place. If the concentration of tryptophan in the cell is such that abundant tryptophanyl-tRNAs exist, translation proceeds down the leader peptide gene. The moving ribosome overlaps regions 1 and 2 of the transcript and allows stem-loop 3–4 to form, as shown in the configuration at the far left of figure 14.16. This stem-loop structure, referred to as the **terminator**, or **attenuator stem**, causes transcription to be terminated. Note that stem-loop 3–4, the terminator stem, followed by a series of uracil-containing bases, is a rho-independent transcription terminator (see chapter 10). Hence, when existing quantities of tryptophan, in the form of tryptophanyl-tRNA, are adequate for translation of the leader peptide gene, transcription is terminated.

Tryptophan Starvation

If the quantity of tryptophanyl-tRNA is lowered, the ribosome must wait at the first tryptophan codon until it acquires a Trp-tRNA^{Trp}. This is shown in the configuration in the middle part of figure 14.16. The stalled ribosome will

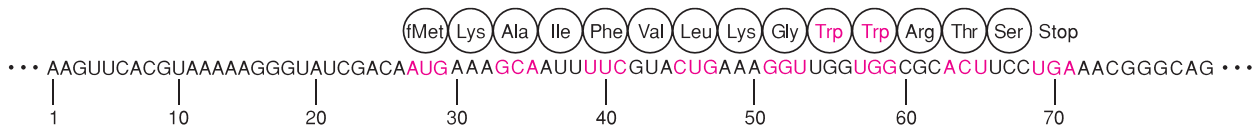


Figure 14.15 Base sequence of the *trp* leader transcript and the amino acids these nucleotides code. Note the presence of adjacent tryptophan codons. (From D. L. Oxender, et al., "Attenuation in the *Escherichia coli* tryptophan operon: Role of RNA secondary structure involving the tryptophan codon region," *Proceedings of the National Academy of Sciences*, 76:5524–28, 1979. Reprinted by permission.)

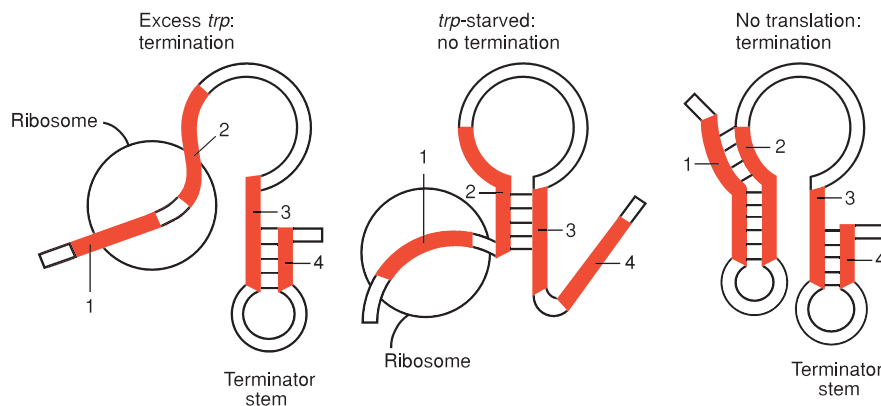


Figure 14.16 Model for attenuation in the *E. coli* *trp* operon. The circle represents the ribosome attempting to translate the leader transcript of figure 14.14. Under conditions of excess tryptophan, the 3–4 stem-loop forms (the terminator stem), terminating transcription. Under conditions of tryptophan starvation, the ribosome is stalled, and stem-loop 2–3 forms, allowing continued transcription. Under general starvation, there is no translation, resulting in the formation of stem-loops 1–2 and 3–4, which again results in the termination of transcription. (From D. L. Oxender, et al., "Attenuation in the *Escherichia coli* tryptophan operon: Role of RNA secondary structure involving the tryptophan codon region," *Proceedings of the National Academy of Sciences*, 76:5524–28, 1979. Reprinted by permission.)

permit stem-loop 2–3 to form, which precludes the formation of the terminator stem-loop (3–4). In this configuration, transcription is not terminated, so that eventually, the whole operon is transcribed and translated, raising the level of tryptophan in the cell. The stem-loop 2–3 structure is referred to as the **preemptor stem**. Note that the preemptor stem is not a rho-independent transcription terminator and thus, without the rho protein present, will not terminate transcription (see chapter 10).

General Starvation

A final configuration is possible, as shown on the far right in figure 14.16. Here, no ribosome interferes with stem formation and, presumably, stem-loops 1–2 and 3–4 (terminator) form. This configuration also terminates transcription because of the terminator stem. It is believed that this configuration occurs if the ribosome is stalled on the 5' side of the *trp* codons, which happens when

the cell is starved for other amino acids. Presumably, it makes no sense to manufacture tryptophan when other amino acids are in short supply. Hence, the cell can carefully bring up the levels of the various amino acids in the most efficient manner.

TRAP Control

The tryptophan operon in bacilli such as *Bacillus subtilis* is also controlled by attenuation, but secondary structure in the mRNA transcript is induced by binding not the ribosome, but a ***trp* RNA-binding attenuation protein (TRAP)**. This protein attaches to the nascent messenger RNA only after the protein binds tryptophan molecules; the result is a terminator stem that forms in the messenger RNA. In the absence of excess tryptophan, TRAP does not bind to the messenger RNA, a preemptor (also called an antiterminator) stem, not the terminator stem, forms, and transcription continues (fig. 14.17a). Recently, the

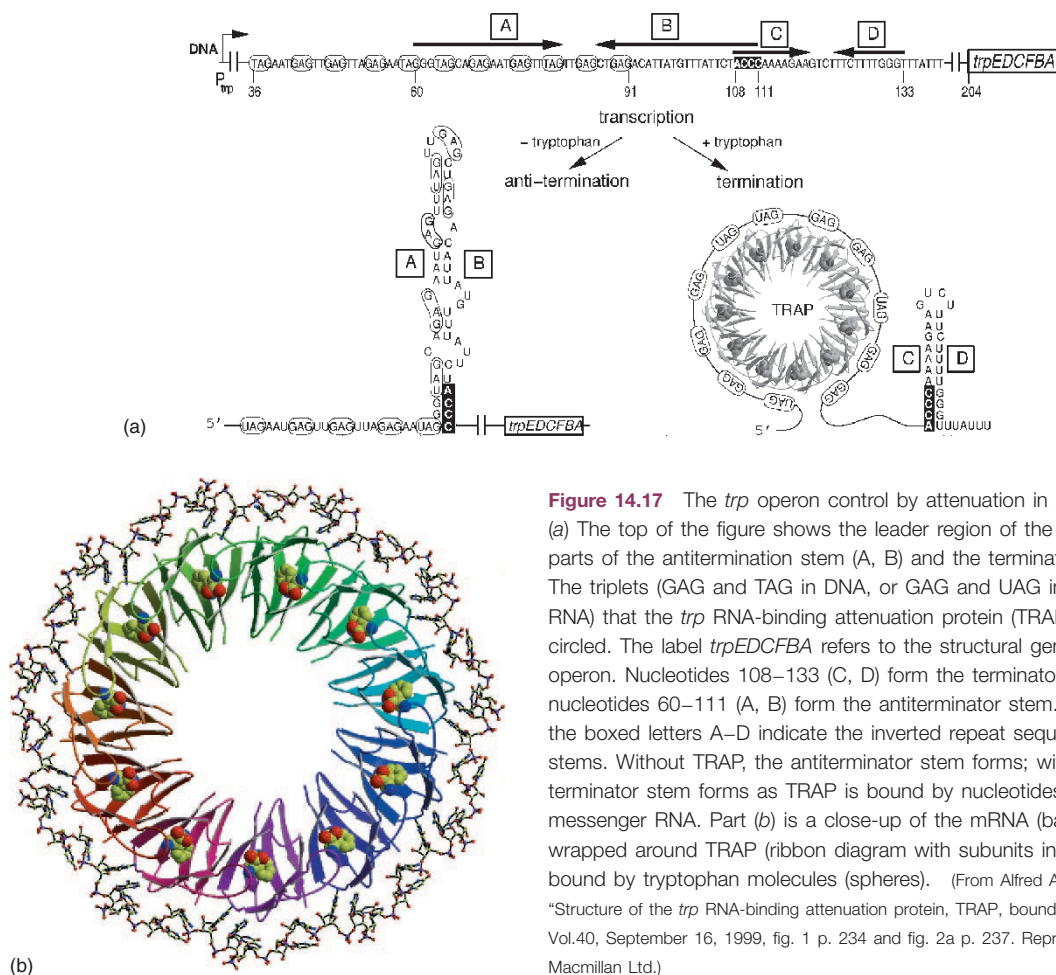


Figure 14.17 The *trp* operon control by attenuation in *Bacillus subtilis*.

(a) The top of the figure shows the leader region of the DNA with the two parts of the antitermination stem (A, B) and the termination stem (C, D). The triplets (GAG and TAG in DNA, or GAG and UAG in the messenger RNA) that the *trp* RNA-binding attenuation protein (TRAP) binds to are circled. The label *trpEDCFBA* refers to the structural genes of the *trp* operon. Nucleotides 108–133 (C, D) form the terminator stem, and nucleotides 60–111 (A, B) form the antiterminator stem. The arrows below the boxed letters A–D indicate the inverted repeat sequences forming the stems. Without TRAP, the antiterminator stem forms; with TRAP, the terminator stem forms as TRAP is bound by nucleotides 36–91 of the messenger RNA. Part (b) is a close-up of the mRNA (ball-and-stick model) wrapped around TRAP (ribbon diagram with subunits in different colors) bound by tryptophan molecules (spheres). (From Alfred A. Antson, et al., "Structure of the *trp* RNA-binding attenuation protein, TRAP, bound to RNA" in *Nature*, Vol.40, September 16, 1999, fig. 1 p. 234 and fig. 2a p. 237. Reprinted by permission of Macmillan Ltd.)

structure of the protein was worked out; it has eleven symmetrical loops, each of which can bind a tryptophan molecule. When TRAP is bound to tryptophan molecules, it can attach to triplets in the messenger RNA transcript, triplets of GAG or UAG. The TRAP wraps the mRNA around itself, forming an elegant pinwheel (fig. 14.17b).

Redundant Controls

Some amino acid operons are controlled only by attenuation, such as the *bis* operon in *E. coli*, in which the leader peptide gene contains seven histidine codons in a row, or the *trp* operon in *B. subtilis*. Redundant control (repression and attenuation) of tryptophan biosynthesis in *E. coli* allows the cell to test both the tryptophan levels (tryptophan is the corepressor) and the tryptophanyl-tRNA levels (in the attenuator control system). The attenuator system also allows the cell to regulate tryptophan synthesis on the basis of the shortage of other amino acids. For example, when there is a shortage of both tryptophan and arginine, operator control allows transcription to begin, but attenuator control terminates transcription because stem-loops 1–2 and 3–4 form (fig. 14.16).

LYTIC AND LYSOGENIC CYCLES IN PHAGE λ

When a bacteriophage infects a cell, it must express its genes in an orderly fashion; some gene products are needed early in infection, and other products are not needed until late in infection. Early genes usually control phage DNA replication; late genes usually determine phage coat proteins and the lysis of the bacterial cell. A phage is most efficient if it expresses the early genes first and the late genes last in the infection process. Also, temperate phages have the option of entering into lysogeny with the cell; here, too, control processes determine which path is taken. One generalization that holds true for most phages is that their genes are clustered into early and late operons, with separate transcriptional control mechanisms for each.

Phage λ is perhaps the best-studied bacteriophage. It has a chromosome of about 48,500 base pairs. Since it is a temperate phage, it can exist either vegetatively or as a prophage, integrated into the host chromosome. This phage warrants our attention because of the interesting and complex way that its life-cycle choice is determined. It is a model system of operon controls. The complexity results from having two conflicting life-cycle choices.

Briefly, the expression of one of the two life-cycle alternatives, lysogenic or lytic cycles, depends on whether two repressors, CI and Cro, have access to op-

erator sites. The CI repressor acts to favor lysogeny: it represses the lytic cycle. The Cro repressor favors the lytic cycle and represses lysogeny. The operator sites, when bound by either CI or Cro, can either enhance or repress transcription. Other control mechanisms are also involved in determining aspects of the λ life cycle, including antitermination and multiple promoters for the same genes.

Phage λ Operons

Phage λ (see fig. 7.21) exhibits a complex system of controls of both early and late operons, as well as controls for the decision of lytic infection versus lysogenic integration. The λ genes are grouped into four operons: left, right, late, and repressor (fig. 14.18). The left and right operons contain the genes for DNA replication and recombination and phage integration. The late operon contains the genes that determine phage head and tail proteins and lysis of the host cell. The sequence of events following phage infection is relatively well known.

The map of λ (fig. 14.18) is a circle, but the λ chromosome has two linear stages in its life cycle (fig. 14.19). It is packed within the phage head in one linear form, and it integrates into the host chromosome to form a prophage in another linear form (fig. 14.19). Those two linear forms do not have the same ends (figs. 14.18 and 14.19b). The mature DNA, which is packed within the phage heads before lysis of the cells, is flanked by *cos* sites (chapter 13). It results from a break in the circular map between the *A* and *R* loci. The prophage is integrated at the *att* site, and the circular map is thus broken there at integration.

The homologous integration sites on both λ and the *E. coli* chromosome consist of a 15 bp core sequence (called “O” in both), flanked by different sequences on both sides in both the bacterium and the phage (fig. 14.20). In the phage, the region is referred to as POP’, where P and P’ (P for phage) are two different regions flanking the O core on the phage DNA. In the bacterium, the region is called BOB’, where B and B’ (B for bacterium) are two different regions flanking the O core on the *E. coli* chromosome. Integration, which is a part of the lysogenic life cycle, requires the product of the λ *int* gene, a protein known as *integrase*, and is referred to as **site-specific recombination**. Later excision of the prophage, during induction, when the phage leaves the host chromosome to enter the lytic cycle, requires both the integrase and the protein product of the neighboring *xis* gene, *excisionase*.

After infection of the *E. coli* cell by a λ phage, the phage DNA circularizes, using the complementarity of the *cos* sites. Transcription begins, and within a very short time the phage is guided toward either entering the lytic cycle and producing virus progeny or entering

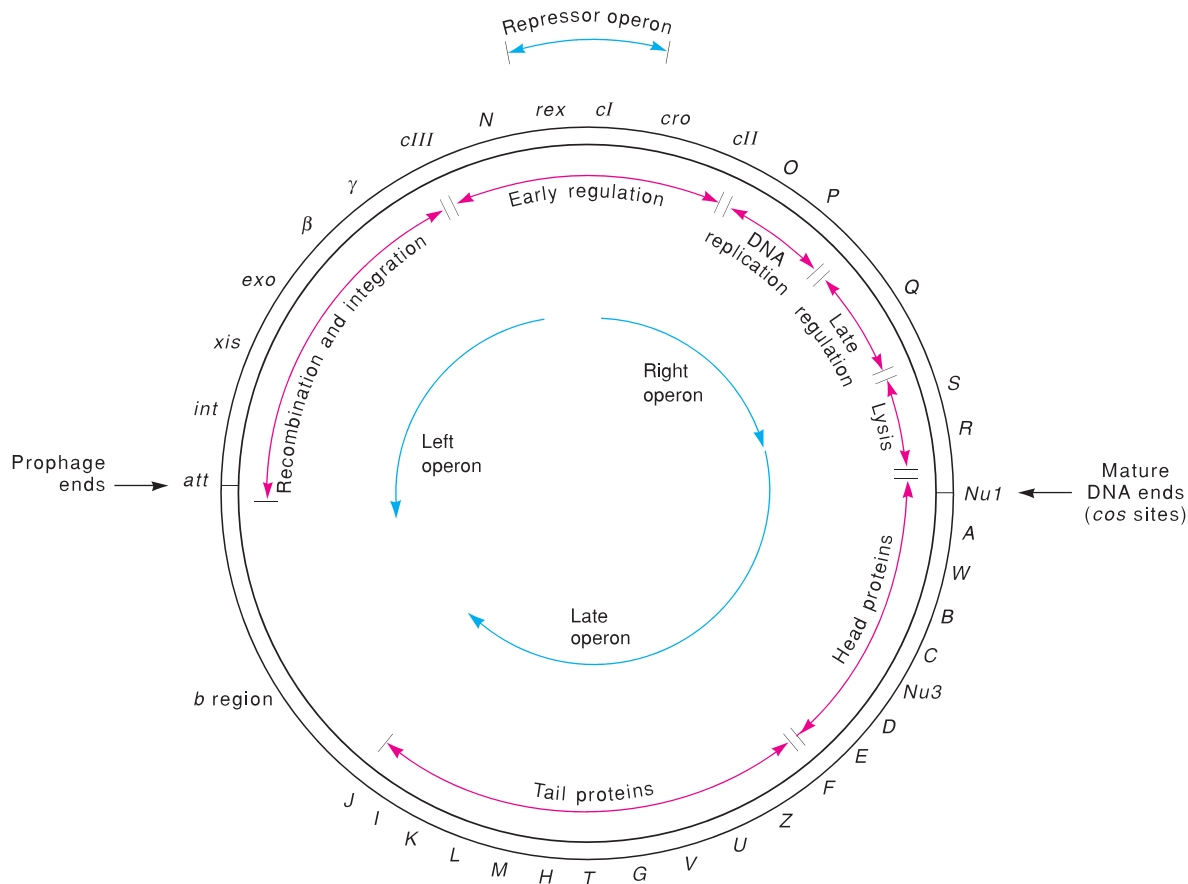
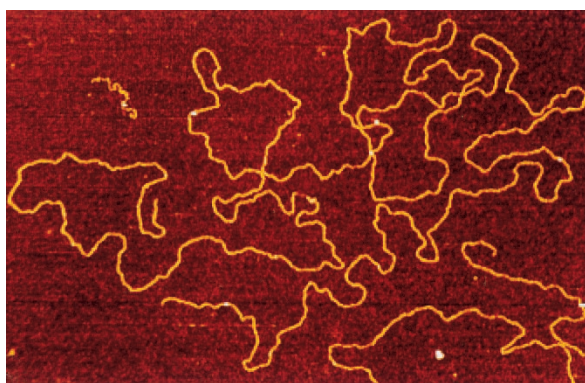
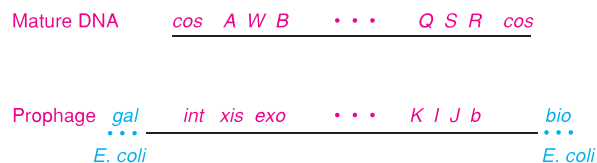


Figure 14.18 Genetic map of phage λ . There are four operons present: the repressor, left, right, and late. The prophage, a linear form integrated into the bacterial chromosome, begins and ends at *att*. The mature phage, another linear form found packed into the phage heads, begins and ends at *Nu1* (*cos* sites).



(a)



(b)

Figure 14.19 The two linear forms of λ phage. (a) False color electron micrograph of the λ chromosome, approximately 16 μm in length. This is the linear form of the phage chromosome found within the phage heads. (b) The mature linear DNA (found within phage protein coats) is flanked by *cos* sites. The prophage is flanked by *E. coli* DNA (*bio* and *gal* loci). ([a] Courtesy of Martin Guthold and Carlos Bustamante, Institute of Molecular Biology and HHMI, University of Oregon.)

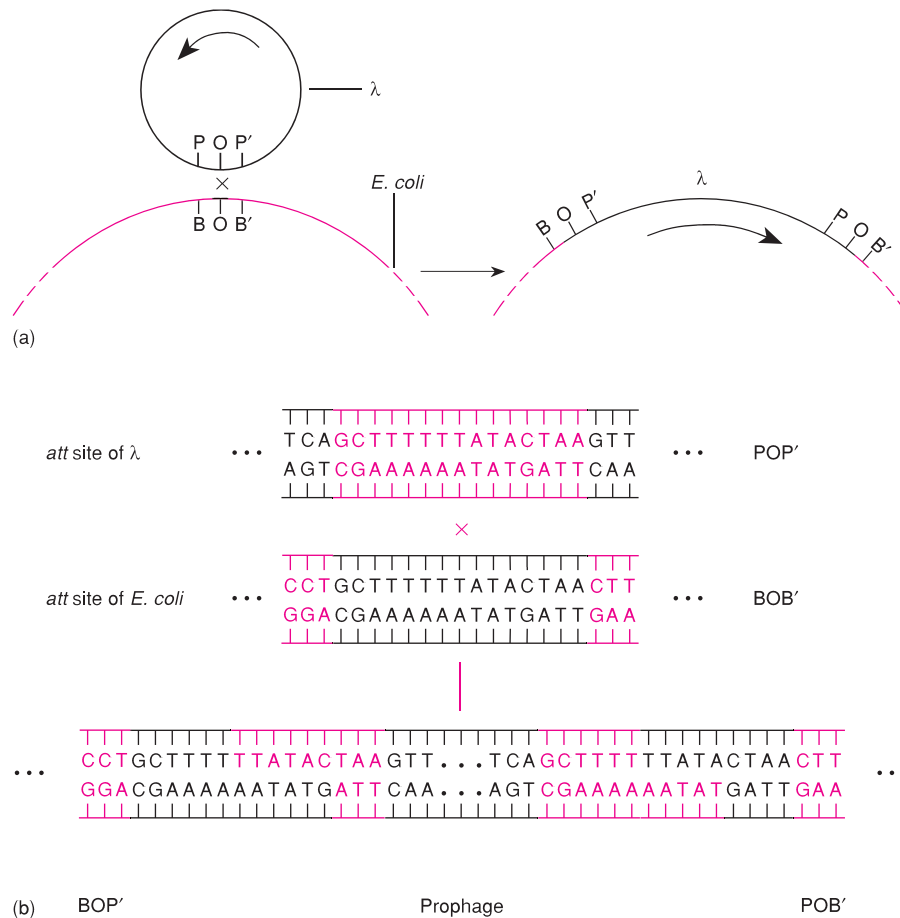


Figure 14.20 Integration of the λ phage into the *E. coli* chromosome requires a crossover between the two attach sites, called POP' (phage) and BOB' (bacteria). (a) General pattern of this site-specific attachment. (b) Nucleotide sequences of the various components.

the lysogenic cycle and integrating into the host chromosome. What events lead up to this “decision” on which path to take?

Early and Late Transcription

When the phage first infects an *E. coli* cell, transcription of the left and right operons begins at the left (p_L) and right (p_R) promoters, respectively. The *N* (left) and *cro* (right) genes are transcribed (fig. 14.21) and then trans-

lated into their respective proteins. Transcription then stops on both operons at rho-dependent terminators (t_{R1} , t_{L1}). Transcription cannot continue until the protein product of the *N* gene is produced. This protein is called an **antiterminator protein**. When it binds at sites upstream from the terminators, called *nutL* and *nutR* (*nut* stands for *N* utilization; *L* and *R* stand for left and right), the polymerase reads through the terminators and continues on to transcribe the left and right operons. (Although it is not completely clear why antitermination

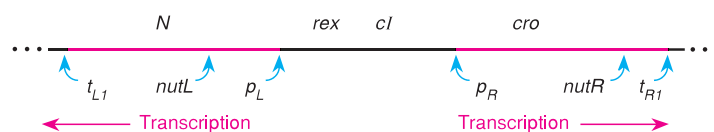


Figure 14.21 Transcription begins at the left and right promoters (p_L , p_R) and proceeds to the left and right terminators (t_{L1} , t_{R1}). Transcription continues through these terminators when the protein product of the *N* gene binds to the *nutL* and *nutR* sites.

has evolved here, it seems to give the phage better control over the timing of events.)

Transcription then continues along the left and right operons through the *cII* and *cIII* genes (see fig. 14.18). Later, if the lytic response is followed, the *Q* gene, which codes for a second antiterminator protein in the right operon, has the same effect on the late operon as the *N* gene did on the two early operons: Without the *Q*-gene product, transcription of the late operon proceeds about two hundred nucleotides and then terminates. With the *Q*-gene product, the late operon is transcribed. Hence, in phage λ , proteins that allow RNA polymerase to proceed past termination signals mediate general control of transcription. If only the previously described events were to transpire, the lytic cycle pathway would always be followed. However, a complex series of events can also take place in the repressor region that may lead to a “decision” to follow the lysogenic cycle instead.

Repressor Transcription

The *cIII*-protein product inhibits a host cell protease, called FtsH, that would break down the *cII*-gene product. The *cII*-gene product binds at two promoters, enhancing their availability to RNA polymerase, just as the CAP-cAMP product enhances the transcription of the *lac* operon. The *cII* protein binds at the promoters for *cI* transcription and for *int* transcription (fig. 14.22). At this point, the phage can still “choose” between either the lytic or the lysogenic cycles. Integrase (the product of the *int* gene)

and *cI* (repressor) proteins are now produced, favoring lysogeny, as well as the *cro*-gene product, the *antirepressor*; which is a repressor of *cI* and therefore favors the lytic pathway. (*Cro* stands for control of repression and other things; the *c* of *cI*, the repressor, stands for “clear,” which is the appearance of λ plaques that have *cI* mutations. These mutants can only undergo lysis without the possibility of lysogeny. Normal λ infections produce turbid plaques, accounted for by lysogenic bacterial growth within the plaques.) We now focus further on the repressor region with its operators and promoters.

Maintenance of Repression

The *cI* gene, with the aid of the *cII*-gene product, is transcribed from a promoter known as p_{RE} , the *RE* standing for repression establishment (fig. 14.23). Once *cI* is transcribed, it is translated into a protein called the λ repressor, which interacts at the left and right operators, o_L and o_R , of the left and right operons. When these operators are bound by *cI* protein, transcription of the left and right operons (and therefore also the late operon) ceases. There are several ramifications of the repression. First, lysogeny can be initiated because the *int* gene has been transcribed at the early stage of infection. Second, since *CII* and *CIII* are no longer being synthesized, *cI* transcription from the p_{RE} promoter stops. However, *cI* can still be transcribed because there is a second promoter, p_{RM} (*RM* stands for repression maintenance), that allows low levels of transcription of the *cI* gene.

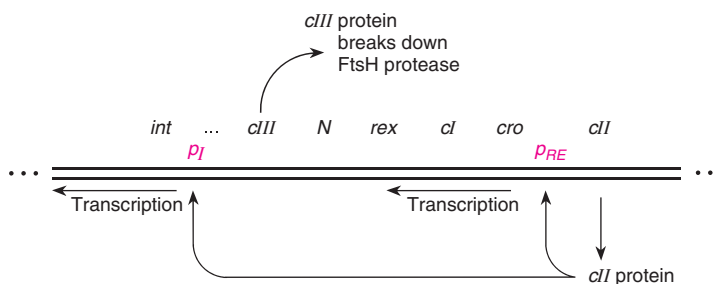


Figure 14.22 The *cII*-gene product of phage λ binds to the *cI* promoter (p_{RE}) and the *int* promoter (p_I), enhancing transcription of those genes. The *cII* protein breaks down the FtsH protease that would normally break down the *cII* protein.

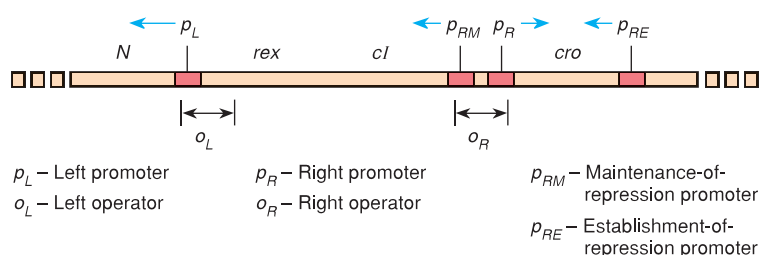


Figure 14.23 Early regulation region of phage λ . Two promoters, p_{RE} and p_{RM} , transcribe the *cI* and *rex* genes. The left operator overlaps the left promoter, and the right operator overlaps both the right promoter and the maintenance-of-repression promoter.

The *cI* gene can further control its own concentration in the cell. When the right and left operators were sequenced, each was discovered to have three sites of repressor recognition (fig. 14.24). On the right operator, for example, the right-most site (O_{R1}) was found to be most efficient at binding repressor. When repressor was bound at this site, the right operon was repressed, and transcription of *cI* was enhanced (again, in a way similar to enhancement of transcription by the binding of CAP-cAMP at the CAP site in the *lac* operon). Excess repressor, when present, however, was also bound by the other two sites within O_R . The foregoing process results in the repression of the *cI* gene itself. Hence, maintenance levels of CI can be kept within very narrow limits.

A third ramification of repression is the prevention of superinfection. That is, bacteria lysogenic for λ phage are protected from further infection by other λ phages because repressor is already present in the cell. Thus, the excess of repressor controls new invading λ phages. (We say that bacterial cells lysogenic for phage λ are immune from infection by additional λ phage.) These bacteria are also protected from infection by T4 phage with *rII* mutants. The *rex*-gene product, the product of the other gene in the repressor operon, controls this protection.

The promoters for maintenance and establishment of repression differ markedly in their control of repressor gene expression. When p_{RE} is active, a very high level of repressor is present, whereas p_{RM} produces only a low level of repressor. The level of repressor is due to the length of the leader RNA transcribed on the 5' side of the *cI* gene. The p_{RE} promoter transcribes a very long leader RNA and is very efficient at translation of the *cI* region. In contrast, the p_{RM} promoter begins transcription at the initiation codon of the protein. This leaderless messenger RNA is translated very inefficiently into CI.

The λ repressor is a dimer of two identical subunits (fig. 14.25). Each subunit is composed of two domains, or

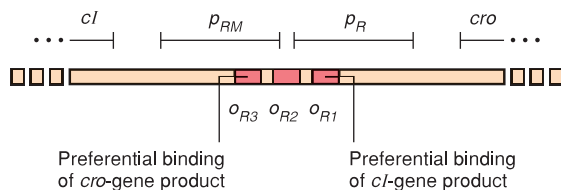
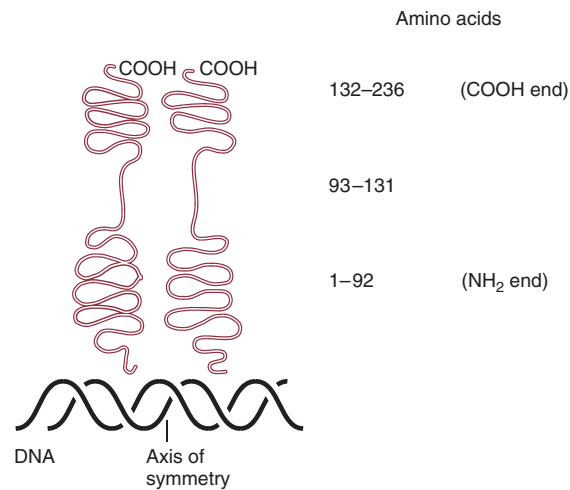
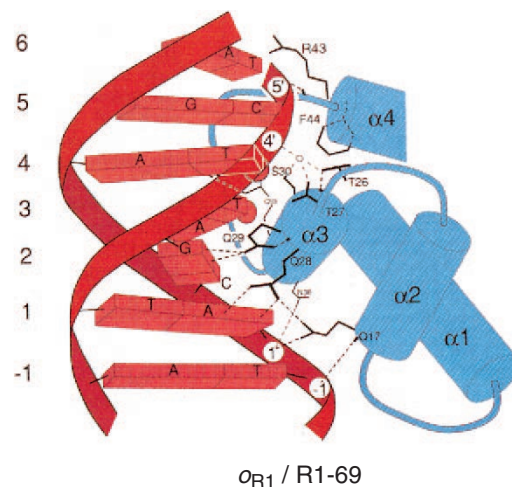


Figure 14.24 The right operator on the phage λ chromosome overlaps the p_{RM} and p_R promoters. There are three repressor recognition sites within the operator: O_{R1} , O_{R2} , and O_{R3} . Preferential binding by the Cro repressor to O_{R3} and the CI repressor to O_{R1} determines whether transcription occurs to the left or the right.



(a)



(b)

Figure 14.25 The λ repressor. (a) The λ repressor is a dimer with each subunit having helical amino- and carboxyl-terminal ends. The helical structure of each amino-terminal end binds in the major DNA groove. (b) Diagram of the interaction of amino acid residues 1–69 (blue) with O_{R1} (red) in the closely related phage 434. Dashed lines are hydrogen bonds. Numbers –1 to 6 and 4' and 5' are phosphate numbers. Amino acids are designated by the single-letter code (fig. 11.1). Small red circles are water molecules. (From D. W. Rodgers and S. C. Harrison, "The complex between phage 434 repressor DNA-binding domain and operator site O_{R3} : Structural differences between consensus and non-consensus half-sites," *Structure*, 1:227–40, Dec. 15, 1993. © Current Biology Ltd.)

“ends.” The carboxyl- and amino-terminal ends are separated by a relatively open region, susceptible to protease attack. The alpha-helical regions of the amino-terminal ends interdigitate into the major groove of the DNA to locate the specific sequences making up the left and right operator sequences. As described earlier for the *lac* operator, o_{R1} , o_{R2} , and o_{R3} each have twofold symmetry.

The binding of the λ repressor in o_{R1} enhances the binding of another molecule of repressor into o_{R2} . Together, they enhance p_{RM} transcription, presumably through contact with RNA polymerase. The repressors also block p_R transcription (see fig. 14.24).

Lysogenic Versus Lytic Response

We have described the mechanism by which λ establishes lysogeny. How then does λ turn toward the lytic cycle? Here, control is exerted by the *cro*-gene product, another repressor molecule that works at the left and right operators in a manner antagonistic to the way the CI repressor works. In other words, using the right operator as an example, *cro*-gene product binds preferentially to the leftmost of the three sites within o_R and represses *cI* but enhances the transcription of *cro* (see fig. 14.24).

The *cro*-gene product can direct the cell toward a lytic response if it occupies the o_R and o_L sites before the λ repressor, or if the λ repressor is removed. From the point of view of phage λ , when would be a good time for the CI repressor to be removed? Thinking in evolutionary terms, we would expect that a prophage might be at an advantage if it left a host's chromosome and began the lytic cycle when it “sensed” damage to the host. In fact, one of the best ways to induce a prophage to enter the lytic cycle is to direct ultraviolet (UV) light at the host bacterium. (Actually, this was how lysogeny was discovered, by French geneticist André Lwoff.) UV light causes damage to DNA and induces several repair systems. One, called SOS repair (see chapter 12), makes use of the protein product of the *recA* gene. Among the activities of this enzyme is to cleave the λ repressor in the susceptible region between domains. The cleaved repressor falls free of the DNA, making the operator sites available for the *cro*-gene product. The lytic cycle then follows.

Initially, however, when the phage first infects an *E. coli* cell, the “decision” for lytic versus lysogenic growth is probably determined by the *cII*-gene product. This protein, as we mentioned, is susceptible to a bacterial protease, which, in turn, is an indicator of cell growth. When *E. coli* growth is limited, its proteases tend to be limited, a circumstance that would favor lysogeny for the phage. It is the *cII* protein that, when active, favors lysogeny and when inactive favors the lytic cycle. Thus, under active bacterial growth, the *cII* protein is

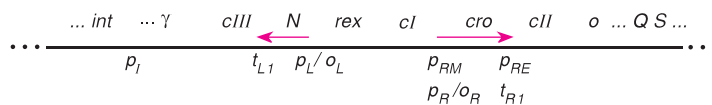
more readily destroyed, it thus fails to enhance *cI* transcription, and lysis follows. When bacteria are not growing actively, the *cII* protein is not readily destroyed, it enhances *cI* transcription, and lysogeny results. Thus, under initial infection, the choice between lysogeny or the lytic cycle depends primarily on the *cII* protein, which gauges the health and activity of the host. After lysogeny is established, it can be reversed by processes that inactivate the *cI* protein, indicating genetic damage to the bacterium (the SOS response) or an abundance of other hosts in the environment (zygotic induction, see chapter 7). In zygotic induction, the lytic cycle is induced during conjugation, presumably when an Hfr cell sends a copy of the λ prophage into an F^- cell. At that point, without repressor present, the prophage can reassess whether to continue lysogeny or enter the lytic cycle.

Not all the details regarding the CI-Cro competition are known, but an understanding of the relationship of lytic and lysogenic life cycles and the nature of DNA-protein recognition has emerged (fig. 14.26 and table 14.1).

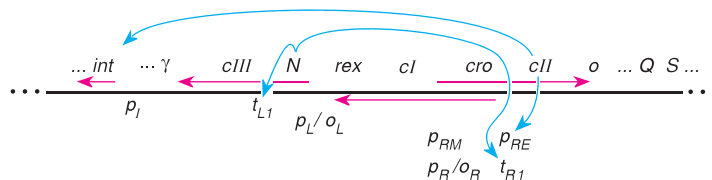
Table 14.1 Elements in Phage λ Infection

Gene Products	
<i>cI</i>	Repressor protein whose function favors lysogeny
<i>cII</i>	Enhances transcription at the p_I and p_{RE} promoters
<i>cIII</i>	Inhibits the <i>FtsH</i> protease
<i>cro</i>	Antirepressor protein that favors lytic cycle
<i>N</i>	Antiterminator acting at <i>nutR</i> and <i>nutL</i>
<i>rex</i>	Protects bacterium from infection by T4 <i>rII</i> mutants
<i>int</i>	Integrase for prophage integration
<i>Q</i>	Antiterminator of late operon
<i>FtsH</i>	Bacterial protease that degrades <i>cII</i> protein
Promoters of	
p_R	Right operon
p_L	Left operon
p_{RE}	Establishment of repression at repressor region
p_{RM}	Maintenance of repression at repressor region
p_R	Late operon
p_I	<i>int</i> gene
Terminators	
t_{R1}	Terminates after <i>cro</i> gene
t_{L1}	Terminates after <i>N</i> gene
Antiterminators	
<i>nutR</i>	In <i>cro</i> gene
<i>nutL</i>	In <i>N</i> gene

- (1) Initial infection. Transcription from p_R and p_L through *cro* and *N*. Termination at t_{R1} and t_{L1} .



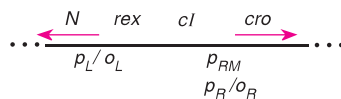
- (2) *N* protein allows antitermination at t_{L1} and t_{R1} . Transcription continues through *cII* and *cIII*. Protein product of *cII* allows transcription at p_I and p_{RE} .



- (3) Repressor and antirepressor (*cl*- and *cro*-gene products) compete for o_R and o_L sites.

Lytic growth

Antirepressor (*cro* protein) gains access to $o_{R3'}$, $o_{R2'}$, $o_{L3'}$, and $o_{L2'}$. Right, left, and late operons transcribed. Repressor region (*cl*, *rex*) repressed.



Lysogeny

Repressor (*cl* protein) gains access to $o_{R1'}$, $o_{R2'}$, $o_{L1'}$, and $o_{L2'}$. Right, left, and late operons repressed. Transcription at p_{RM} enhanced.

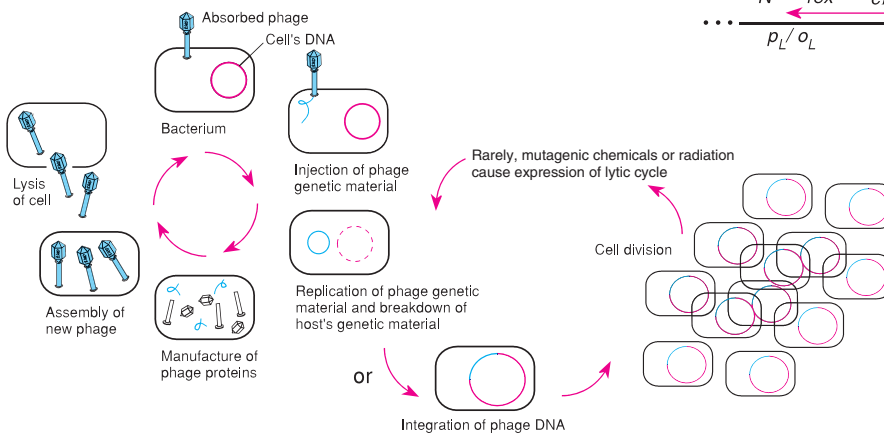
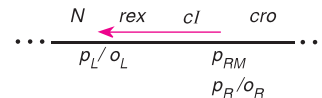


Figure 14.26 Summary of regulation of phage λ life cycles. (1) In the initial infection, transcription begins in *cro* and *N* but terminates shortly thereafter at left and right terminators. (2) The product of the *N* gene allows transcription through the initial terminators; in essence, all genes can now be transcribed. (3) Lysogeny will occur if the *cl* protein gains access to the right and left operators; the lytic cycle will prevail if the *cro*-gene product gains access to those two operators.

TRANSPOSABLE GENETIC ELEMENTS

Up until this point, we have thought of the genome in fairly conservative terms. If we map a gene today, we expect to see it in the same place tomorrow. However, our discovery of mobile genetic elements has modified that view to some extent. We now know that some segments of the genome can move readily from one place to another. Their moving can have an effect on the phenotype of the organism, primarily at the transcriptional level. We thus begin our discussion of mobile genetic elements here, and we conclude it in chapter 16, because mobile elements also affect the phenotypes of eukaryotes.

IS Elements

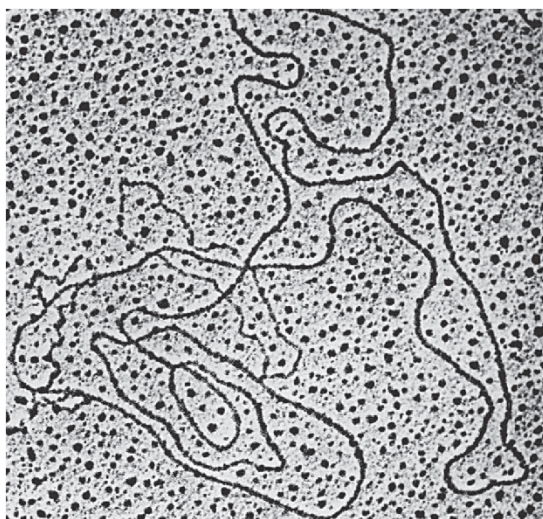
Transposable genetic elements, transposons, or even *jumping genes*, are regions of the genome that can move from one place to another. In some cases, transposition is *conservative*: the transposons move without copying themselves. They are liberated from the donor site by double-strand breaks in the DNA. In other cases, transposition is *replicative*: a copy of the transposon is inserted while the original stays in place. This mechanism involves only single-strand breaks of the DNA at the donor site.

Barbara McClintock first discovered transposable elements in corn in the 1940s (see chapter 16); they were discovered in prokaryotes in 1967, where they first showed up as **polar mutants** in the galactose operon of *E. coli*. No

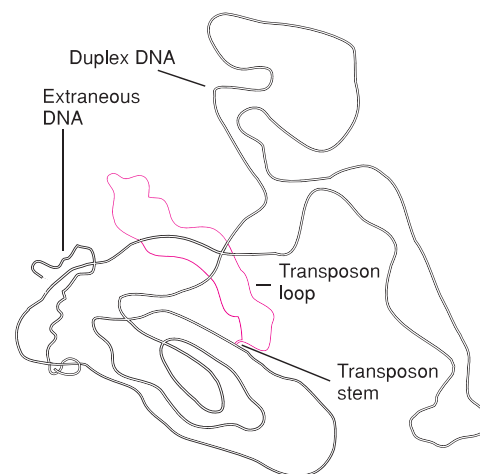
genes of the operon were expressed past the point of the polar mutation. This effect was explained by assuming that the transposon brought with it a transcription stop signal. The presence of an inserted piece of DNA in these polar mutants was verified by heteroduplex analysis (fig. 14.27).

The first transposable elements discovered in bacteria were called **insertion sequences** or **IS elements**. It turns out that these are the simplest transposons. The IS elements consist of a central region of about 700 to 1,500 base pairs surrounded by an inverted repeat of about 10 to 30 base pairs, the numbers depending on the specific IS element. Presumably, the inverted repeats signal the transposing enzyme that it is at the ends of the IS element. The central region of the IS element contains a gene or genes for the transposing event (usually genes for transposase and resolvase enzymes); the relatively small IS elements carry no bacterial genes (fig. 14.28).

The target site that the transposable element moves to is not a specific sequence, as with the *att* site of λ . It becomes a direct repeat flanking the IS element only after insertion, giving rise to a model of insertion (fig. 14.29). The target site is cut in a staggered fashion, leaving single-stranded ends. The IS element is then inserted between the single-stranded ends. Repair processes convert the two single-stranded tails to double-stranded segments and, hence, to direct flanking repeats. When DNA is sequenced, the pattern of a direct flanking repeat surrounding an inverted repeat, with a segment in the middle, signals the existence of a transposable element. Currently, we know of more than fifteen families, including a total of over five hundred known members, of IS elements.



(a)



(b)

Figure 14.27 Heteroduplex analysis revealing a transposon. (a) Two plasmids were hybridized, one with and one without a transposon. (b) The transposon is seen as a single-stranded loop (red); it has nothing to pair with in the heteroduplex. ([a] Courtesy of Richard P. Novick, M.D.)

Figure 14.28 An IS element (IS5) inserted into a target site in a bacterial chromosome creates a direct repeat on either side of the IS element. The explanation is shown in figure 14.29. An inverted repeat (red) is seen as the same sequence read inwards from outside on the upper strand (*left*) and the lower strand (*right*). (Reprinted with permission from *Nature*, Vol. 297, M. Kroger and G. Hoborn, "Structural Analysis of Insertion Sequence IS5." Copyright © 1982 Macmillan Magazines Limited.)

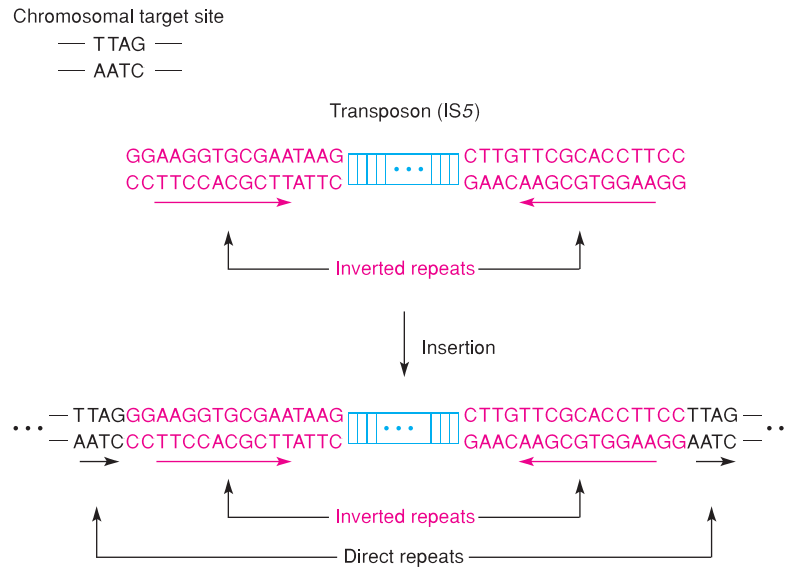
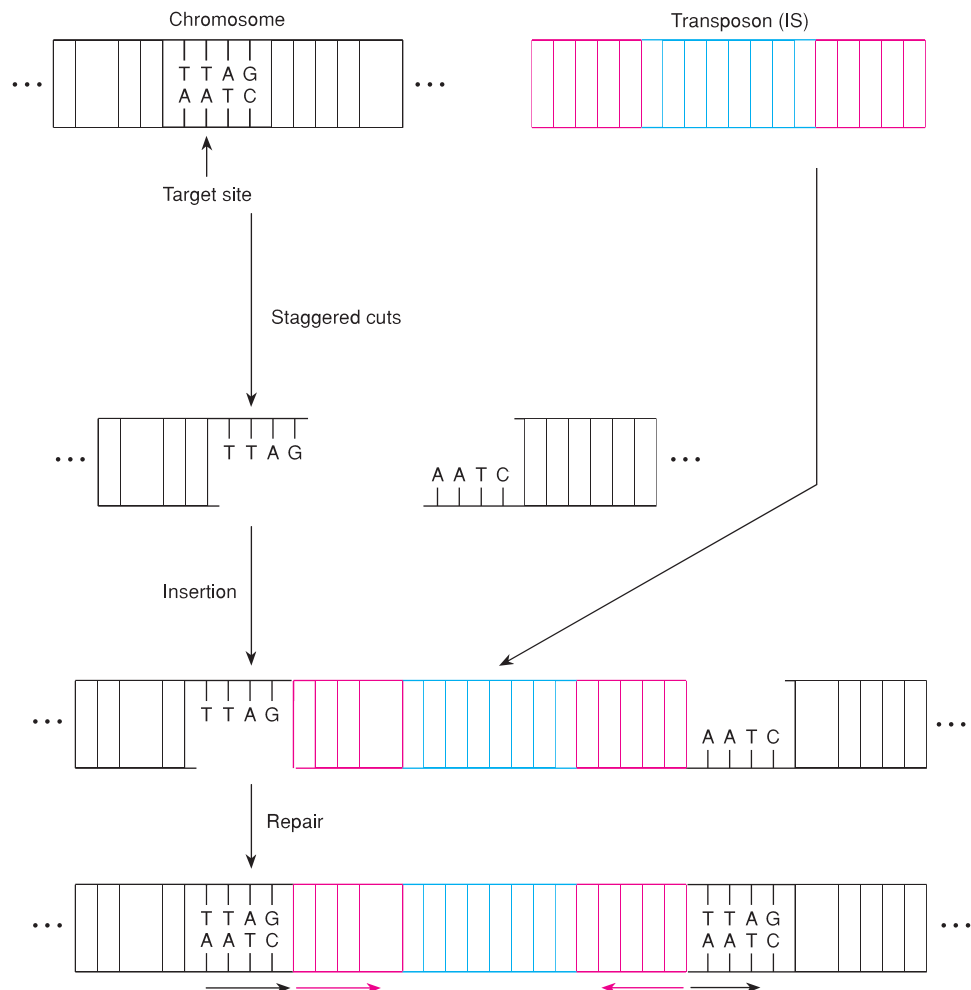


Figure 14.29 Insertion of an IS element (IS5 of fig. 14.28) results in a direct flanking repeat surrounding the transposon in the host chromosome. This occurs because the insertion takes place at a point in which a staggered cut is made in the host DNA, leaving complementary regions on either side of the transposon. Repair replication results in two copies of the flanking sequence.



Composite Transposons

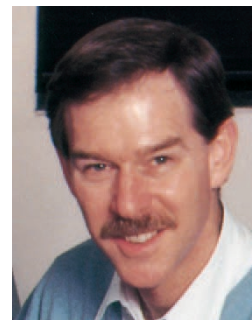
After the discovery of IS elements, a more complex type of transposable element, a **composite transposon**, was discovered. A composite transposon consists of a central region surrounded by two IS elements. The central region usually contains bacterial genes, frequently antibiotic resistance loci. For example the composite transposon *Tn10* contains the genes for transposase and resolvase, as well as the bacterial gene for β -lactamase, which confers resistance to ampicillin (fig. 14.30). Arrangements of composite transposons can vary quite a bit. The IS elements at the two ends can be identical or different; they can be in the same or different orientations; they can be similar to known IS elements or different from any freely existing IS elements. In the latter case, they are called IS-like elements.

Two IS elements can transpose virtually any region between them. In fact, composite transposons most likely came into being when two IS elements became located near each other. We can see this very clearly in a simple experiment. In figure 14.31, there is a small plasmid constructed with transposon *Tn10* in it. The “re-

verse” transposon, consisting of the two IS elements and the plasmid genes, or the normal transposon, could each transpose.

Mechanism of Transposition

Transposition can come about by several mechanisms; however, it does not use the normal recombination machinery of the cell (see chapter 12). One model, by J. Shapiro, explains the fact that many transposons in the



J. A. Shapiro (1943–).
(Courtesy of Dr. J. A. Shapiro.)

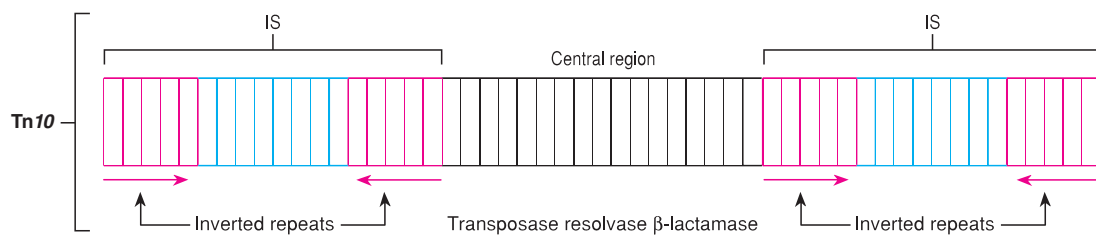


Figure 14.30 A composite transposon consists of a central region flanked by two IS elements. Transposon *Tn10* contains the transposase and resolvase enzyme genes as well as the bacterial gene β -lactamase, which protects the cell from the antibiotic ampicillin.

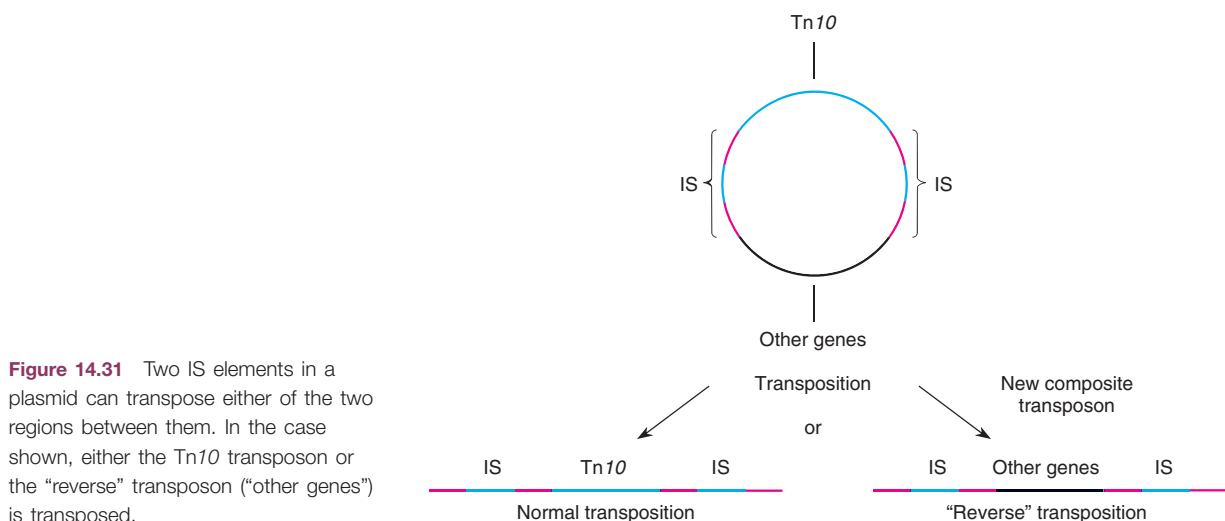


Figure 14.31 Two IS elements in a plasmid can transpose either of the two regions between them. In the case shown, either the *Tn10* transposon or the “reverse” transposon (“other genes”) is transposed.

process of transposition go through a **cointegrate** state (fig. 14.32), in which there is a fusion of two elements. During the process of transposition (in this case from one plasmid to another), an intermediate cointegrate stage is formed, made up of both plasmids and two copies of the transposon. Then, through a process called *resolution*, the cointegrate is reduced back to the two original plasmids, each now containing a copy of the transposon.

A diagram of Shapiro's mechanism appears in figure 14.33. At first, staggered cuts are made in the donor and recipient DNA molecules (fig. 14.33a and b). Then non-homologous ends are joined so that only one strand of the transposon connects them (fig. 14.33c). This process is presumably controlled by the transposon-coded *transposase* enzyme. Repair-DNA replication now takes place to fill in the single-stranded segments. The result is a cointegrate of the two plasmids with two copies of the transposon. The last step is a recombination event at a homologous site within the two transposons. This is catalyzed by a *resolvase* enzyme, which resolves the cointegrate into the original two plasmids, each with a copy of the transposon (fig. 14.33e).

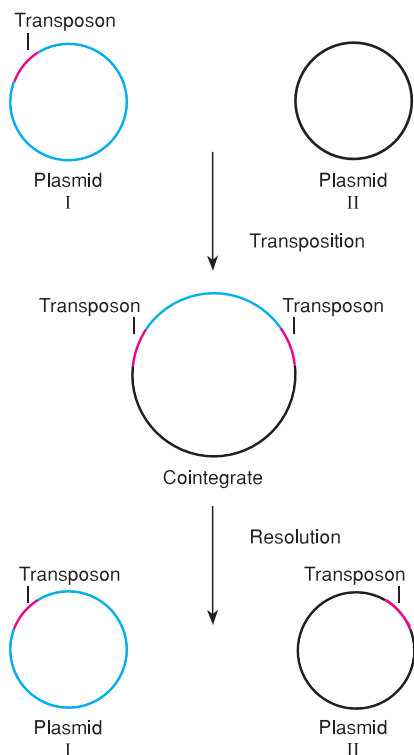


Figure 14.32 Transposition frequently goes through an intermediate cointegrate stage. In this case, the transposon is copied from one plasmid to another, with an intermediate stage consisting of a single large plasmid.

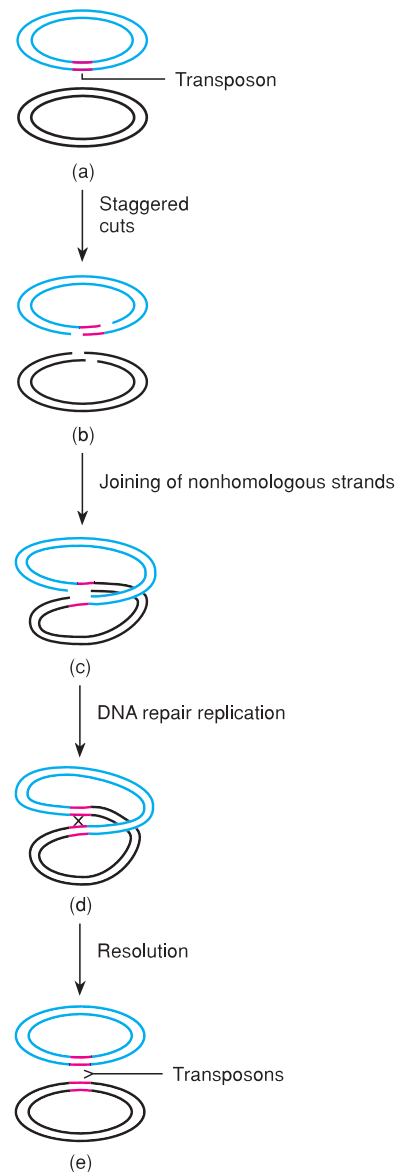


Figure 14.33 The Shapiro mechanism of transposition. Staggered cuts are made at the site of transposon insertion and at either side of the transposon itself (a and b). Nonhomologous single strands join, resulting in two single-stranded copies of the transposon in the cointegrate (c). Repair replication of these single strands produces two copies of the transposon (d). A crossover at the transposon resolves the cointegrate into two plasmids, each with a copy of the transposon (e).

Phenotypic and Genotypic Effects of Transposition

Transposition can have several effects on the phenotype and genotype of an organism. If transposition takes place into a gene or its promoter, it can disrupt the expression of that gene. Depending on the orientation of a transposon, it can prevent the expression of genes. A transposon can also cause deletions and inversions.

Direct repeats on a chromosome can come about, for example, by the sequential transposition of the same IS or transposon, in the same orientation. Pairing followed by recombination results in a deletion of the section between the repeats (fig. 14.34). In the case of inverted repeats, pairing followed by recombination results in an inversion of the section between the repeats.

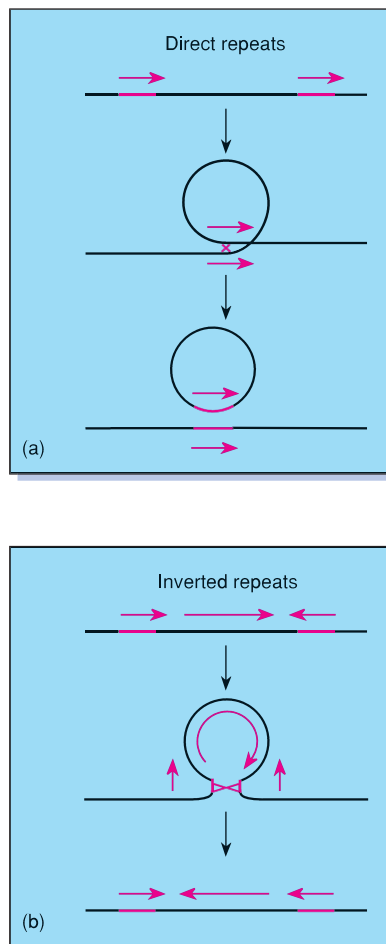


Figure 14.34 Pairing and recombination in DNA repeats. (a) Direct repeats can result in deletion (in the form of a circle) due to a single crossover. (b) Inverted repeats can result in an inversion of the region between the repeats due to a crossover.

A well-known case of transposon orientation controlling a phenotype in bacteria occurs in *Salmonella typhimurium*. The flagella of this bacterium occur in two types. Any particular bacterium has either type 1 or type 2 flagella (called phase 1 or phase 2 flagella). The difference is in the flagellin protein the flagella are composed of. Phase 1 flagella are determined by the *H1* gene and phase 2 flagella are determined by the *H2* gene. The change from one phase to another occurs at a rate of about 10^{-4} per cell division. After extensive genetic analysis, the following scheme was suggested and later verified using recombinant DNA techniques.

The *H1* and *H2* genes are at separate locations on the bacterial chromosome (fig. 14.35). *H2* is part of an operon that also contains the *rH1* gene, the repressor of *H1*. The promoter of this operon lies within a transposon upstream of the operon. When the promoter is in the proper orientation, the *H2* operon is expressed, resulting in phase 2 flagella. The *rH1* gene product represses the *H1* gene (fig. 14.35a). If the inverted repeat ends of the transposon undergo recombination, the transposon is inverted (see fig. 14.34), moving the promoter into an incorrect orientation for the transcription of the *H2* operon. No *H1* repressor is made, so the *H1* gene is expressed (fig. 14.35b).

As N. Kleckner has summarized, transposons can have marked effects on the phenotype by their actions in transposition and by the fact that they may carry genes valuable to the cell. However, they can also exist without any noticeable consequences. This fact has led some evolutionary geneticists to suggest that transposons are an evolutionary accident that, once created, are self-maintaining. Since they may exist without a noticeable benefit to the host's phenotype, transposons have been

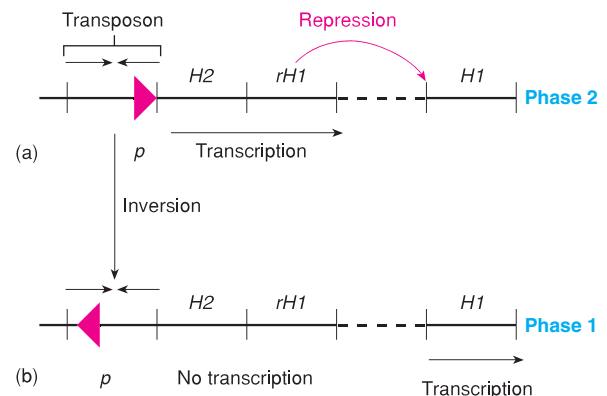


Figure 14.35 Arrangement of flagellin genes on the *Salmonella* chromosome. The promoter (*p*) is within a transposon. In one orientation (a), the *H2* operon is transcribed, which results in *H2* flagellin and *rH1* protein, the repressor of the *H1* gene. In the second orientation (b), the *H2* operon is not transcribed, resulting in uninhibited transcription of the *H1* gene.



Nancy Kleckner (1947–).
(Courtesy of Nancy Kleckner.
Photo by Stu Rosner.)

referred to as **selfish DNA**. In recent theoretical and experimental studies, however, some scientists have suggested that transposons improve the evolutionary fitness of the bacteria that have them (see chapter 20).

OTHER TRANSCRIPTIONAL CONTROL SYSTEMS

Transcription Factors

Phage T4

Phage T4, a relatively large phage with seventy-three genes, has transcription controlled by particular RNA polymerase specificity factors. Like phage λ , phage T4 has early, middle, and late genes, genes that need to be expressed in a particular order. Early T4 genes have promoters whose specificity of recognition depends on the sigma factor of the host (σ^{70} of *E. coli*). Two products of early phage genes are the AsiA and MotA proteins. AsiA binds to the -35 recognition region of σ^{70} , preventing transcription from both host genes and early phage genes. AsiA is thus called an **anti-sigma factor**, a protein that interferes with a sigma factor. Middle phage promoters do not have -35 recognition regions but do bind MotA. Host RNA polymerase bound with the σ^{70} -AsiA complex recognizes these promoters. Finally, late phage genes have promoters that depend on the phage-encoded sigma factor σ^{gp55} . Some proteins are needed both early and late in the infection process; they are specified by genes that have promoters recognized by different specificity factors.

Heat Shock Proteins

A response to elevated temperature, found in both prokaryotes and eukaryotes, is the production of heat shock proteins (see chapter 10). In *E. coli*, elevated temperatures cause the general shutdown of protein synthesis concomitant with the appearance of at least seventeen heat shock proteins. These proteins help protect the cell against the consequences of elevated temperature; some are molecular chaperones (see chapter 11). The production of these

proteins is the direct result of the gene product of the *hspR* gene, which codes for σ^{32} . The normal sigma factor is σ^{70} , the product of the *rpoD* gene; the heat shock genes have promoters recognized by σ^{32} rather than σ^{70} . Heat causes the *hspR* gene to become active, as well as stabilizing σ^{32} . From DNA sequence data, the difference in promoters between normal genes and heat shock genes seems to lie in the -10 consensus sequence (Pribnow box). In normal genes, it is TATAAT; in heat shock protein genes, it is CCC-CATXT, in which X is any base.

Promoter Efficiency

In addition to the mechanisms previously described, there are other ways to regulate the transcription of messenger RNA. One is to control the efficiency of various processes. For example, we know that the promoter sequence of different genes in *E. coli* differs. Since the affinity for RNA polymerase is different for the different sequences, the rate of initiation of transcription for these genes also varies. The more efficient promoters are transcribed at a greater rate than the less efficient promoters. An example is the promoter of the *i* gene of the *lac* operon. This promoter is for a constitutive gene that usually produces only about one messenger RNA per cell cycle. However, mutants of the promoter sequence are known that produce up to fifty messenger RNAs per cell cycle. Here, then, the transcriptional rate is controlled by the efficiency of the promoter in binding RNA polymerase. Efficiency can be controlled by the direct sequence of nucleotides (i.e., differences from the consensus sequence) or by the distance between consensus regions. For example, promoters vary in the number of bases between the -35 and -10 sequences. Seventeen seems to be the optimal number of bases separating the two. Presumably, more or fewer than seventeen reduces the efficiency of transcription.

TRANSLATIONAL CONTROL

When considering control of gene expression, it is important to remember that all control mechanisms are aimed at exerting an influence on either the amount, or the activity, of the gene product. Therefore, in addition to transcriptional controls, which influence the amount of messenger RNA produced, there are also translational controls affecting how efficiently the messenger RNA is translated. (Attenuator control—see fig. 14.16—can also be viewed as translational control because the environment is tested by translation even though attenuation results in the cessation of transcription.) In prokaryotes, translational control is of lesser importance than transcriptional control for two reasons. First, messenger RNAs are extremely unstable; with a lifetime of only

about two minutes, there is little room for controlling the rates of translation of existing messenger RNAs because they simply do not last very long. Second, although there are some indications of translational control in prokaryotes, such control is inefficient—energy is wasted synthesizing messenger RNAs that may never be used.

Translational control can be exerted on a gene if the gene occurs distally from the promoter in a polycistronic operon. The genes that are transcribed last appear to be translated at a lower rate than the genes transcribed first. The three *lac* operon genes, for instance, are translated roughly in a ratio of 10:5:2. This ratio is due to the polarity of the translation process. That is, in prokaryotes, translation is directly tied to transcription—a messenger RNA can have ribosomes attached to it well before transcription ends. Thus, genes at the beginning of the operon are available for translation before genes at the end. In addition, exonucleases seem to degrade messenger RNA more efficiently from the 3' end. Presumably, natural selection has ordered the genes within operons so that those producing enzymes needed in greater quantities will be at the beginning of an operon.

Translation can also be regulated by RNA-RNA hybridization. RNA complementary to the 5' end of a messenger RNA can prevent the translation of that messenger RNA. The regulating RNA is called **antisense RNA**. In figure 14.36, the messenger RNA from the *ompF* gene in *E. coli* is prevented from being translated by complementary base pair binding with an antisense RNA called *micF* RNA (*mic* stands for *mRNA-interfering complementary RNA*). The *ompF* gene codes for a membrane component called a *porin*, which, as the name suggests, provides pores in the cell membrane for transport of materials. Surprisingly, a second porin gene, *ompC*, seems to be the source of the *micF* RNA. Transcription of the opposite DNA strand (the one not normally transcribed) near the promoter of the *ompC* gene yields the antisense

RNA. One porin gene thus seems to regulate the expression of another porin gene, for reasons that are not completely understood. Antisense RNA has also been implicated in such phenomena as the control of plasmid number and the control of transposon Tn10 transposition. Control by antisense RNA is a fertile field for gene therapy because antisense RNA can be artificially synthesized and then injected into eukaryotic cells.

A third translational control mechanism consists of the efficiency with which the messenger RNA binds to the ribosome. This is related to some extent to the sequence of nucleotides at the 5' end of the messenger RNA that is complementary to the 3' end of the 16S ribosomal RNA segment in the ribosome (the Shine-Dalgarno sequence). Variations from the consensus sequences demonstrate different efficiencies of binding and, therefore, the initiation of translation occurs at different rates.

The redundancy in the genetic code can also play a part in translational control of some proteins since different transfer RNAs occur in the cell in different quantities. Genes with abundant protein products may have codons that specify the more common transfer RNAs, a concept called **codon preference**. In other words, certain codons are preferred; they specify transfer RNAs that are abundant. Genes that code for proteins not needed in abundance could have several codons specifying the rarer transfer RNAs, which would slow down the rate of translation for these genes. The codon distribution of the phage MS2 in table 14.2 shows that every codon is used except the UGA stop codon. (The numbers in the table refer to the incidence of a particular codon in the phage genome.) However, the distribution is not random for all amino acids. For example, the amino acid glycine has two common codons and two rarer codons. The same holds for arginine but not, for example, valine.

Finally, translational control can be exerted at the ribosome. When an uncharged transfer RNA finds its way into

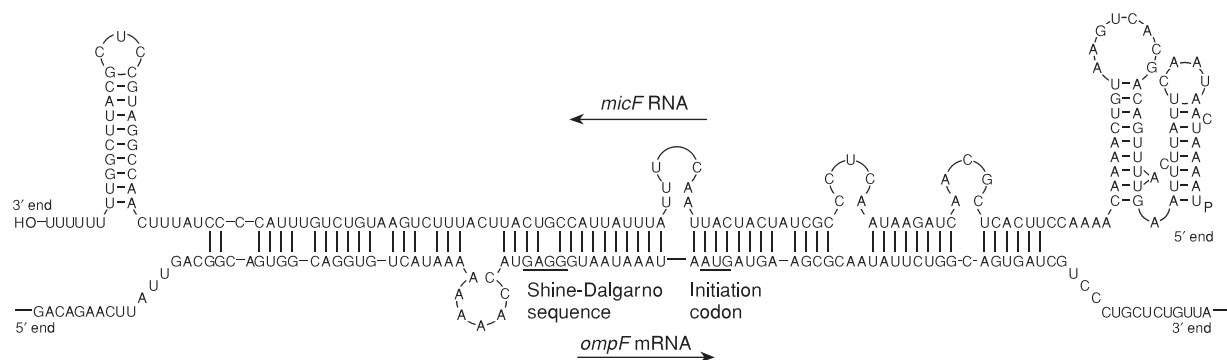


Figure 14.36 Complementarity between the RNA of the *ompF* gene and antisense RNA, or *micF* RNA. The region of overlap includes the Shine-Dalgarno sequence and the initiation codon, effectively preventing ribosome binding and translation of the *ompF* RNA. Notice the stem-loops on each side of the overlap. (Reproduced, with permission, from the *Annual Review of Biochemistry*, Volume 55, ©1986 by Annual Reviews, Inc.)

the A site of the ribosome, a likely event under amino acid starvation, it causes an **idling reaction** in the ribosome, which entails the production of the nucleotide guanosine tetraphosphate (5'-ppGpp-3'; fig. 14.37). This is part of a control mechanism called the **stringent response**. A protein called the **stringent factor**, the product of the *relA* gene, produces guanosine tetraphosphate (ppGpp), originally called *magic spot* because of its sudden appearance on chromatograms. (The gene is called *rel* from the **re-laxed mutant**, which does not have the stringent response.) The stringent factor is associated with the ribosome, where ppGpp is synthesized, although it is not one of the structural proteins of the ribosome. The SpoT protein, the product of the *spoT1* gene, breaks down ppGpp; thus, the concentration of ppGpp is regulated.

The ppGpp interacts with RNA polymerase, causing an almost complete cessation of the transcription of ribosomal RNA; thus, no energy is wasted synthesizing ribosomes when translation is not possible. However, many amino acid-synthesizing operons require ppGpp for transcription; ppGpp thus inhibits ribosomal RNA production and enhances the production of enzymes to synthesize amino acids, all when the cell is starved for amino acids.

One other thing a ribosome can do when faced with amino acid shortages is to slide past *hungry* codons, codons for which a charged transfer RNA is not available. At that point, the growing peptide chain will be attached to the last charged transfer RNA to enter the ribosome, the one now in the peptidyl (P) site. The complex of the transfer RNA and the ribosome slides down the messenger RNA until it encounters the next codon for the transfer RNA. At this point, it is hoped, the next codon en-

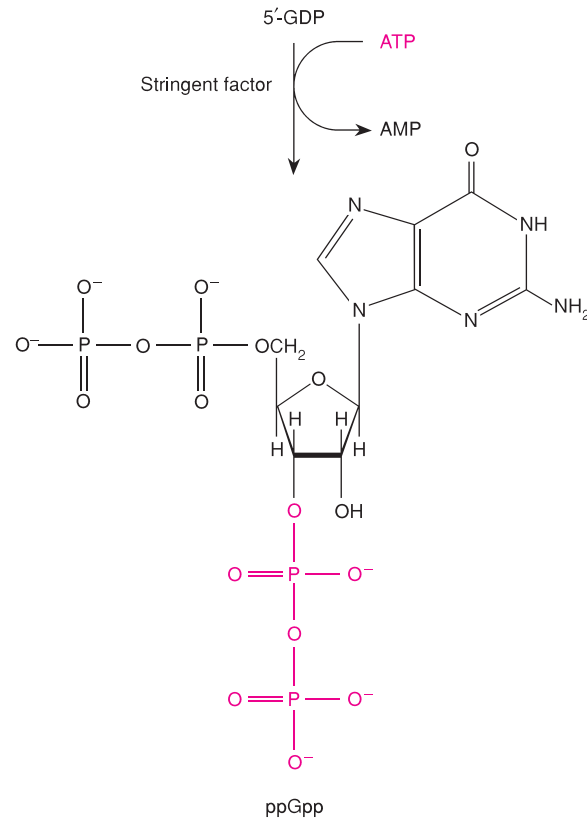


Figure 14.37 The idling reaction. The stringent factor catalyzes the conversion of GDP to 5'-ppGpp-3'. The added pyrophosphate groups come from ATP.

Table 14.2 Codon Distribution in MS2, an RNA Virus

First Position	Second Position								Third Position
	U	C	A	G	U	C	A	G	
U	Phe	10	Ser	13	Tyr	8	Cys	7	U
	Phe	13	Ser	10	Tyr	13	Cys	4	C
	Leu	11	Ser	10	stop	1	stop	0	A
	Leu	4	Ser	13	stop	1	Trp	14	G
C	Leu	10	Pro	7	His	4	Arg	13	U
	Leu	14	Pro	3	His	4	Arg	11	C
	Leu	13	Pro	6	Gln	10	Arg	6	A
	Leu	6	Pro	5	Gln	16	Arg	4	G
A	Ile	8	Thr	14	Asn	11	Ser	4	U
	Ile	16	Thr	10	Asn	23	Ser	8	C
	Ile	7	Thr	8	Lys	12	Arg	8	A
	Met	15	Thr	5	Lys	17	Arg	6	G
G	Val	13	Ala	19	Asp	18	Gly	17	U
	Val	12	Ala	12	Asp	11	Gly	11	C
	Val	11	Ala	14	Glu	9	Gly	4	A
	Val	10	Ala	8	Glu	14	Gly	4	G

countered in the aminoacyl (A) site will code for a charged transfer RNA present.

POSTTRANSLATIONAL CONTROL

Feedback Inhibition

Even after a gene has been transcribed and the messenger RNA translated, a cell can still exert some control over the functioning of the enzymes produced if the enzymes are allosteric proteins. We have discussed the activation and deactivation of operon repressors (e.g., *lac*, *trp*) owing to their allosteric properties. Similar effects occur with other proteins. The need for posttranslational control is apparent because of the relative longevity of proteins as compared with RNA. When an operon is repressed, it no longer transcribes messenger RNA; however, the messenger RNA that was previously transcribed has been translated into protein, and this protein is still functioning. Thus, during operon repression, it would also be efficient for the cell to control the activity of existing proteins.

An example of posttranslational control occurs with the enzyme aspartate transcarbamylase, which catalyzes the first step in the pathway of pyrimidine biosynthesis in *E. coli* (fig. 14.38). An excess of one of the end products of the pathway, cytidine triphosphate (CTP), inhibits the functioning of aspartate transcarbamylase. This method of control is called **feedback inhibition** because a product of the pathway is the agent that turns the pathway off.

Aspartate transcarbamylase is an allosteric enzyme. Its active site is responsible for the condensation of carbamyl phosphate and L-aspartate (fig. 14.38). However, it also has regulatory sites that have an affinity for CTP. When CTP is bound in a regulatory site, the conformation of the enzyme changes, and the enzyme has a lowered affinity for its normal substrates; recognition of CTP inhibits the condensation reaction the enzyme normally carries out (fig. 14.39). Thus, allosteric enzymes provide a mechanism for control of protein function after the protein has been synthesized.

Protein Degradation

A final control affecting the amount of gene product in a cell is control of the rate at which proteins degrade. The normal life spans of proteins vary greatly. For example, some proteins last longer than a cell cycle, whereas others may be broken down in minutes. Several models have been suggested for control of protein degradation, including the **N-end rule** and the **PEST hypothesis**.

According to the N-end rule, the amino acid at the amino-, or N-terminal, end of a protein is a signal to proteases that control the average length of life of a protein.

In recent experiments, the life span of the β -galactosidase protein was determined with almost complete predictability based on its modified N-terminal amino acid. Protein life spans range from two minutes for those with N-terminal arginine to greater than twenty hours for those with N-terminal methionine or five other amino acids (table 14.3).

According to the PEST hypothesis, protein degradation is determined by regions rich in one of four amino acids: proline, glutamic acid, serine, and threonine. (The one-letter abbreviations of these four amino acids are P, E, S, and T, respectively.) Proteins that have these regions tend to degrade in less than 2 hours. In one study of thirty-five proteins with half-lives of between 20 and 220 hours, only three contained a PEST region. We see that not only are different proteins programmed to survive for varying lengths of time in the cell, but that programming seems to be based on the N-terminal amino acid as well as various regions rich in the PEST amino acids.

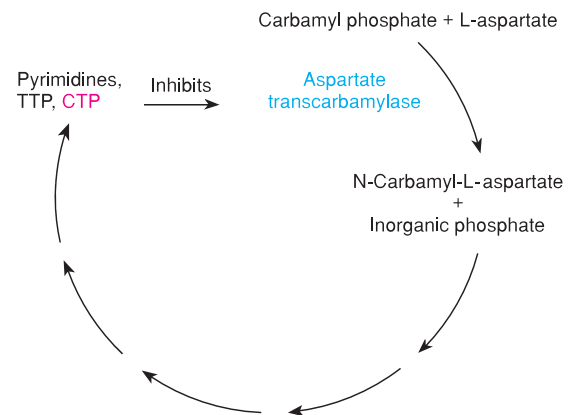


Figure 14.38 Aspartate transcarbamylase catalyzes the first step in pyrimidine biosynthesis. An end product, cytidine triphosphate (CTP), inhibits the enzyme.

Table 14.3 Relationship Between N-Terminal Amino Acid and Half-Life of *E. coli* β -Galactosidase Proteins with Modified N-Terminal Amino Acids

N-Terminal Amino Acid	Half-Life
Met, Ser, Ala, Thr, Val, Gly	>20 hours
Ile, Glu	30 minutes
Tyr, Gln	10 minutes
Pro	7 minutes
Phe, Leu, Asp, Lys	3 minutes
Arg	2 minutes

Source: Data from Bachmair, et al., *Science*, 234:179–86, 1986.

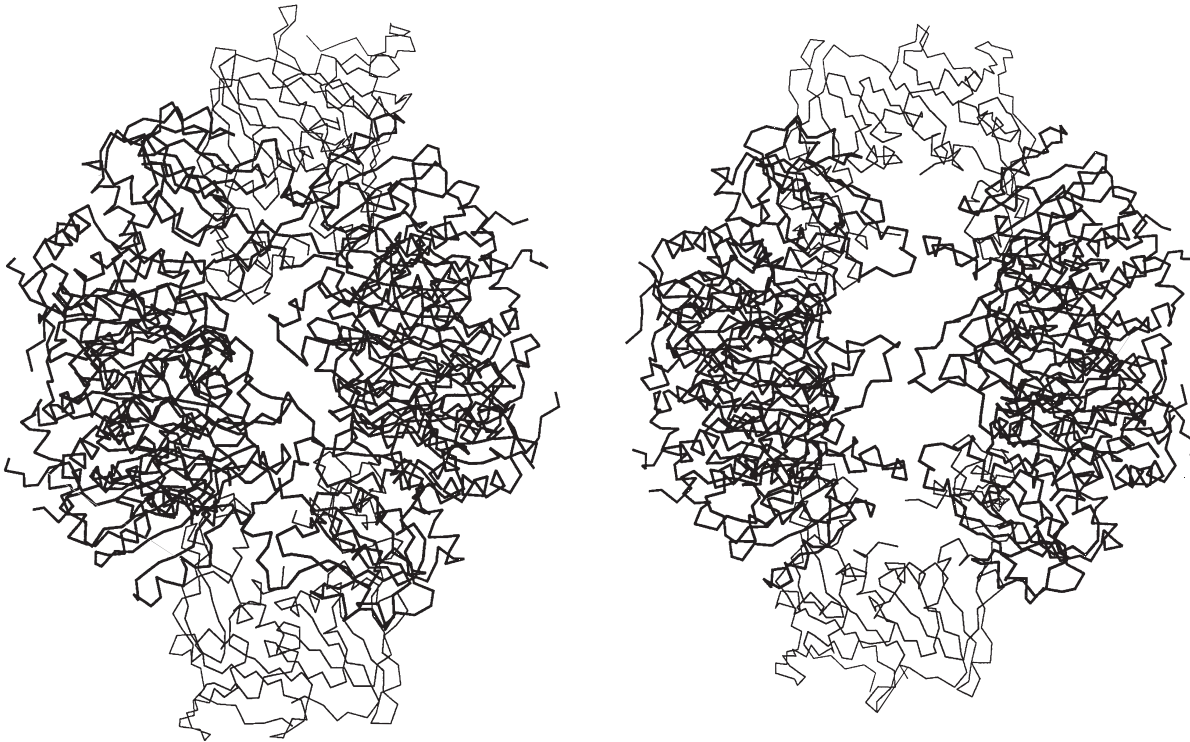


Figure 14.39 Aspartate transcarbamylase. *Left:* the enzyme bound with cytidine triphosphate (CTP). *Right:* the enzyme without CTP. Notice the difference in shape with and without CTP. With CTP, the enzyme literally closes up. (Courtesy of Evan R. Kantrowitz.)

S U M M A R Y

STUDY OBJECTIVE 1: To study the way in which inducible and repressible operons work 406–414

Most bacterial genes are organized into operons, which can either be repressed or induced. Transcription begins in inducible operons, such as *lac*, when the metabolite that the operon enzymes act upon appears in the environment. The metabolite (or a derivative), the inducer, combines with the repressor (the product of the independent regulator gene) and renders the repressor nonfunctional. In the absence of the inducer, the repressor binds to the operator, a segment between the promoter and the first gene of the operon. When in place, the repressor blocks transcription. After combining with the inducer, the repressor diffuses from the operator, and transcription proceeds.

All operons responsible for the breakdown of sugars in *E. coli* are inducible. In the presence of glucose, other inducible sugar operons (such as the arabinose and galactose operons) are repressed, even if their sugars appear in the environment. This process is called *catabolite repression*. Cyclic AMP and a catabolite activator protein (CAP) enhance the

transcription of the nonglucose sugar operons. Glucose lowers the level of cyclic AMP in the cell and thus prevents the enhancement of transcription of these other operons.

Repressible operons, such as the *trp* operon in *E. coli*, have the same basic traits as an inducible operon—polycistronic transcription controlled by an operator site between the promoter and the first structural gene. However, the repressor protein, controlled by an independent regulator gene, is functional in blocking transcription only after it has combined with the corepressor. This corepressor is the end product of the operon's pathway or some form of the end product (tryptophan in the *trp* operon).

STUDY OBJECTIVE 2: To examine attenuator control in bacteria 415–418

Amino acid-synthesizing operons often have an attenuator region. The ability of a ribosome to translate a leader peptide gene determines the secondary structure of the messenger RNA transcript. If the ribosome can translate the leader peptide gene, there must be adequate quantities of

the amino acid present, and a terminator stem and loop form in the messenger RNA, causing termination of transcription. In the *trp* operon of bacilli, a protein that binds tryptophans serves the same purpose.

STUDY OBJECTIVE 3: To analyze the control of the life cycle of phage λ 418–424

Control of gene expression in λ phage is complex. The “decision” for lytic versus lysogenic response is determined by competition between two repressors, *CI* and *Cro*.

STUDY OBJECTIVE 4: To determine the way in which transposable genetic elements transpose and control gene expression in bacteria 425–430

Transposons are mobile genetic elements; copies of them can be inserted at other places in the genome. Their ends are inverted repeats. Upon insertion, they are flanked by short, direct repeats. They can be simple (IS elements) or

complex. Their presence can cause inversions or deletions. The flagellar phase in *Salmonella* is controlled by the orientation of a transposon.

STUDY OBJECTIVE 5: To look at other transcriptional and posttranscriptional mechanisms of control of gene expression in bacteria and phages 430–434

Affinity of early and late operons in phages for different sigma factors are another form of transcriptional control, as seen in phage T4 transcription and the transcription of heat shock proteins. Translational control can be exercised through a gene's position in an operon (genes at the beginning are transcribed most frequently), through redundancy of the genetic code, or through a stringent response that shuts down most transcription during starvation. Posttranslational control is primarily regulated by feedback inhibition. The N-terminal amino acid or particular regions within the proteins program the rate of protein degradation.

S O L V E D P R O B L E M S

PROBLEM 1: How could you determine whether the genes for the breakdown of the sugar arabinose are under inducible control in *E. coli*?

Answer: *Inducible* means that the genes to break down the substrate—arabinose, a five-carbon sugar, in this case—are not active in the absence of the inducer (again, arabinose). Therefore, in the absence of arabinose in the cells' environment, the arabinose utilization enzymes should not be active within the bacterial cells, but after arabinose is added to the medium, the enzymes should be present. We thus need to assay the contents of the cells before and after arabinose is added to the medium, performing the assay after the cells are broken open and the DNA destroyed so as not to confound the experiment. Using a standard biochemical analysis for arabinose, we should find that the bacterial cell is incapable of metabolizing arabinose before induction but capable of metabolizing it afterward. If the cells were capable of metabolizing arabinose in both cases, we would say that arabinose utilization is constitutive. If the cells were incapable of utilizing arabinose in both cases, we would conclude that the bacterium is incapable of using the sugar arabinose as an energy source. (In fact, arabinose utilization is inducible.)

PROBLEM 2: Why would the RecA protein of *E. coli* cleave the λ repressor?

Answer: Since the cleaving of the λ repressor is a signal to begin the lytic phase of the life cycle of the phage, it seems odd that the lysogenized bacterial cell would be

an accomplice to its own destruction. However, the phenomenon makes much more sense if we realize that the RecA protein has several other functions critically important to the bacterial cell (see chapter 12). The λ phage has evolved the ability to take advantage of the existence of the RecA protein by evolving a repressor sensitive to it. Evolutionary biologists view this as “co-evolution,” two interacting organisms evolving to take advantage of or minimize properties of the other. The bacterium, however, might be at a disadvantage. Since RecA has many functions involving interactions with other proteins, it may be highly limited in how it can change. This is one plausible explanation as to why RecA liberates phage λ .

PROBLEM 3: What are the differences in action of the λ promoters p_{RE} and p_{RM} ?

Answer: The promoters p_{RE} and p_{RM} are both promoters of the repressor operon of phage λ . Transcription from these promoters allows production of the *ci* repressor protein, the repressor that favors lysogeny. Initially, the promoter p_{RE} is activated. For it to be a transcription site, it must be activated by the product of the *cII* gene, which lies in the right operon. This promoter, p_{RE} , produces a messenger RNA with a long leader that is translated very efficiently. Once the repressor binds at the operators of the left and right operons, the *cII* gene is no longer transcribed, and therefore p_{RE} is no longer a site for transcription. However, the repressor gene can still be transcribed from the p_{RM} promoter, which does not need the product

of the *cII* gene. This promoter produces a transcript with no leader and thus is translated very inefficiently. At that point, however, only a very small quantity of repressor is needed to maintain lysogeny. Thus, the two promoters are the sites for the initiation of the repressor operon under different circumstances: one early in the infection stage and one after lysogeny is under way.

PROBLEM 4: What are the phenotypes of the following partial diploids for the *lac* operon in *E. coli* in the presence and absence of lactose?

- (F') $i^+ o^+ p^+ z^- / i^- o^+ p^+ z^+$ (chromosome)
- (F') $i^+ o^c p^+ z^+ / i^+ o^+ p^- z^+$ (chromosome)

Answer: Consider one DNA molecule at a time. If one DNA molecule can never make the enzyme, it can be ignored. In (a), the plasmid DNA (F') will never make enzyme (it is z^-), and the chromosomal DNA will never make repressor (it is i^-). The functional repressor in the plasmid (i^+) will bind to both DNAs, and hence the chromosomal operon will not be transcribed in the absence of lactose and will be induced to transcribe in the presence of lactose. In (b), the plasmid DNA (F') will always be transcribing (operator constitutive) because the repressor can never bind the operator (o^c); hence, the operon will be transcribed all the time. The chromosomal DNA can never make RNA (it is p^-).

EXERCISES AND PROBLEMS *

LAC OPERON (INDUCIBLE SYSTEM)

- Are the following *E. coli* cells constitutive or inducible for the *z* gene?
 - $i^+ o^+ z^+$
 - $i^- o^+ z^+$
 - $i^- o^c z^+$
 - $i^+ o^c z^+$
 - $i^s o^+ z^+$
 - $i^Q o^+ z^+$
- Determine whether the following *lac* operon merozygotes are inducible or constitutive for the *z* gene.
 - $i^+ o^+ z^+ / F' i^+ o^+ z^+$
 - $i^- o^+ z^+ / F' i^+ o^+ z^-$
 - $i^+ o^+ z^+ / F' i^- o^+ z^-$
 - $i^- o^c z^- / F' i^+ o^+ z^+$
 - $i^- o^+ z^- / F' i^+ o^c z^+$
- You have isolated a repressor for an inducible operon and have determined that it has two different binding sites, one for the inducer and one for the operator. Mutants of the repressor result in three different phenotypes as far as binding is concerned. What are these phenotypes?
- An *E. coli* strain is isolated that produces β -galactosidase (*lac z*) and permease (*lac y*) constitutively. Provide two possible mutations that could cause this phenotype, and then describe how each mutation would behave in a partial diploid in which the second operon is wild-type for the entire *lac* system.
- You have isolated two *E. coli* mutants that synthesize β -galactosidase constitutively.

- If these mutants affect different functions, in what two functions could they be defective?
 - You can make a partial diploid of the mutants with the wild-type. What result do you expect for each mutant?
- A hypothetical operon has a sequence of sites, Q R S T U, in the promoter region, but the exact location of the operator and promoter consensus sequences have not been identified. Various deletions of this operator region are isolated and mapped. Their locations appear as follows, with a "/" representing a deleted region.

	Q	R	S	T	U
Deletion					
1	////////				
2		////////			
3			////////		
4			//////////		
5				////////	

Deletions 3 and 4 are found to produce constitutive levels of RNA of the operon, and deletion 1 is found to never make RNA. Where are the operator and promoter consensus sequences probably located?

CATABOLITE REPRESSION

- Describe the role of cyclic AMP in transcriptional control in *E. coli*.
- Operon systems exert negative control by acting through inhibition. The CAP system exerts positive control because it acts through enhancement of transcription. Describe how an operon could work if it were dependent only upon positive control.

* Answers to selected exercises and problems are on page A-16.

9. J. Beckwith isolated point mutations that were simultaneously uninducible for the *lac*, *ara*, *mal*, and *gal* operons, even in the absence of glucose. Provide two different functions that could be missing in these mutants.

TRP OPERON (REPRESSIBLE SYSTEM)

10. Construct a merozygote of the *trp* operon in *E. coli* with two forms of the first gene (*e* gene: e_1, e_2) in the operon. Describe the types of *cis* and *trans* effects that are possible, given mutants of any component of the operon. Can this repressible system work for any type of operon other than those that control amino acid synthesis?
11. The tryptophan operon is under negative control; it is on (transcribing) in the presence of low levels of tryptophan and off in the presence of excess tryptophan. The symbols *a*, *b*, and *c* represent the gene for tryptophan synthetase, the operator region, and the repressor—but not necessarily in that order. From the following data, in which superscripts denote wild-type or defective, determine which letter is the gene, the repressor, and the operator (+ is tryptophan synthetase activity; – is no activity).

Strain	Genotype	Tryptophan Absent	Tryptophan Present
1	$a^- b^+ c^+$	+	+
2	$a^+ b^+ c^-$	+	+
3	$a^+ b^- c^+$	–	–
4	$a^+ b^- c^+ / a^- b^+ c^-$	+	+
5	$a^+ b^+ c^- / a^- b^- c^-$	+	–
6	$a^+ b^+ c^- / a^- b^- c^+$	+	–
7	$a^- b^+ c^+ / a^+ b^- c^-$	+	+

12. The histidine operon is a repressible operon. The corepressor is charged tRNA^{His}, and its gene is not part of the operon. For the following mutants, tell whether the enzymes of the operon will be made; then tell whether each mutant would be *cis*-dominant in a partial diploid.
- RNA polymerase cannot bind the promoter.
 - The repressor-corepressor complex cannot bind operator DNA (the operator has the normal sequence).
 - The repressor cannot bind charged tRNA^{His}.

TRP OPERON (ATTENUATOR-CONTROLLED SYSTEM)

13. Describe the interaction of the attenuator and the operator control mechanisms in the *trp* operon of *E. coli* under varying concentrations of tryptophan in the cell. How does attenuator control react to shortages of other amino acids?

LYTIC AND LYSOGENIC CYCLES IN PHAGE λ

14. What is the fate of a λ phage entering an *E. coli* cell that contains quantities of λ repressor? What is the fate of the same phage entering an *E. coli* cell that contains quantities of the *cro*-gene product?
15. Describe the fate of λ phages during the infection process with mutants in the following genes: *cI*, *cII*, *cIII*, *N*, *cro*, *att*, *Q*.
16. What is the fate of λ phages during the infection process with mutants in the following areas: *O_{RI}*, *O_{R3}*, *P_L*, *P_{RE}*, *P_{RM}*, *P_R*, *t_{LI}*, *t_{RI}*, *nutL*, *nutR*?
17. What are the three different physical forms that the phage λ chromosome can take?
18. How does ultraviolet light (UV) damage induce the lytic life cycle in phage λ ?
19. The λ prophage is sometimes induced into the lytic life cycle when an Hfr lysogen (lysogenic cell) conjugates with a nonlysogenic F[–] cell. How might induction come about in this instance?
20. A temperature-sensitive mutant of the λ *cI* gene has been isolated. At 30° C the *cI* repressor binds λ DNA, but it cannot bind DNA at 42° C (it denatures). What is the consequence of incubating *E. coli* that are lysogenic for this λ mutant at 42° C?
21. The mutant in problem 20 is heated to 42° C for five minutes, cooled to 30° C, and grown for one hour so that the cells divide several times. The temperature is then raised to 42° C, and you wait for lysis. Many of the cells are not lysed and are in fact able to form colonies. Explain these results.

TRANSPOSABLE GENETIC ELEMENTS

22. Why are IS elements sometimes referred to as “selfish DNA”?
23. What are the differences among an IS element, a transposon, an intron, a plasmid, and a cointegrate?
24. Describe the Shapiro model of transposition. What are the roles of transposase, DNA polymerase I, ligase, and resolvase?
25. Why are transposons flanked by direct repeats?
26. How do transposons induce deletions? inversions?
27. Describe how a transposon controls the expression of the flagellar phase in *Salmonella*.
28. What is a polar mutation? What can cause it?

OTHER TRANSCRIPTIONAL CONTROL SYSTEMS

29. List the steps from transcription through translation to enzyme function, noting all the points at which control could be exerted. (See also TRANSLATIONAL CONTROL and POSTTRANSLATIONAL CONTROL)

30. What are the advantages of transcriptional control over translational control? (See also TRANSLATIONAL CONTROL)
31. How are heat shock proteins induced?
32. In phage T4, the genes *rIIA* and *rIIB* lie adjacent to each other on the T4 chromosome. During the early phase of infection, *rIIA* and *rIIB* products are present in equimolar amounts. In the late phase of infection, the amount of *rIIB* protein is ten to fifteen times higher than that of *rIIA* protein. Nonsense mutations (mutations to a stop codon) in *rIIA* eliminate early but not late *rIIB* transcription. In the mutants that contain small deletions near the end of *rIIA*, the amount of *rIIA* product is always equal to the amount of *rIIB* product, regardless of the time of infection. Based on this information, devise a map of the *rII* region. Include the location(s) of the promoter(s).

TRANSLATIONAL CONTROL

33. What is the stringent response? How does it work? What is an idling reaction?
34. What is antisense RNA? How does it work? What is the obvious source of this regulatory RNA? How could this RNA be used to treat a disease clinically?

POSTTRANSLATIONAL CONTROL

35. What is feedback inhibition? What other roles do allosteric proteins play in regulating gene expression?
36. What controls the rate of degradation of proteins?

C R I T I C A L T H I N K I N G Q U E S T I O N S

1. From an evolutionary perspective, why do you think *Escherichia coli* evolved a CAP system of positive control of gene expression? Why not just metabolize any and all sugars in the environment as they appear?
2. Why might some proteins and messenger RNAs produced in *Escherichia coli* be degraded so quickly?

Suggested Readings for chapter 14 are on page B-13.

15

THE EUKARYOTIC CHROMOSOME

STUDY OBJECTIVES

1. To examine the arrangement of DNA and proteins comprising the eukaryotic chromosome 440
2. To look at the nature of centromeres and telomeres in eukaryotic chromosomes 453
3. To analyze the nature of the DNA in eukaryotic chromosomes 457

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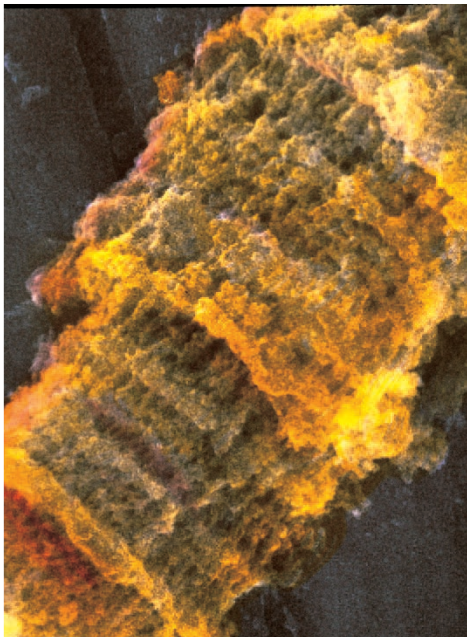
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Artificially colored scanning electron micrograph of part of a polytene salivary gland chromosome from a fruit fly (*Drosophila*), revealing the underlying banding pattern. (© Professors P. Motta and T. Naguro/SPL/Photo Researchers, inc.)

In chapter 14, we looked at the control of gene expression in prokaryotes and bacteriophages. Compared to eukaryotes, bacteriophages and prokaryotes are relatively simple. Of fundamental importance is that, in these lower forms, the operon model of induction and repression of transcription is a unifying theme for control of gene expression. Despite nuances such as catabolite repression and attenuator control, the operon model provides a relatively clear picture of how genes are turned on and off in phages and prokaryotes. This model does not exist for eukaryotes. In attempting to elucidate models for control of gene expression in eukaryotes, we must take one very important factor into account: the complexity of the structure of the eukaryotic chromosome. In this chapter, we cover the current understanding of how these very large structures are organized.

THE EUKARYOTIC CELL

Eukaryotes and prokaryotes are the two superkingdoms of organisms. The following comparisons, using *E. coli* as a general model for prokaryotes, show how much more complex eukaryotes are:

1. An *E. coli* chromosome contains approximately 4.2×10^6 base pairs of DNA. The haploid human genome contains nearly one thousand times as much DNA.
2. Eukaryotic DNA is in the form of nucleoprotein, a DNA-histone protein complex. Although a few histonelike proteins have been found in *E. coli*, its chromosomal DNA is not complexed with protein to anywhere near the same extent.
3. An *E. coli* cell has very little internal structure. Eukaryotes have a number of internal organelles and an extensive lipid membrane system, including the nuclear envelope itself.
4. An *E. coli* cell is small (0.5 to 5.0 μm in length for bacteria). Eukaryotic cells are generally larger than prokaryotes (10 to 50 μm in length for animal tissue cells; box 15.1).
5. The messenger RNA of *E. coli* is translated while it is being transcribed. Eukaryotic messenger RNA is modified within the nucleus before it is transported out for translation in the cytoplasm.
6. Almost no messenger RNA isolated from eukaryotic cells, including the messenger RNA of animal viruses, has been found to be polycistronic (containing many genes). Most prokaryotic messenger RNAs are polycistronic.
7. Most *E. coli* genes are parts of inducible or repressible operons; there are almost no operons in eukaryotes.
8. *E. coli* exists as a simple, single cell. Although some prokaryotes do aggregate, sporulate, and show a few

other limited forms of differentiation, they are primarily one-celled organisms. And, although some eukaryotes are single-celled (e.g., yeast), the essence of eukaryotes is differentiation. In human beings, a zygote gives rise to every other cell type in the body in a relatively predictable manner.

To fully appreciate the complexity of eukaryotes, we begin by looking at the eukaryotic chromosome. In the next chapter, we look at the patterns of development in eukaryotes and some mechanisms of control of gene expression.

THE EUKARYOTIC CHROMOSOME



DNA Arrangement

Evidence that the eukaryotic chromosome is **uninemic**—that is, contains one double helix of DNA—comes from several sources. The best data are provided by radioactive-labeling studies, first done by J. Taylor and his colleagues in 1957. If a eukaryote is allowed to undergo one DNA replication in the presence of tritiated (^3H -) thymidine, each of the daughter chromatids would be expected to contain a double helix with one unlabeled DNA template strand and one labeled strand of newly synthesized bases (fig. 15.1). This configuration is expected on the basis of semiconservative replication, with each chromatid containing one double helix. A second round of DNA replication, in the absence of ^3H -thymidine, should produce chromosomes in which one chromatid would have unlabeled DNA and one would have labeled DNA. Figure 15.2 shows the chromosomes after this second replication in nonlabeled media. As expected, one chromatid of every pair is labeled and one is not.

In another kind of experiment, R. Kavenoff, L. Klotz, and B. Zimm demonstrated that *Drosophila* nuclei contained pieces of DNA of the size predicted from their DNA content, based on the premise that each chromosome contains one DNA molecule. They isolated the DNA and measured the size of the largest DNA molecules using the *viscoelastic* property of DNA, the rate at which



Ruth Kavenoff (1944–).
(Courtesy of Dr. Ruth
Kavenoff.)

BOX 15.1

Generally, eukaryotic cells are large, and prokaryotic cells are small. For example, an average eukaryotic cell is about 50 μm in diameter, whereas an average bacterium is about 5 μm in length. The average virus is about 0.05 μm in diameter. These size differences occur because eukaryotic cells have complex substructures and internal architecture that prokaryotic cells lack. Since we believe that prokaryotic cells depend on diffusion to exchange materials with the environment, they would have to be small. And viruses, intracellular parasites, would of necessity be very small. There are, of course, exceptions.

In 1999, a team of scientists from Germany, Spain, and the United States isolated large sulfur bacteria off the Namibian coast of Africa and named them *Thiomargarita namibiensis*, the sulfur pearl of Namibia. These bacteria can be almost half a millimeter in diameter, the size of the period at the end of this sentence (fig. 1). Based on the sequence of 16S ribosomal DNA, these bacteria were shown to be close relatives of other marine sulfur bacteria. They are almost one hundred times the volume of the bacteria previously believed to be largest, *Epuliscium fisbelsoni*, known only from the intestine of the brown surgeonfish.

The smallest prokaryotes are the Mycoplasmas, at about 0.2 μm in diameter, rivaling the viruses in size. They are animal pathogens and decomposing organisms. The smallest eukaryote, *Ostreococcus tauri*, a green alga found in the plankton, was discovered in 1994 from a water sample in a French lagoon on the Mediterranean Sea by a group of French sci-

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Small Is Small?

entists. These organisms are less than 1 μm in diameter. Scientists believe that the lower limit on the size of a cell (not counting viruses) is about 200 nm (0.2 μm), based on the size of DNA and ribosomes that a cell must contain.

With *T. namibiensis* as the largest prokaryote, we note that the largest eukaryotic cell with a single nucleus is most likely the ostrich egg. The largest organisms are the blue whale, *Balaenoptera musculus*, weighing in

at 118,000 kilograms; giant redwood trees, *Sequoiadendron giganteum*, 100 meters tall and weighing 5.5 million kilograms; a quaking aspen clone, *Populus tremuloides*, weighing 6 million kilograms; and *Armillaria bulbosa*, a fungus. In 1992, three scientists from the University of Toronto and Michigan Technological University, using restriction fragment length polymorphisms (RFLPs) and polymerase chain reaction (PCR) techniques, showed that the huge hyphal mass of this tree-root colonizing fungus growing in a forest in northern Michigan was a single organism. It covered about eight hectares, probably weighed more than 10,000 kilograms, and probably has existed for more than 1,500 years.

Although we don't want to get distracted by the oddities and extremes of nature, size differences are remarkable.

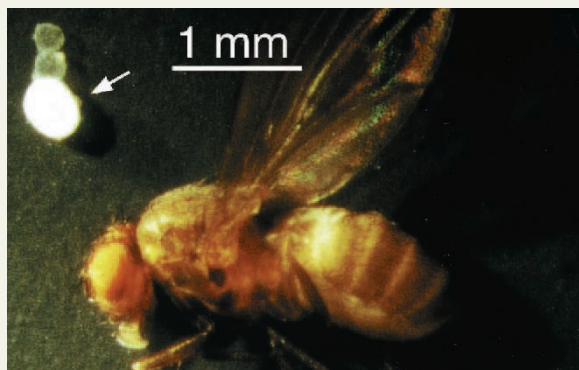


Figure 1 The bacterium *Thiomargarita namibiensis* shown with a fruit fly (*Drosophila virilis*, 3 mm in length) for size comparison. The arrow points to a single bacterial cell, 0.5 mm wide, bright with sulfur inclusions. Above the cell are empty sheaths of dead bacteria. (From H.N. Schulz, et al., "Dense populations of a giant sulfur bacterium in Namibian Shelf Sediments" in *Science*, Vol. 284, pp. 493–95, April 16, 1999. Reprinted by permission of the American Association for the Advancement of Science.)

stretched molecules relax. From other sources, primarily UV absorbance studies, it was estimated that the largest *Drosophila* chromosome had about 43×10^9 daltons of DNA. Results from the viscoelastic measurements indicated the presence of DNA molecules of between 38 and

44×10^9 daltons. Viscoelastic measurements of inversions, which changed the ratio of the arms but not the overall size of the chromosome, yielded similar results. However, a translocation that radically changed the size of the chromosome to 59×10^9 daltons resulted in an

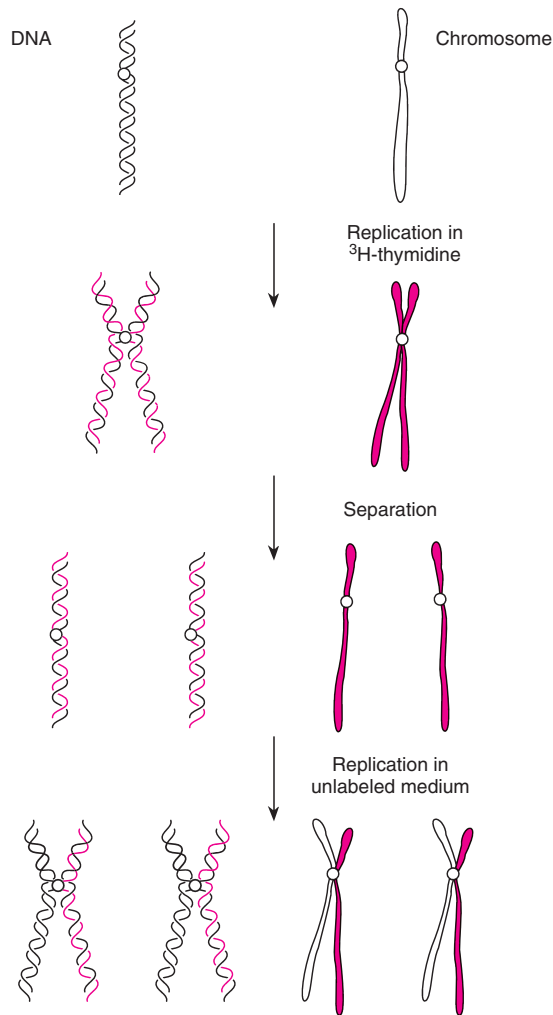


Figure 15.1 Radioactive labeling of a uninematic eukaryotic chromosome following semiconservative replication. Replication occurs first in the presence of ^3H -thymidine and then in its absence. Red represents labeling. After the second round of replication, one chromatid of each chromosome is labeled, whereas the other is not, confirming that there is only one DNA molecule per chromatid and that the chromosome is thus uninematic.

equivalent change in the viscoelastic estimates to between 52 and 64×10^9 daltons.

The conclusion from these studies is that the largest *Drosophila* chromosome, and by extension every eukaryotic chromosome, contains a single DNA molecule running from end to end, encompassing both arms. The viscoelastic values were corroborated by carefully isolating and measuring the lengths of long DNA molecules, an especially difficult task given DNA's propensity to break.



Figure 15.2 Second metaphase in hamster cells in culture after one replication in the presence of ^3H -thymidine followed by one in nonradioactive medium, verifying the uninematic nature of the eukaryotic chromosome. Cases in which the label apparently switches from one chromatid to the other are caused by sister chromatid exchanges (at arrows). (Source: G. Marin and D. M. Prescott, "The frequency of sister chromatid exchanges following exposure to varying doses of ^3H -thymidine or X-ray," *Journal of Cell Biology*, 21, (1964): 159–67, by copyright permission of the Rockefeller University Press.)

The longest molecule that the investigators found was 1.2 cm long, equivalent to between 24 and 32×10^9 daltons (fig. 15.3), close to the predicted size. Thus, the evidence is in complete concordance with the simple uninematic model of eukaryotic chromosomal structure (box 15.2).

Nucleoprotein Composition



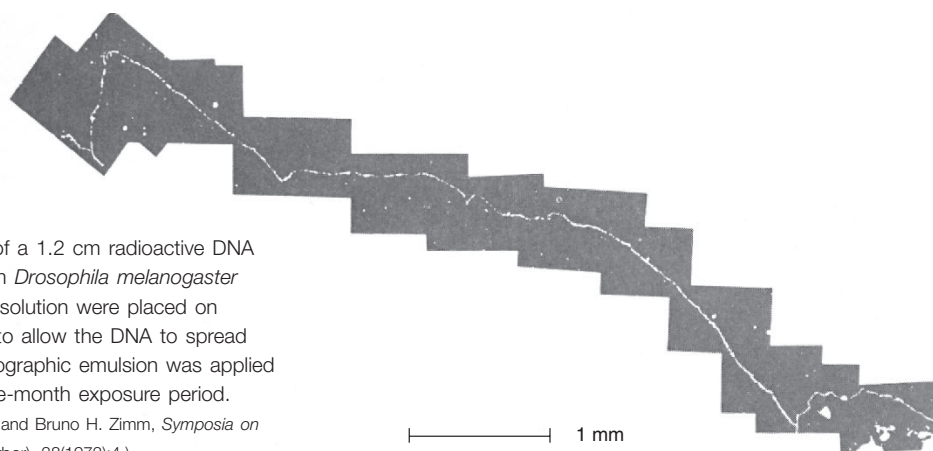
Nucleosome Structure

Since each eukaryotic chromosome consists of a single, relatively long piece of duplex DNA, the average diploid cell contains many of these long pieces of DNA. For chromosomes to be properly distributed to each daughter cell during mitosis and meiosis, they must be condensed into structures that are more easily managed. Wrapping the DNA around "spools" of protein constitutes the first step in a series of coiling and folding processes that eventually result in the fully compacted chromosome we see at metaphase.

Interphase nuclei can be disrupted by placing them in a hypotonic liquid such as water. When this happens, chromatin material is released. When this material is observed under the electron microscope, small particles

Figure 15.3 Autoradiograph of a 1.2 cm radioactive DNA molecule carefully isolated from *Drosophila melanogaster* chromosomes. Drops of DNA solution were placed on microscope slides, then tilted to allow the DNA to spread slowly down the slide. A photographic emulsion was applied and later developed after a five-month exposure period.

(From Ruth Kavenoff, Lynn C. Klotz, and Bruno H. Zimm, *Symposia on Quantitative Biology* (Cold Spring Harbor), 38(1973):4.)



called **nucleosomes** can be seen (fig. 15.4). These are the spools that the DNA is wrapped around. They are made of **histone** proteins and associated DNA (table 15.1). The histones, a group of arginine- and lysine-rich basic proteins, have been well characterized. They are especially well suited to bind to the negatively charged DNA (table 15.2).

When chromatin is treated with micrococcal nuclease, individual nucleosomes can be isolated, indicating that the DNA between nucleosomes is accessible to digestion. The results of these studies indicate that a length of 168 base pairs (bp) of DNA, the core DNA, is intimately associated with the nucleosome, and another 50 to 75 base pairs, depending on species, connects the nucleosomes (linker DNA; fig. 15.5). When the quantities of the various histones were measured, there were two each of histones H2A, H2B, H3, and H4 per nucleosome and only one molecule of histone H1. Reconstitution and degradation studies have indicated that histone H1 is not a necessary component in the formation of nucleosomes. We believe that histone H1 is associated with the linker DNA as it enters and emerges from the nucleosome (fig. 15.6), although its exact position is not known with certainty. Histone H1 may be more off center and internally located than illustrated. The term **chromatosome** has been suggested for the core nucleosome plus the H1 protein, a unit that includes approximately 168 base pairs of DNA. Nucleosomes, then, are a first-order packaging of DNA; they reduce its length and undoubtedly make the coiling and contraction required during mitosis and meiosis more efficient (fig. 15.7).

When DNA is replicated, twice as many nucleosomes are needed since one double helix becomes two. Recent studies indicate that a parental nucleosome is partly disassembled during DNA replication and reassembled on one or the other daughter strand, apparently randomly. The other DNA strand has a new nucleosome constructed

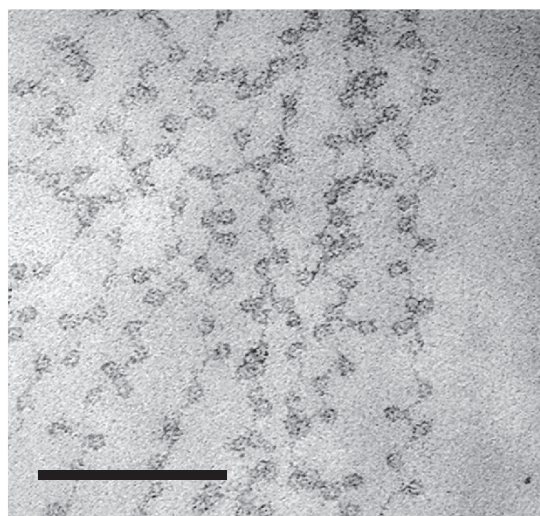


Figure 15.4 Electron micrograph of chromatin fibers. Photo shows nucleosome structures (*spheres*) and connecting strands of DNA called linkers. The bar is 100 nm long. (Source: D. E. Olins and A. L. Olins, "Nucleosomes: The structural quantum in chromosomes," *American Scientist*, 66: 704–11, November 1978. Reproduced by permission.)

Table 15.1 The Constituency of Calf Thymus Chromatin

Constituent	Relative Weight*
DNA	100
Histone proteins	114
Nonhistone proteins	33
RNA	1

* Weight relative to 100 units of DNA.

BOX 15.2

To facilitate the creation of recombinant genomic libraries, for mapping purposes, and for other reasons, it is useful to be able to isolate individual human chromosomes. To these ends, several methods have been developed to isolate chromosomes. Here we discuss a high-speed sorting method based on fluorescent staining and flow cytometry.

DNA can be treated with several fluorescent dyes. Chromosomes can then be recognized individually by their relative fluorescent intensities. The dyes Hoechst 33258 and chro-

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Methods*High-Speed Chromosomal
Sorting*

momycin A3 are a valuable combination because they respond to different wavelengths of light and they bind DNA differently. Hoechst binds preferentially to DNA rich in adenine and thymine, whereas chromomycin

binds preferentially to DNA rich in guanine and cytosine. Thus, since every human chromosome has a unique ratio of bases, the relative intensity of each chromosome is different when fluoresced.

Chromomycin fluoresces in the presence of a laser tuned to 458 nm, and Hoechst fluoresces in the presence of a UV laser. The chromosomes can be identified when their relative fluorescence in the two lasers is plotted, producing a *flow karyotype* (fig. 1). Modern flow cytometry techniques then allow the isolation of these identified chromosomes.

In practice, chromosomes are isolated in large numbers from cells that have been arrested in metaphase by treatment with colcemid, which inhibits spindle formation. These chromosomes are then purified in buffer and treated with the two dyes. The chromosomes are separated at high speed (two hundred chromosomes per second) in a flow cytometry device (fig. 2). As the chromosome-containing buffer passes through the laser beams, identification is made. The liquid is then forced to form minute droplets (215,000 per second) by passing through a vibrator. Specific droplets carrying the identified chromosomes are then charged, either positively or negatively, and passed between deflection plates. Positively charged droplets pass one way, and negatively charged droplets pass the other way, thus allowing the simultaneous isolation of two different chromosomes. At a rate of two hundred chromosomes per second, it

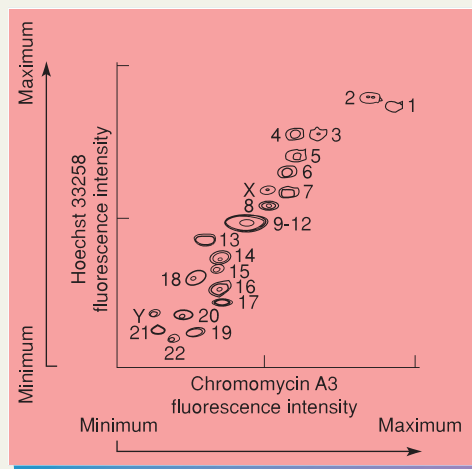


Figure 1 Flow karyotype of human chromosomes at very high resolution, measured under low-speed sorting (fifteen to thirty-five chromosomes per second). The ordinate is Hoechst 33258 fluorescence intensity, and the abscissa is chromomycin A3 fluorescence intensity. All chromosomes are resolved except numbers 9–12. (Reprinted with permission from J. W. Gray, et al., "High-Speed Chromosome Sorting," *Science*, 238:323–329, 1987. Copyright © 1987 American Association for the Advancement of Science.)

of histones from the cellular pool with the help of proteins called **chromatin assembly factors**; at least three of these factors are known.

For example, in fruit flies, a protein complex called the **replication-coupling assembly factor** assembles new nucleosomes. In addition, a protein complex called **condensin** is needed for the condensation of interphase

chromosomes to mitotic chromosomes. This complex includes two **SMC proteins** (for structural maintenance of chromosomes) and two non-SMC proteins. SMC proteins also aid other chromosomal activities, such as mitotic segregation, sister-chromatid adhesion, dosage compensation, and recombination. Thus, a diverse array of proteins is involved in creating nucleosomes and chromato-

is possible to isolate 0.1 g of DNA in less than an hour; 0.1 g of DNA is adequate for library construction and represents about 5×10^5 average chromosomes.

The technique is not perfect. During isolation, debris and clumps of chromosomes are produced that

cause contamination problems. Then, some chromosomes are so similar in their fluorescence that they are hard to separate. This is true, for example, for chromosomes 9 to 12. Also, chromosome 21 is hard to separate because its fluorescence tends to fall into the debris area.

Some of these problems, however, can be overcome by using hybrid cell lines of hamsters, for example, containing only one human chromosome. It is much easier to isolate the human chromosome from the hybrid line. Purity values of 90% are not unreasonable, with some in excess of 95%.

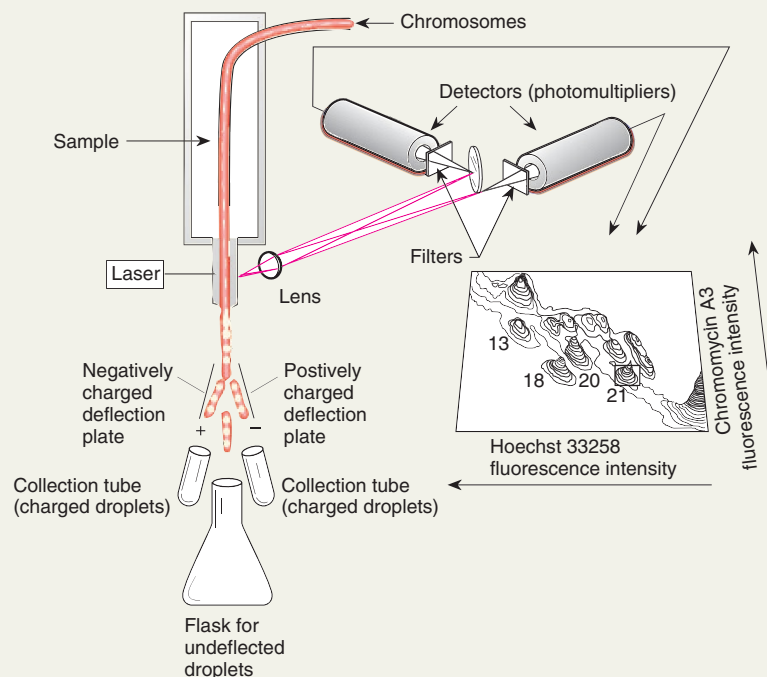


Figure 2 The flow cytometry device used to separate chromosomes at high speed. A buffer with chromosomes enters the device. Lasers cause fluorescence that is analyzed with the aid of the photomultiplier tubes. Droplet formation is induced by vibration, and, based on a flow rate of 50 m/sec, appropriate drops are charged. Charged drops are then separated by charged deflection plates and collected. Uncharged droplets pass through. (Reprinted with permission from J. W. Gray, et al., "High-Speed Chromosome Sorting," *Science*, 238:323–329, 1987. Copyright © 1987 American Association for the Advancement of Science.)

somes, condensing interphase chromosomes, and performing numerous other activities of chromosomes.

Dosage compensation has recently been associated with a change in nucleosome structure. The inactivated X chromosome appears to have a different type of histone present. Histone H2A is replaced by a variant called mH2A. The details of this mechanism are under study.

Nucleosomes apparently play a major role in controlling gene expression; DNA with nucleosomes has a much lower transcription rate than DNA without nucleosomes. It makes sense that the positions of nucleosomes can provide or prevent access to promoters. There are regions of the DNA, known as **nuclease-hypersensitive sites**, that appear to be nucleosome

free. These sites, usually multiples of a nucleosomal region of about two hundred base pairs, are particularly sensitive to digestion by different nucleases. When these regions are isolated, they usually have sequences that control functions in replication, transcription, or other activities of DNA. For example, numerous promoter regions in *Drosophila*, mouse, and human DNA are in nuclease-hypersensitive sites. Hence, some specific DNA sequences are kept free of nucleosomes, and these sequences appear to be recognized by various enzymes such as RNA polymerase. In many other cases, however, nucleosomes do appear to cover promoters and repress transcription. For transcription to occur in these cases, some form of **chromatin remodeling** must take place.

Two general classes of proteins are involved in chromatin remodeling. First are proteins that acetylate the N-terminal tails of the histones, a process that may cause the nucleosomes to bind the DNA less tightly and thus make it available for attachment of transcription factors. These enzymes are called **histone acetyl transferases (HATs)**. Deacetylating enzymes have the reverse effect: They act to repress transcription. Second, a class of ATP-dependent proteins such as the SWI/SNF complex in yeast also affect chromatin remodeling. (Some workers called the proteins SWI because they were involved in mating type *switching*, and others called them SNF for sucrose *nonfermenting*.) The SWI/SNF complex is a group of eleven proteins involved in transcription activation in many genes, presumably allowing transcription

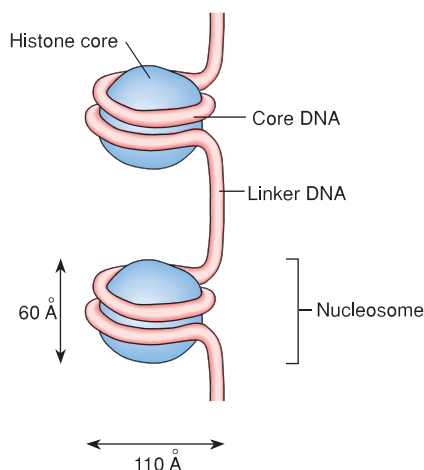


Figure 15.5 The eukaryotic chromosome is associated with histone proteins to form nucleosomes. The protein core is wrapped with 1.7 loops of DNA and connected with a length of DNA called a linker.

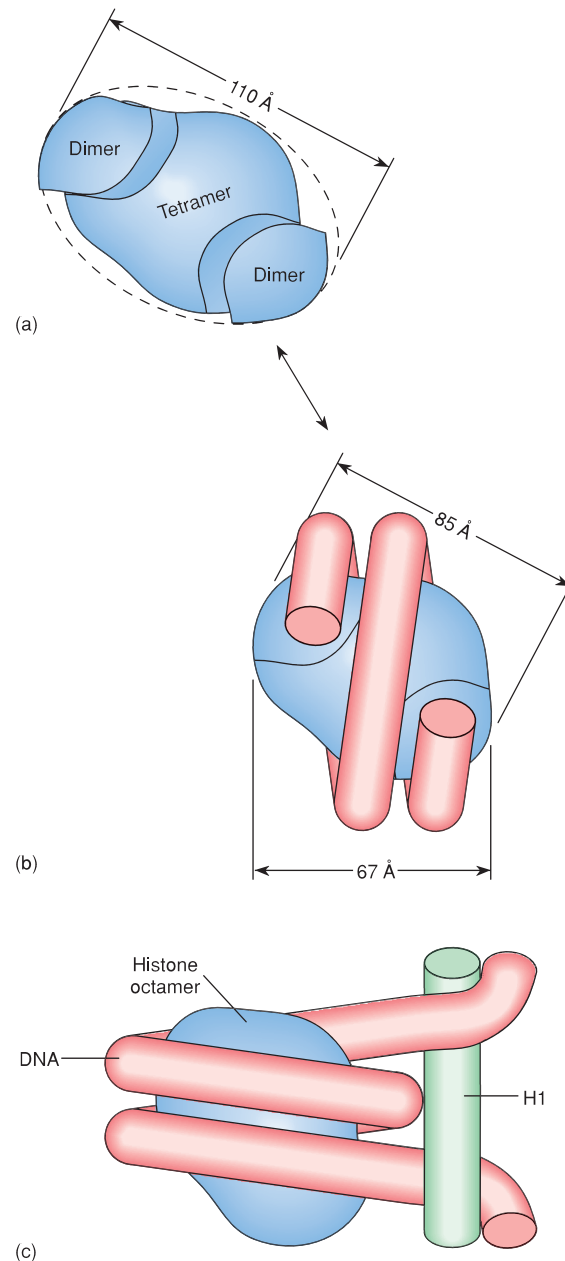


Figure 15.6 Nucleosome structure. (a) Schematic comparison of the eight histones comprising the nucleosome in salt solution. A dimer consists of one H2A and one H2B histone molecule; a tetramer consists of two H3 and two H4 histones. (b) DNA fits in surface grooves on the more compacted structure found in physiological conditions. (c) The diagram shows the presumed position of the H1 histone, encompassing 166 base pairs of DNA.

Table 15.2 Composition of Histones

Fraction	Class	Number of Amino Acids	Percentage of Basic Amino Acids
H1	Very lysine rich	213	30
H2A	Lysine, arginine rich	129	23
H2B	Moderately lysine rich	125	24
H3	Arginine rich	135	24
H4	Arginine, glycine rich	102	27



Figure 15.7 Nucleosome core particle at 2.8 Å resolution. Shown are 146 base pairs of DNA (brown and turquoise) and the eight histone protein chains (purple: H3; green: H4; yellow: H2A; and red: H2B). Note the protein tails (N-terminal ends) of the histone polypeptides extending out of the nucleosome. On the left is the view down the DNA helix, and on the right is the perpendicular view. (From Karolin Luger, et al., "Crystal structure of the nucleosome core particle at 2.8 Å resolution" in *Nature*, 389:251–260, September 18, 1997, fig. 1a p. 252. Reprinted by permission of Macmillan, Ltd.)

factors to access promoters by remodeling chromatin. These proteins are able to reposition a nucleosome on DNA by sliding the nucleosome down the DNA.

We thus conclude that although nucleosomes serve as a general, first-order packing mechanism in eukaryotic DNA, they can be positioned precisely and can attenuate transcription. It is interesting to note that once transcription begins, RNA polymerase apparently moves along nucleosomal DNA by translocation of the histones by 75 to 80 base pairs without disrupting the nucleosome itself. This seems to be accomplished by the RNA polymerase moving the DNA and then re-forming the nucleosome in its wake (fig. 15.8).

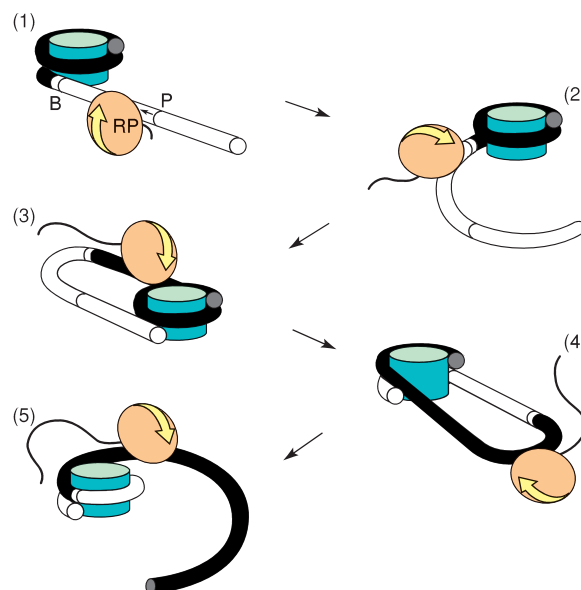


Figure 15.8 RNA polymerase steps around a nucleosome without disrupting it. (1) The RNA polymerase begins at a promoter (*P*) and heads for the nucleosomal DNA (filled in), whose border is noted with a line and the letter *B*. As the polymerase encounters the nucleosome, it begins to unwind the DNA from the histones (2). The displaced DNA then reencounters the histones (3), about seventy-five to eighty base pairs upstream from the original point of nucleosome formation. The polymerase continues on its way (4 and 5), and the nucleosome re-forms in its displaced position without disrupting the histones or ever fully losing contact with the core DNA. (From Vasily M. Studitsky, et al., "A histone octamer can step around a transcribing polymerase without leaving the template," *Cell*, 76: 371–82, January 28, 1994. Copyright © 1994 by Cell Press. Reprinted by permission.)

Higher-Order Structure of Chromatin

Since the nucleosome has a width of only 110 Å, and metaphase chromosomes appear to be constructed of a fiber having a diameter of about 2,400 Å (fig. 15.9), several additional levels of chromatin compaction lead to the metaphase chromosome. Various experiments, which change the ionic strength the chromatin is subjected to, indicate that the 110 Å DNA spontaneously forms a 300 Å, solenoidlike fiber with increased ionic strength. It seems that this fiber results from the coiling of the nucleosomal DNA (fig. 15.10). This 300 Å fiber is not, however, the final form of the DNA. We can account for the contraction of the 300 Å fiber to the 2,400 Å fiber found in metaphase chromosomes by the formation of a second solenoidlike structure from the winding of the 300 Å fiber (fig. 15.11).

If the histones are removed from a chromosome, the DNA billows out, leaving a proteinaceous structure termed a **scaffold** (fig. 15.12). This scaffold structure is formed from **nonhistone proteins**; two of them predominate, namely SC1 and SC2. SC1 has been identified

as topoisomerase II. It would not be unreasonable to expect several hundred different proteins, many in minute quantities, to be associated with the chromosome and involved in replication, repair, and transcription.

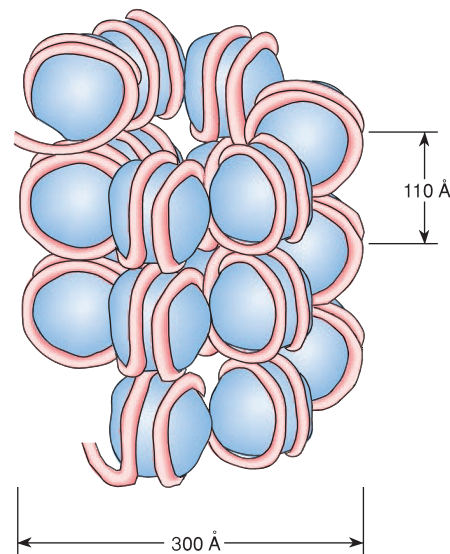


Figure 15.10 Solenoid model for the formation of the 300 Å chromatin fiber. Nucleosomal DNA wraps in a helical fashion, forming a hollow core. Although histone H1 is not shown, it is known to be on the inside of the solenoid.



Figure 15.9 Chinese hamster chromosome. Note the fibers making up the chromosome; they are approximately 2,400 Å in diameter. Magnification 11,800X. (Source: Courtesy of Dr. Hans Ris.)

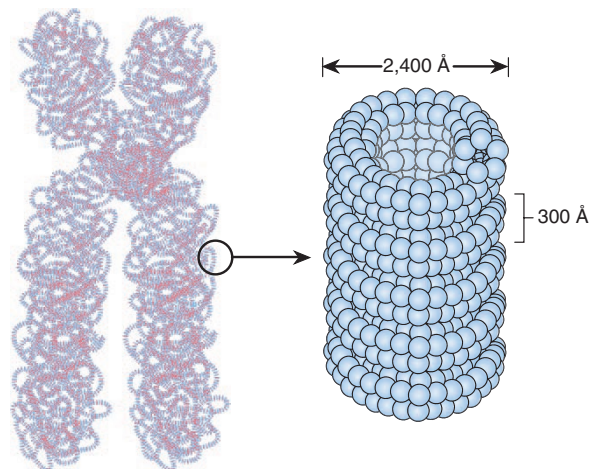


Figure 15.11 The 2,400 Å fiber of the eukaryotic chromosome is a hollow, solenoidlike structure. It is formed by the coiling of the 300 Å fiber, which itself is a solenoid.

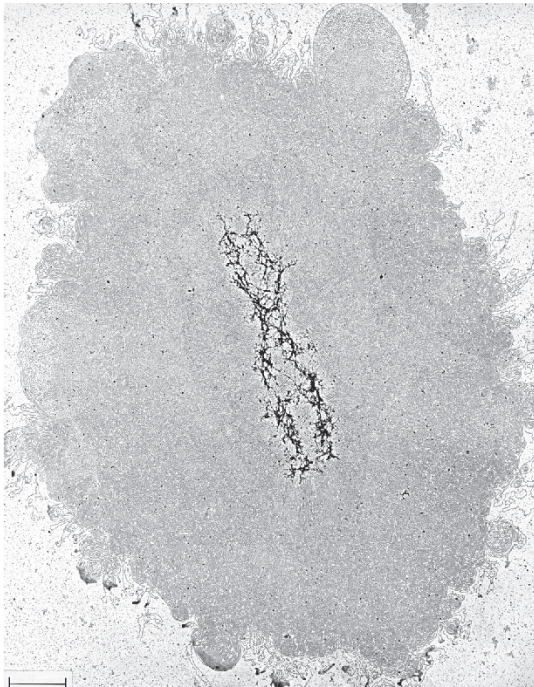


Figure 15.12 Scaffold protein. When the histones are removed from a eukaryotic chromosome, a fibrous scaffold remains. The DNA loops out from this scaffold. The bar is 2 μm long. (J. Paulson and U. Laemmli, "The structure of histone-depleted metaphase chromosomes," *Cell*, 12:817–28, 1977. Micrograph courtesy of James R. Paulson.)

Polyteny, Puffs, and Balbiani Rings

Drosophila's salivary glands, as well as some other tissues of *Drosophila* and other diptera, contain giant banded chromosomes (see fig. 6.12) that result from the replication of the chromosomes and the synapsis of homologues without cell division (endomitosis). These chromosomes consist of more than one thousand copies of the same chromatid and appear as alternating dark bands and lighter interband regions. The dark bands are referred to as *chromomeres*. Also seen are diffuse areas called **chromosome puffs** (fig. 15.13). Chromosome puffs are also referred to as **Balbiani rings**. These rings were originally defined as puffs in the midge, *Chironomus*, whose polytene chromosomes were discovered by E. G. Balbiani in 1881. Currently, the term applies to all puffs, or at least the larger puffs, in all species with polytene chromosomes.

The structure of the polytene chromosome can be explained by the diagram in figure 15.14. Dark bands (chromomeres) are due to tight coiling of the 300 Å fiber; light interband regions are due to looser coiling. The figure

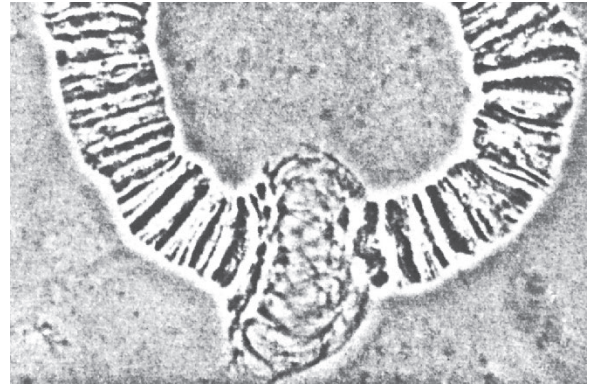


Figure 15.13 A chromosome puff on the left arm of chromosome 3 of the midge *Chironomus pallidivittatus*. (Jan-Erik Edström, et al., *Developmental Biology* 91:131–37, 1982, Figure 1B, Academic Press.)

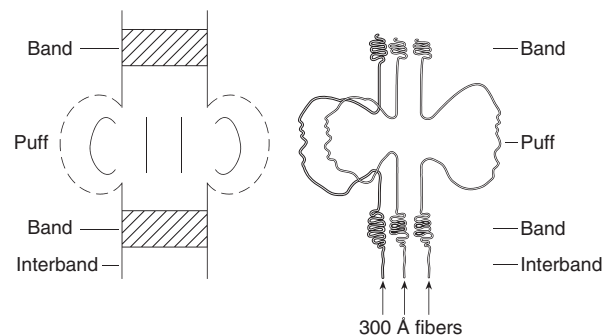


Figure 15.14 Polytene chromosome with bands and a puff. Three of the approximately one thousand synapsed chromatids are shown diagrammatically on the right.

also shows how chromosome puffs would come about as fibers unfold in regions of active transcription.

Staining with reagents specific for RNA, such as toluidine blue, or autoradiography with tritiated (^3H) uridine, have been used to demonstrate that active transcription is going on in the puffs but not in neighboring regions of the polytene chromosomes. The messenger RNA isolated from cells with puffs has also been shown to hybridize only to the puffed regions of the chromosomes. Thus, these regions of the DNA are complementary to the messenger RNA (fig. 15.15) and represent areas of active transcription. Modern recombinant DNA techniques have also shown that many puffs probably represent the transcription of only one gene, although there are exceptions.

Puffs generally fall into four categories. *Stage-specific puffs* appear during a certain stage of development, such

as molting. *Tissue-specific puffs* are active in one tissue but not another. (In dipteran larvae, tissues other than the salivary glands, such as the midgut and Malpighian tubules, have polytene chromosomes.) *Constitutive puffs* are active almost all the time in a specific tissue. And *environmentally induced puffs* appear after some environmental change, such as heat shock (fig. 15.16). In *Drosophila*, about 80% of the puffs are stage specific; in *Chironomus*, only about 20% are. For example, at the time of molt in insects, the hormone ecdysone is secreted by the prothoracic gland. At the same time, many puff patterns change (fig. 15.17). Similar changes in puff patterns can be induced by the injection of ecdysone. Hence, molting, a stage-specific developmental sequence, is related to a sequential transcription sequence in the chromosomes.

Lampbrush Chromosomes

Lampbrush chromosomes, which occur in amphibian oocytes, are so named because their looped-out configuration has the appearance of a brush for cleaning lamps, now a relatively uncommon household item (fig. 15.18). The loops of the lampbrush chromosomes are covered by an RNA matrix and are the sites of active transcription. Presumably, the loops are unwindings of the single chromosome, similar to the unwindings in the polytene chromosome shown in figure 15.14. Thus, under certain circumstances, such as in polytene chromosomal puffs



Figure 15.15 Hybridization at a *Chironomus tentans* salivary gland chromosome puff. The chromosomal DNA is hybridized with labeled RNA (black dots) transcribed from the locus. The activity of the locus is forming the puff. (Reprinted by permission from B. Lambert, "Repeated DNA sequences in a Balbiani ring," *Journal of Molecular Biology*, 72:65–75, 1972. Copyright by Academic Press, Inc. (London) Ltd.)

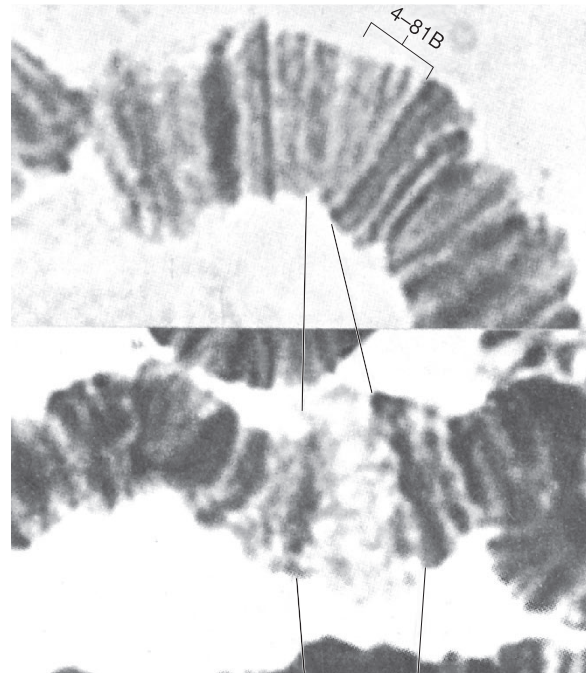


Figure 15.16 Puff 4-81B of the salivary gland in *Drosophila hydei* is induced by heat shock (37° C for one-half hour). At the top, normal activity. At the bottom, temperature shock in vitro, resulting in the puff. (Source: H. D. Berendes, et al., "Experimental puffs in salivary gland chromosomes of *Drosophila hydei*," *Chromosoma* [Berl.] 16:35–46, Fig. 4a–b, 1965. © Springer-Verlag.)

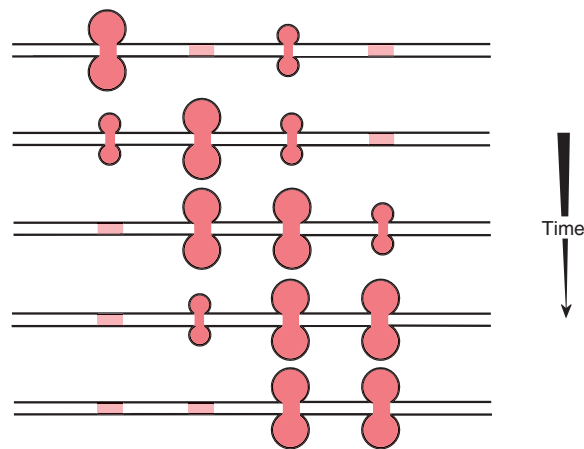


Figure 15.17 Puff patterns on a segment of a *Chironomus tentans* salivary gland chromosome during molt. As time proceeds, puffs appear and disappear and change in size.

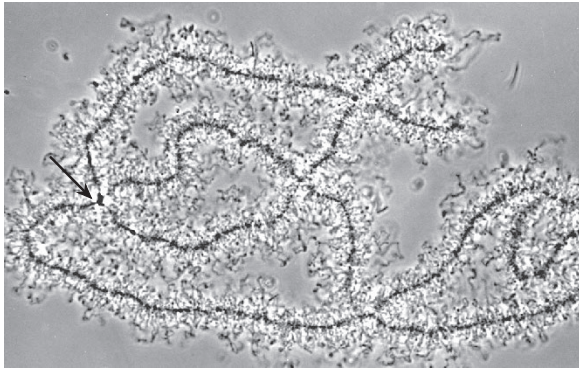


Figure 15.18 Lampbrush chromosome of the newt, *Notophthalmus viridescens*. Centromere is at the left (arrow); the two long homologues are held together by three chiasmata. Magnification 238 $\times$. (Source: Joseph G. Gall, figure 2 in D. M. Prescott, ed., *Methods in Cell Physiology*, vol. 2 [New York: Academic Press, 1966], 39. Reproduced by permission.)

and in lampbrush chromosomes, active transcription can be seen in the light microscope. Since only certain bands puff at any one moment in polytene chromosomes, and since the loops of lampbrush chromosomes are of various sizes (with some regions not looped at all), we have evidence of specific transcription. However, we have no indication, so far, of the nature of the control of that transcription.

Chromosomal Banding

Several chromosomal staining techniques reveal consistent banding patterns. By means of these patterns, all of the human chromosomes can be differentiated (see fig. 5.1). Of possibly greater importance is the fact that these staining techniques have provided some insight into the structure of the chromosome. The techniques for staining the C, G, and R chromosomal bands will serve as an illustration.

G-bands are obtained with **Giemsa stain**, a complex of stains specific for the phosphate groups of DNA. Treatment of fixed chromatin with trypsin or hot salts brings out the G-bands. Giemsa stain enhances banding that is already visible in mitotic chromosomes. The banding pattern is caused by the arrangement of chromomeres. Under careful observation, the major G-bands prove to consist of many smaller chromomeres. This banding appearance has led D. Comings to suggest the mechanism of chromosomal folding shown in figure 15.19.

C-bands are Giemsa-stained bands after the chromosomes are treated with NaOH. The **C** is for “centromere,” because these bands represent constitutive heterochro-

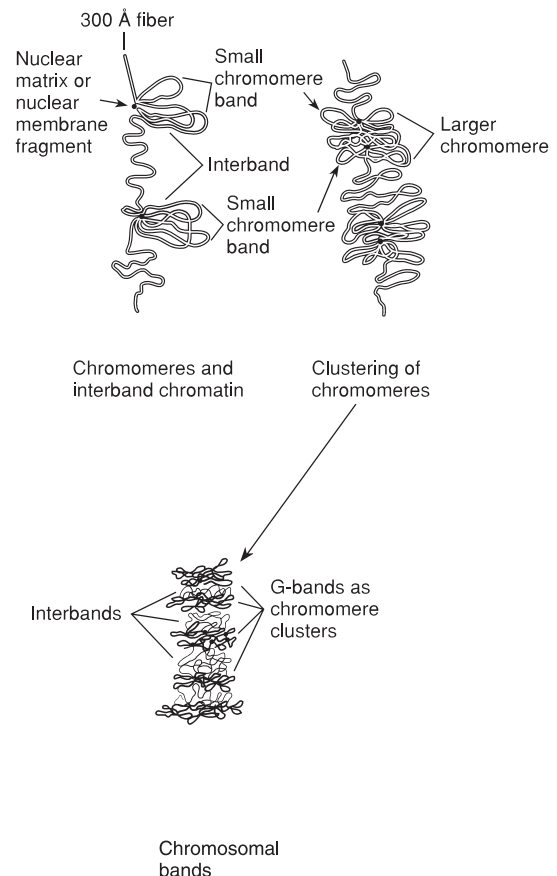


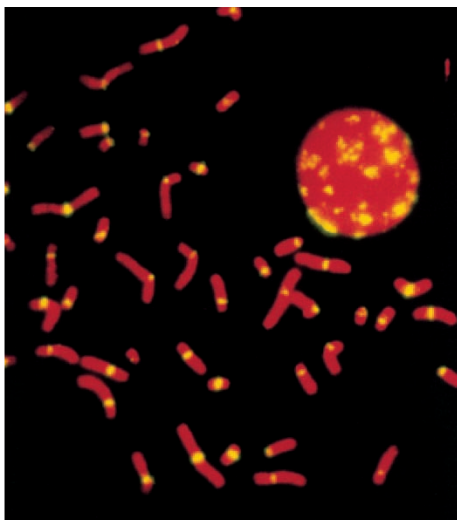
Figure 15.19 Model of eukaryotic (mammalian) chromosomal banding. G-bands are chromomere clusters, which result from the contraction of smaller chromomeres. These, in turn, result from looping of the 300 Å fiber. (Reproduced with permission, from the *Annual Review of Genetics*, Volume 12, © 1978 by Annual Reviews, Inc.)

matin surrounding the centromeres (fig. 15.20). The DNA is also usually satellite rich. **Satellite DNA** differs in **buoyant density** from the major portion of cellular DNA. When eukaryotic DNA is isolated and centrifuged in CsCl, forming a density gradient, the majority of the DNA forms one band in the gradient at a single buoyant density. The buoyancy is determined by the G-C content of the DNA. However, smaller secondary bands are also usually present, indicating regions of DNA having sequences different from the majority of the cell's DNA (fig. 15.21). DNA isolated this way is referred to as **satellite DNA** because of the secondary, or satellite, bands formed in the density gradient. As we will see, this DNA is found primarily around centromeres and consists of numerous repetitions of a short sequence.

R-bands are visible with a technique that stains the regions between G-bands. The chromosomes are fixed, stained with Giemsa, and then viewed with a phase contrast microscope. Since the dark-light pattern is the opposite of the G-band pattern, these bands are called *reverse bands*.



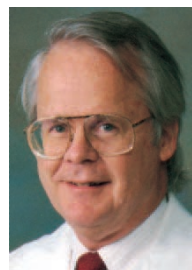
(a)



(b)

Figure 15.20 (a) C banding of chromosomes from a cell in the bone marrow of the house mouse, *Mus musculus*. The arrow indicates that the Y chromatids have already separated into two chromosomes. (b) Yellow fluorescence indicates a satellite DNA probe in human chromosomes (centromeres). ([a] B. Vig, "Sequence of centromere separation: Role of centromeric heterochromatin," *Genetics*, 102:795–806, 1982. [b] Photograph Courtesy of Oncor, Inc. Gaithersburg, Maryland.)

From the information gleaned from these staining techniques, D. Comings distinguished between three basic chromatin types: euchromatin, constitutive heterochromatin, and intercalary heterochromatin (table 15.3). Presumably, the only chromatin involved in transcription is **euchromatin**. **Constitutive heterochromatin** surrounds the centromere and is rich in



David E. Comings (1935–).
(Courtesy Dr. David E. Comings.)

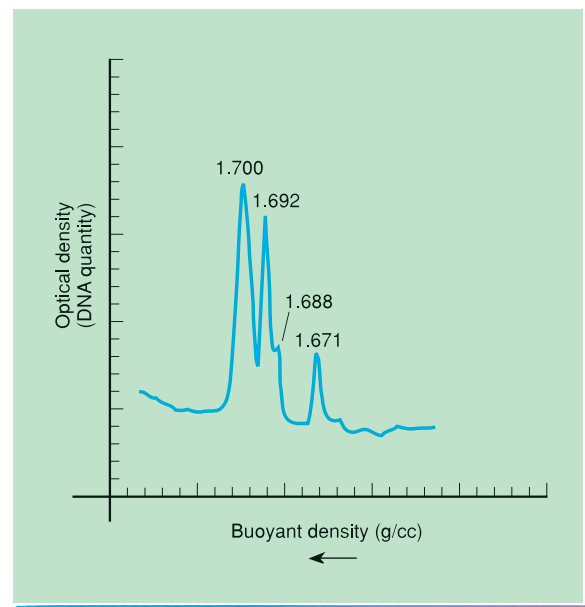


Figure 15.21 Satellite DNA in *Drosophila virilis*. The quantity of DNA is graphed against the buoyant density (g/cc), resulting in four peaks. The large peak (at left) is the major DNA component of the cell; the other three bands are satellite DNA. The left-most of the satellite peaks (1.692) is DNA with a repeating sequence of ACAAAC; the middle satellite peak (1.688) is a sequence of ATAAAC; and the right-most satellite peak (1.671) has a sequence of ACAAAT. (From Joseph G. Gall, et al., *Cold Spring Harbor Laboratory Symposia on Quantitative Biology*, 38:417–21. Copyright © 1974 Cold Spring Harbor Laboratory, Cold Spring Harbor, NY. Reprinted by permission.)

Table 15.3 The Three Major Types of Chromatin in Eukaryotic Chromosomes

	Euchromatin	Centromeric Constitutive Heterochromatin	Intercalary Heterochromatin
Relation to bands	In R-bands	In C-bands	In G-bands
Location	Chromosome arms	Usually centromeric	Chromosome arms
Condition during interphase	Usually dispersed	Condensed	Condensed
Genetic activity	Usually active	Inactive	Probably inactive
Relation to chromomeres	Interchromomeric	Centromeric chromomere	Intercalary chromomeres

satellite DNA. **Intercalary heterochromatin** is found throughout the chromosome. Thus, it becomes apparent that the eukaryotic chromosome is a relatively complex structure.

Centromeres and Telomeres

Centromeres

Two regions of the eukaryotic chromosome have specific functions—the centromere and the telomeres. The centromere is involved in chromosomal movement during mitosis and meiosis, whereas the telomeres terminate the chromosomes. As we pointed out in chapter 3, the terms *centromere* and *kinetochore*, while occasionally used interchangeably, are distinct. The kinetochore is the interface between the visible constriction in the chromosome (the centromere) and the microtubules of the spindle. The kinetochore of higher organisms (e.g., mammals) contains proteins and some RNA. Microscopically, it is a trilaminar structure, attached to chromatin at the inner layer and to microtubules at the outer layer (fig. 15.22).

Most of our knowledge of the genetics of centromeres has come from work in yeast (*Saccharomyces cerevisiae*). Cells did not maintain most artificially created yeast plasmids because they were lost during mitosis. However, plasmids were isolated that did replicate normally during cell division. Presumably, they contained centromeres, allowing them to replicate and move in synchrony with the host's chromosomes. Further genetic engineering made it possible to isolate smaller and smaller regions that could serve as centromeres. After sequencing the centromeres of fifteen of the sixteen yeast chromosomes, it was possible to conclude that the centromere from yeast is about 250 base pairs long with three consensus regions (fig. 15.23); we are defining a centromere as a sequence of DNA called the *CEN* locus or CEN region. Recent data indicate that this region may contain a single, modified nucleosome associated with

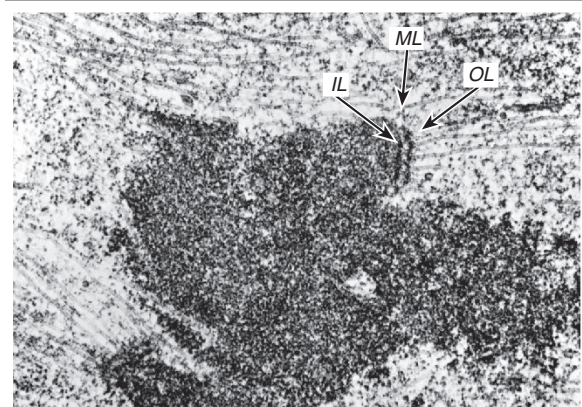


Figure 15.22 The kinetochore of a metaphase chromosome of the rat kangaroo. *IL*, *ML*, and *OL* refer to *inner*, *middle*, and *outer layers*, respectively, of the kinetochore. Note the microtubules attached to the kinetochore and the large mass of dark-staining chromatin making up most of the figure. Magnification 30,800 $\times$. (From B. R. Brinkley and J. Cartwright, Jr., *J. Cell Biology*, 50:416–31, 1971.)

region II. The 250 base-pair length of the CEN regions of yeast chromosomes is about 200 Å, the same as the diameter of a microtubule, indicating that only one microtubule attaches to each centromere during mitosis or meiosis in a yeast cell. This region is called a **point centromere** (fig. 15.24).

Higher eukaryotes have larger centromeric regions that attach more microtubules. These regions are referred to as **regional centromeres** (see figs. 15.22 and also 3.12). Regional centromeres range from nineteen to one hundred kilobases (kb; 19,000–100,000 bases) with unique and satellite (repeated sequence) DNA that is heterochromatic and may include expressed genes. We know much less about regional centromeres than we do about point centromeres.

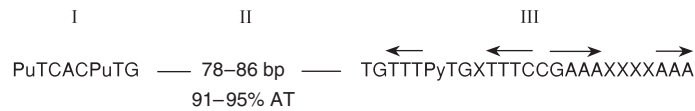


Figure 15.23 Consensus sequence for the three regions (I–III) of fifteen yeast centromeres. *Pu* represents any purine, *Py* represents any pyrimidine, and *X* represents any base. The arrows appear over inverted repeat sequences. (Source: Data from L. Clarke and J. Carbon, "The structure and function of yeast centromeres," *Annual Review of Genetics*, 19:29–56, 1985.)

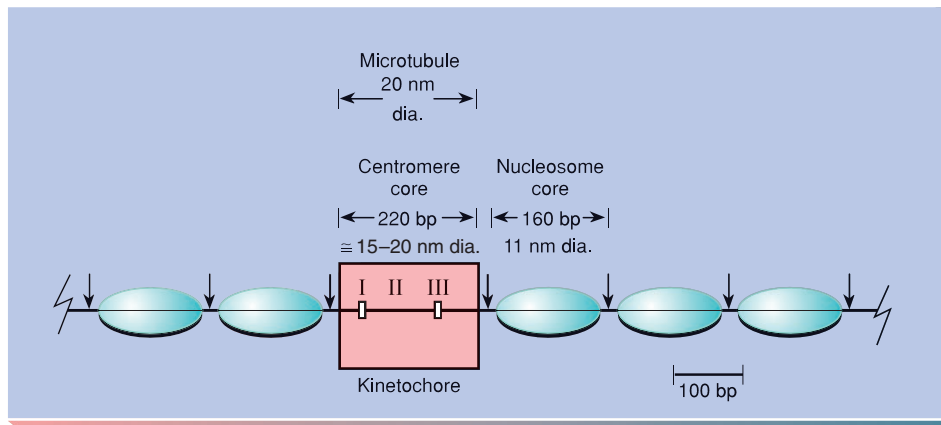


Figure 15.24 Schematic view of a yeast centromeric region. The arrows are the nuclease-hypersensitive sites. A microtubule is about the same width as the centromeric region. (With permission, from the *Annual Review of Genetics*, Volume 19 © 1985 by Annual Reviews www.AnnualReviews.org)

Telomeres

Since eukaryotic chromosomes are linear, each has two ends, referred to as **telomeres**, that not only mark the termination of the linear chromosome but also have several specific functions (fig. 15.25). Telomeres must prevent the chromosomal ends from acting in a "sticky" fashion, the way that broken chromosomal ends act (see chapter 8). In other words, chromosomal ends must not elicit a DNA repair response (see chapter 12). Telomeres must also prevent the ends of chromosomes from being degraded by exonucleases and must allow chromosomal ends to be properly replicated.

Most telomeres isolated so far are repetitions of sequences of five to eight bases. In human beings, the telomeric sequence is TTAGGG, repeated 300 to 5,000 times at the end of each chromosome. The human telomere was discovered by R. Moyzis and his colleagues when they probed the highly repetitive segment of human DNA. (Highly repetitive DNA, as its name implies, consists of numerous copies of a single sequence and usually comprises the satellite components of the cell's DNA; see next section.) When a probe for this sequence was applied to human chromosomes, the sequence was

found at the tip of each chromosome in roughly the same quantity (fig. 15.26). This is a highly conserved sequence, found in all vertebrates studied as well as in unicellular trypanosomes. Similar sequences are found in various other eukaryotes (table 15.4); the first sequence was isolated by E. Blackburn and J. Gall in 1978.

When a linear DNA molecule is replicated, the 3' → 5' strand can be replicated to the end (see chapter 9). The 5' → 3' strand, however, is replicated with RNA primers that are then degraded, leaving a short gap on the progeny strand (fig. 15.27). It is always the G-rich strand of telomeric DNA that ends up single-stranded, forming a 3' overhang of twelve to sixteen nucleotides. Thus, the normal replication process of a linear DNA molecule leaves an incomplete terminus. Hence, scientists suspected that there would be a unique mechanism for the replication of telomeres.

Telomeric sequences appear to be added *de novo* without, DNA template assistance by an enzyme called **telomerase**, discovered by E. Blackburn and her colleagues. This was seen when telomeres from another species were engineered into yeast cells. After a cell cycle, the yeast telomeric sequence had been added on at